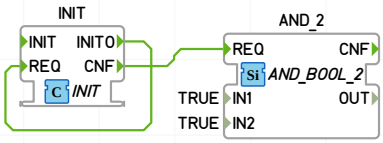


Uebung_000

ADD



Uebung_000b

AND



Uebung_001

DigitalInput_I1 auf DigitalOutput_Q1

Uebung_001c

DigitalInput_I1 auf DigitalOutput_Q1 --> Eingang abfragen bei Boot.

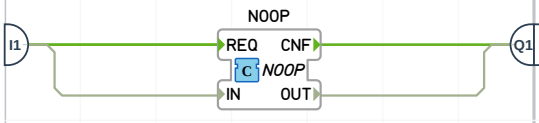


Uebung_001c2

DigitalInput_I1 negiert auf DigitalOutput_Q1 --> Eingang abfragen bei Boot.

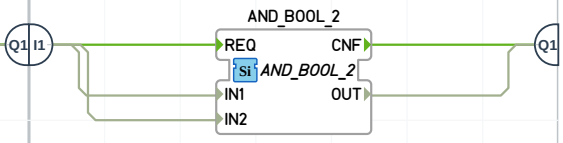
Uebung_001c3

DigitalInput_I1 auf DigitalOutput_Q1 --> Eingang abfragen bei Boot.



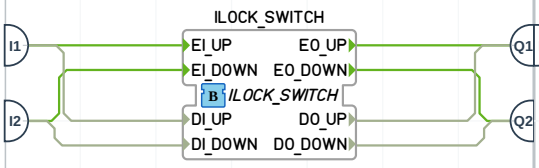
Uebung_001c4

DigitalInput_I1 auf DigitalOutput_Q1 --> Eingang abfragen bei Boot.



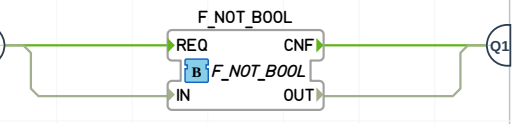
Uebung_001d

DigitalInput_I1 auf DigitalOutput_Q1



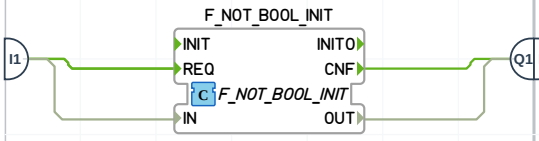
Uebung_001d2

DigitalInput_I1/2 auf DigitalOutput_Q1/2, als Alternative(Verriegelt) mit Interlock, ohne ECC



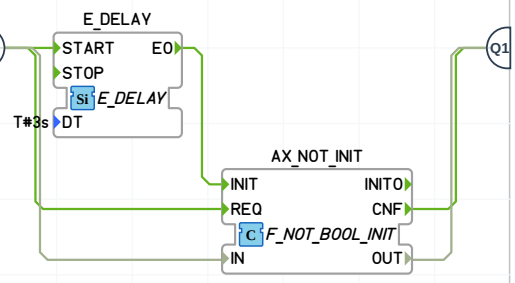
Uebung_001e

DigitalInput_I1 negiert auf DigitalOutput_Q1



Uebung_001f

DigitalInput_I1 negiert mit INIT auf DigitalOutput_Q1



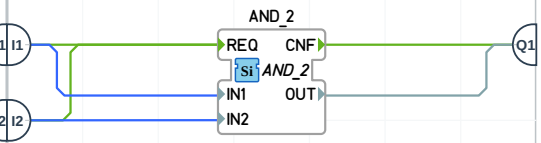
Uebung_001g

DigitalInput_I1 negiert mit INIT und Delay auf DigitalOutput_Q1



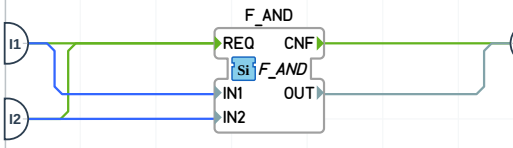
Uebung_002

DigitalInput_I1 auf DigitalOutput_Q1/_Q2



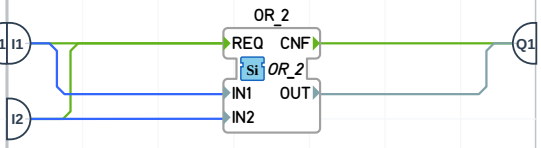
Uebung_002a

DigitalInput_I1/_I2 mit AND auf DigitalOutput_Q1



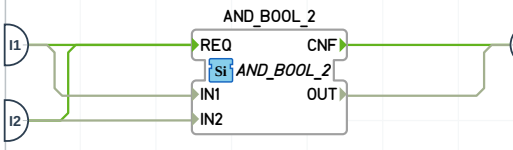
Uebung_002a2

DigitalInput_I1/_I2 mit AND (alt) auf DigitalOutput_Q1



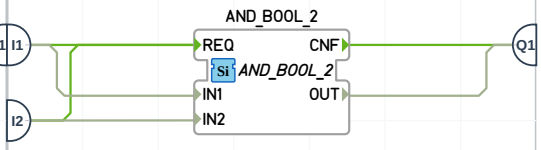
Uebung_002a3

DigitalInput_I1/_I2 mit OR auf DigitalOutput_Q1



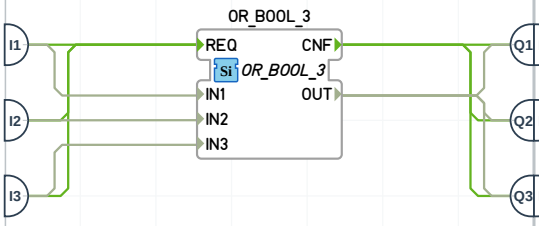
Uebung_002a4

DigitalInput_I1/_I2 mit AND_BOOL auf DigitalOutput_Q1



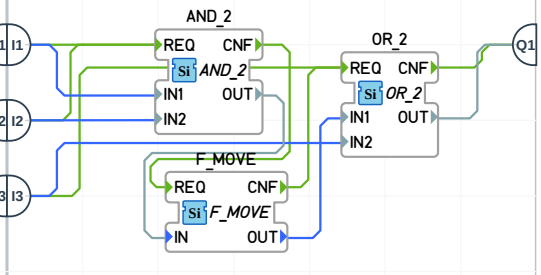
Uebung_002a4b

DigitalInput_I1/_I2 mit AND_BOOL und Negate auf DigitalOutput_Q1



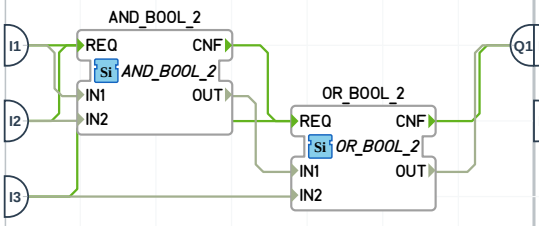
Uebung_002a5b

DigitalInput_I1–3 mit OR auf DigitalOutput_Q1–3



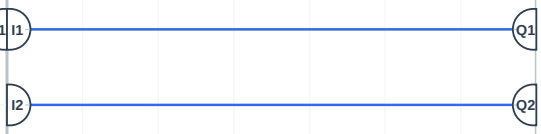
Uebung_002b2

DigitalInput_I1/_I2/_I3 auf DigitalOutput_Q1; mit MOVE



Uebung_002b3

DigitalInput_I1/_I2/_I3 auf DigitalOutput_Q1; ohne MOVE



Uebung_003

DigitalInput_I1/_I2 auf DigitalOutput_Q1/_I2 – flach



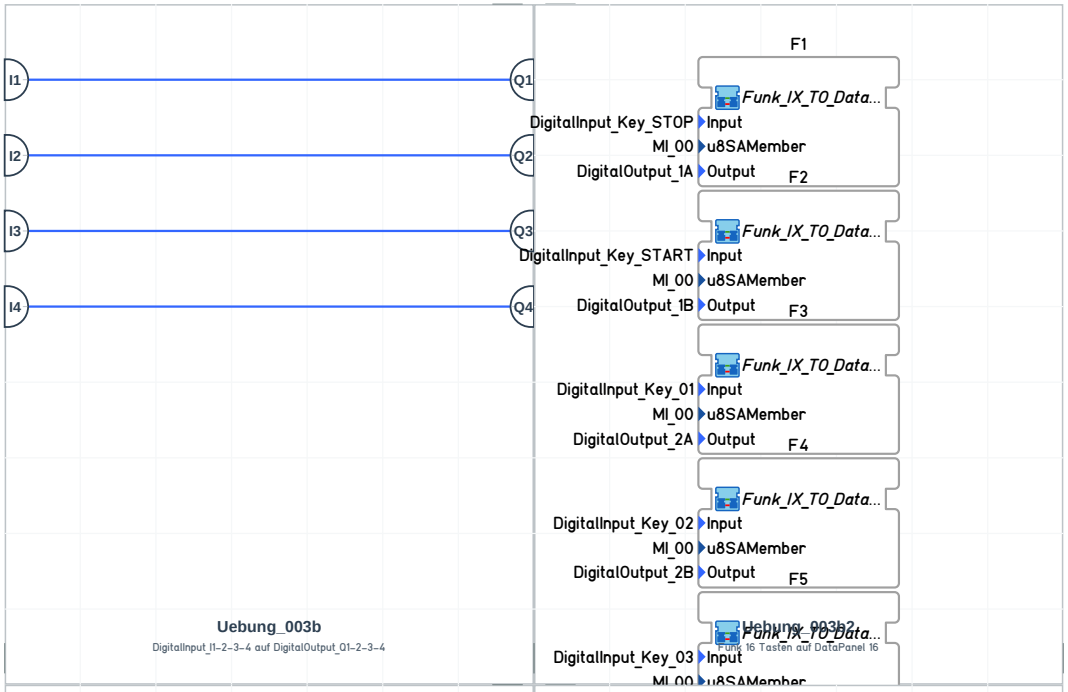
Uebung_003a

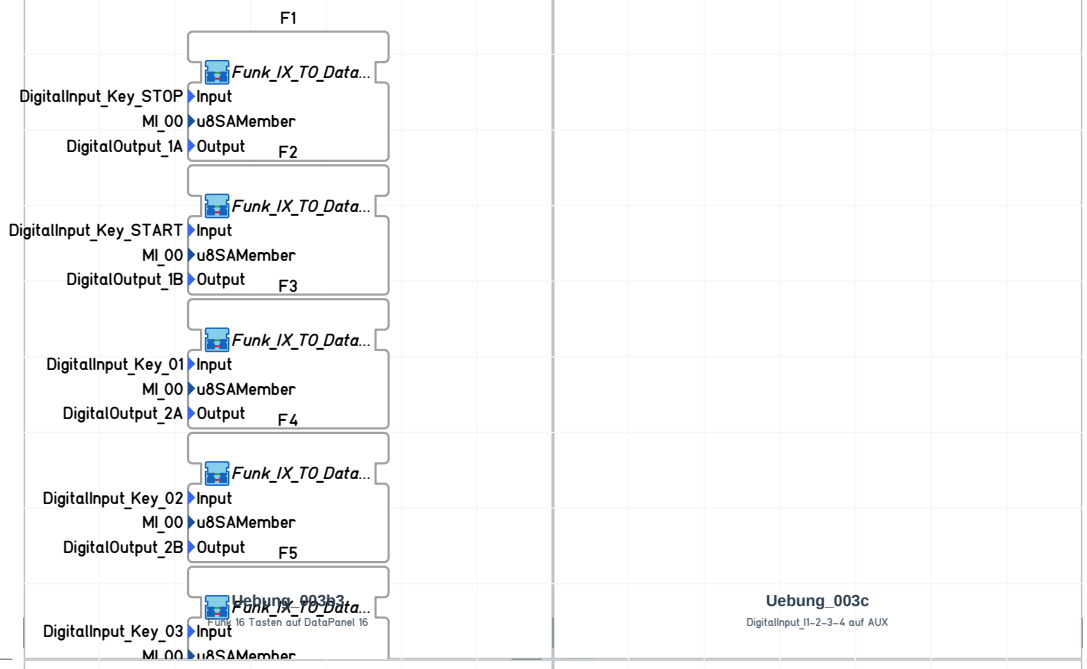
DigitalInput_I1/_I2 auf DigitalOutput_Q1/_I2 – mit typed Subapp

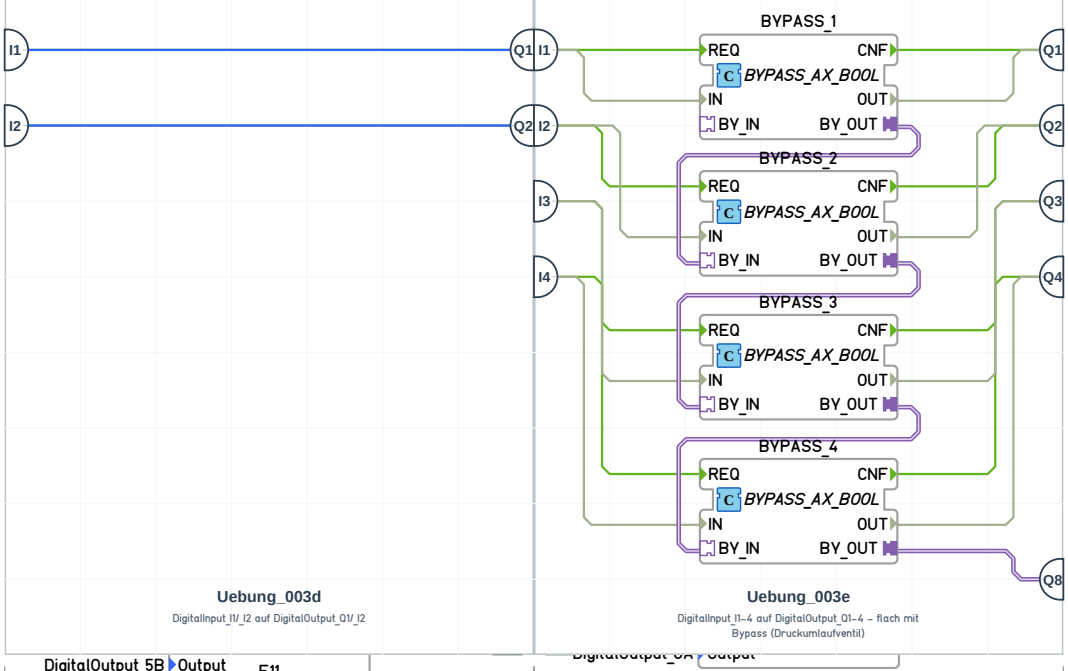


Uebung_003a0

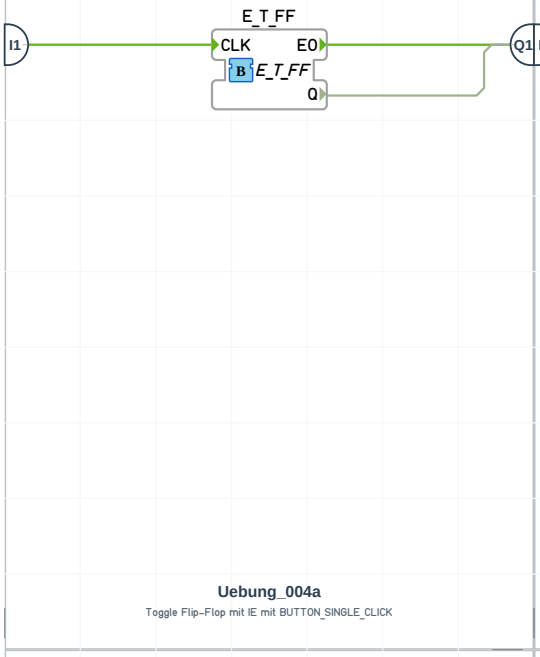
DigitalInput_I1/_I2 auf DigitalOutput_Q1/_I2 – mit untyped Subapp


I1
I2
I3
I4
Q1
Q2
Q3
Q4
F1
Funk_IX_TO_Data...
DigitalInput_Key_STOP
Input
MI_00
u8SAMember
DigitalOutput_1A
Output
F2
Funk_IX_TO_Data...
DigitalInput_Key_START
Input
MI_00
u8SAMember
DigitalOutput_1B
Output
F3
Funk_IX_TO_Data...
DigitalInput_Key_01
Input
MI_00
u8SAMember
DigitalOutput_2A
Output
F4
Funk_IX_TO_Data...
DigitalInput_Key_02
Input
MI_00
u8SAMember
DigitalOutput_2B
Output
F5
Uebung_003b
DigitalInput_I1-2-3-4 auf DigitalOutput_Q1-2-3-4


F1
Funk_IX_TO_Data...
DigitalInput_Key_STOP
Input
MI_00
u8SAMember
DigitalOutput_1A
Output
F2
Funk_IX_TO_Data...
DigitalInput_Key_START
Input
MI_00
u8SAMember
DigitalOutput_1B
Output
F3
Funk_IX_TO_Data...
DigitalInput_Key_01
Input
MI_00
u8SAMember
DigitalOutput_2A
Output
F4
Funk_IX_TO_Data...
DigitalInput_Key_02
Input
MI_00
u8SAMember
DigitalOutput_2B
Output
F5
Uebung_003b2
Funk_IX_TO_Data...
Funk 16 Tasten auf DataPanel 16
DigitalInput_Key_03
Input
MI_00
u8SAMember
Uebung_003c
DigitalInput_I1-2-3-4 auf AUX

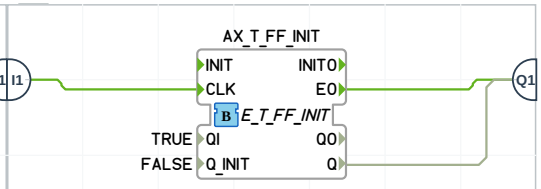

I1
I2
Q1
Q2
BYPASS_1
REQ
CNF
C
BYPASS_AX_BOOL
IN
OUT
BY_IN
BY_OUT
BYPASS_2
BYPASS_3
BYPASS_4
I3
I4
Q3
Q4
Q8
Uebung_003d
DigitalInput_I1/_I2 auf DigitalOutput_Q1/_I2
Uebung_003e
DigitalInput_I1-4 auf DigitalOutput_Q1-4 - flach mit Bypass (Druckumlaufventil)


DigitalOutput_5B
Output
F11
Funk_IX_TO_Data...
DigitalInput_Key_09
Input



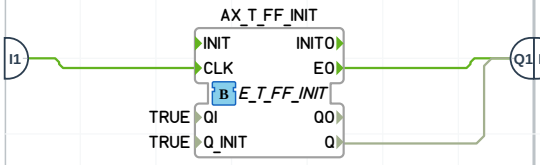
Uebung_004a

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK



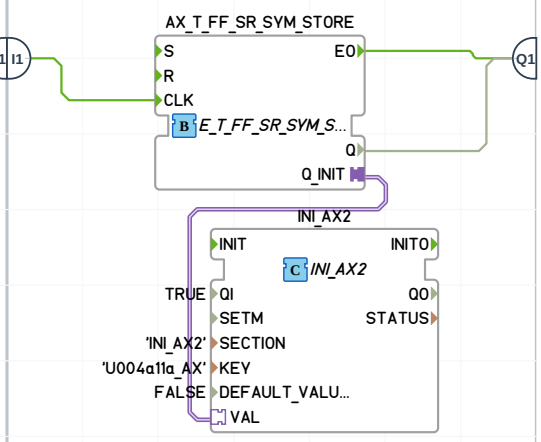
Uebung_004a10a

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK und INIT auf FALSE



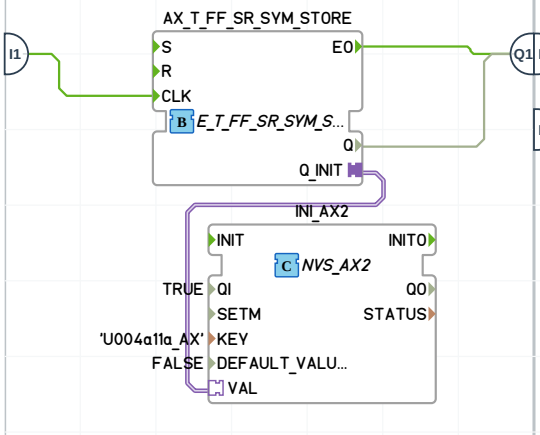
Uebung_004a10b

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK und INIT auf TRUE



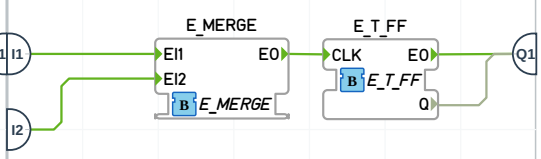
Uebung_004a11a

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK und STORE (INI)



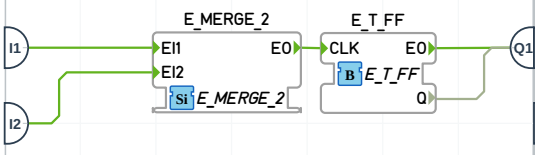
Uebung_004a11b

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK und STORE (NVS)



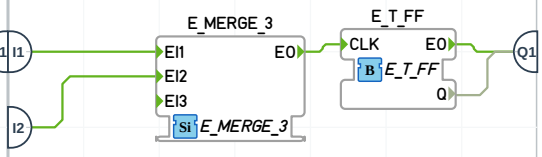
Uebung_004a2

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK mit E_MERGE



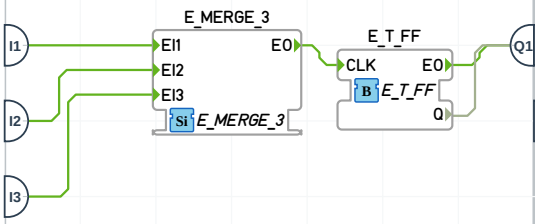
Uebung_004a2_2

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK mit E_MERGE



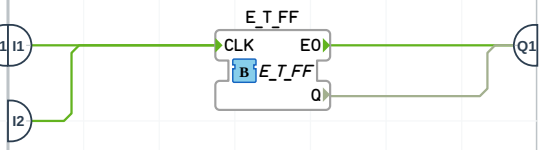
Uebung_004a2_3

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK mit E_MERGE



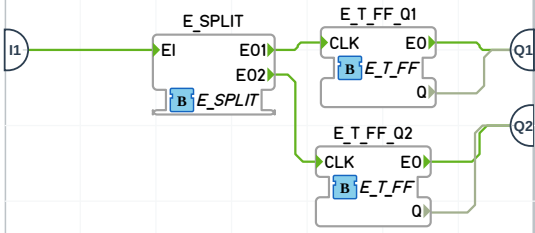
Uebung_004a2_3b

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK mit E_MERGE_3 (3 Taster)



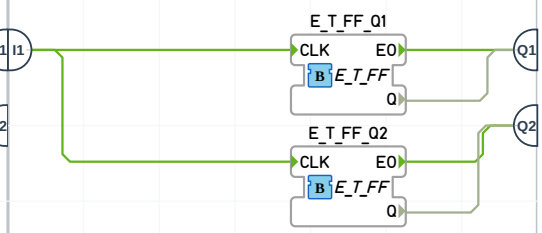
Uebung_004a3

wie Uebung_004a2 aber ohne E_MERGE



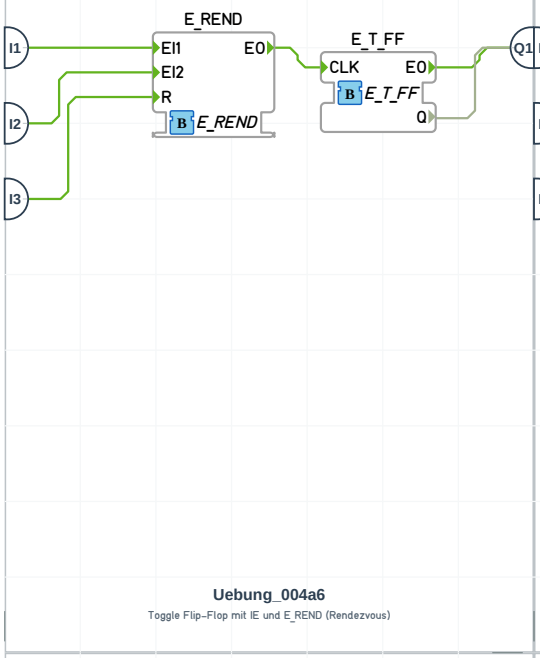
Uebung_004a4

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK mit E_SPLIT



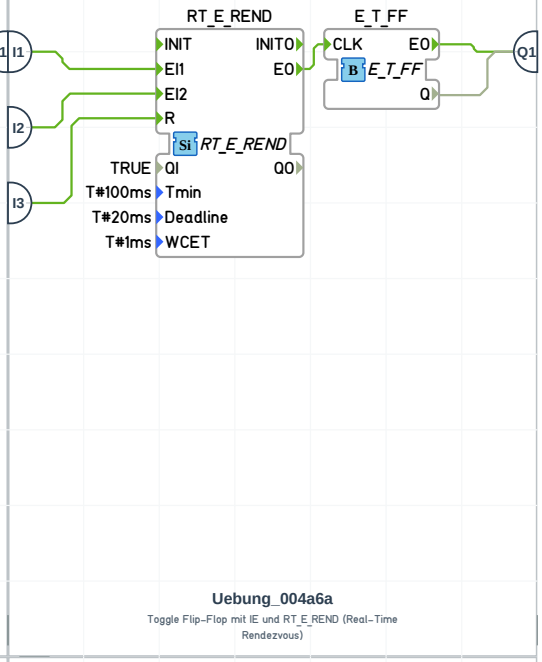
Uebung_004a5

wie Uebung_004a4 aber ohne E_SPLIT



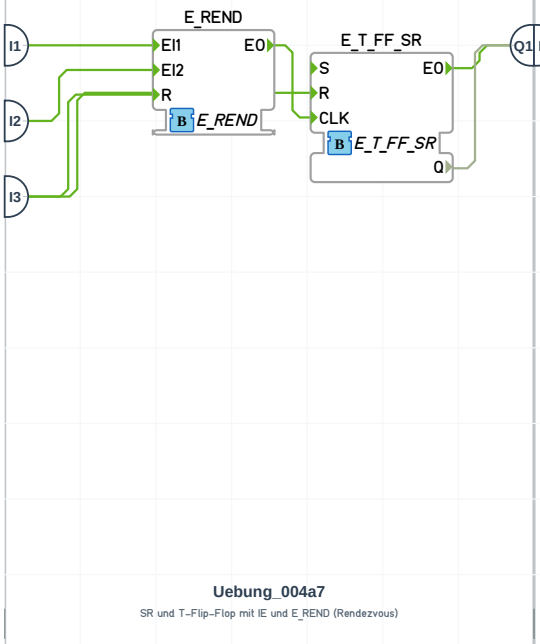
Uebung_004a6

Toggle Flip-Flop mit IE und E_REND (Rendezvous)



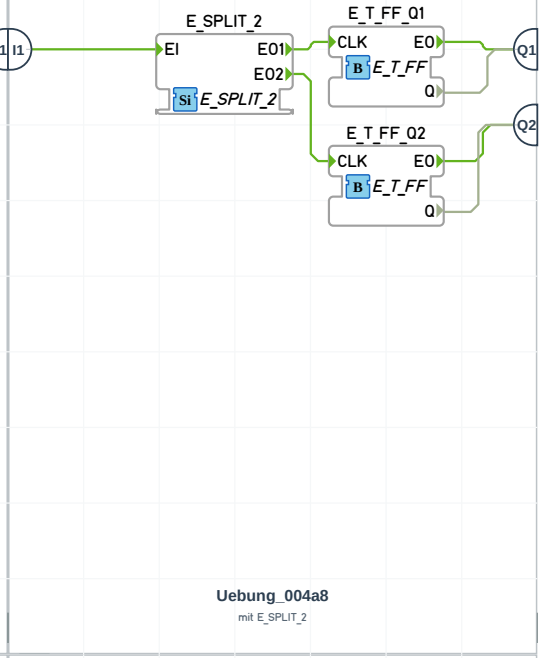
Uebung_004a6a

Toggle Flip-Flop mit IE und RT_E_REND (Real-Time Rendezvous)



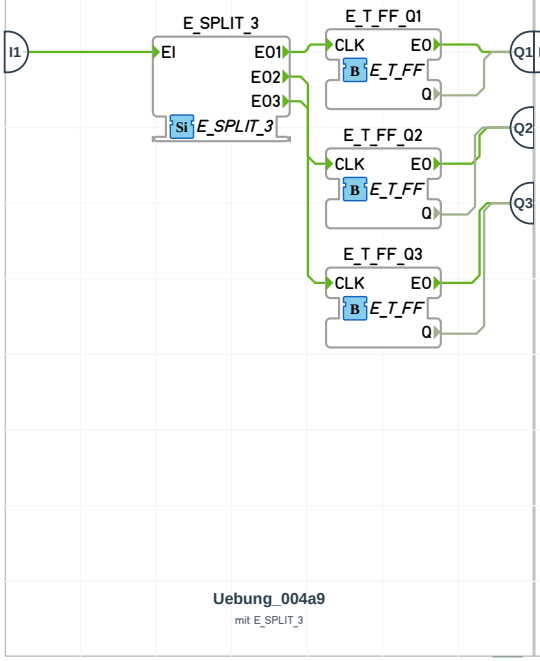
Uebung_004a7

SR und T-Flip-Flop mit IE und E_REND (Rendezvous)



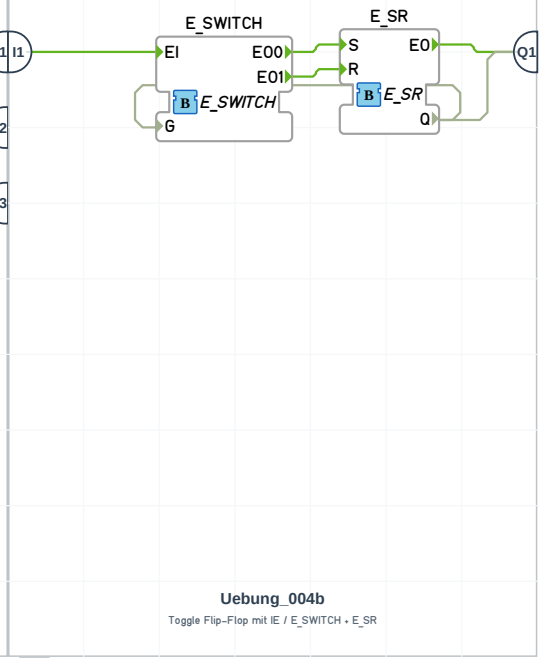
Uebung_004a8

mit E_SPLIT_2



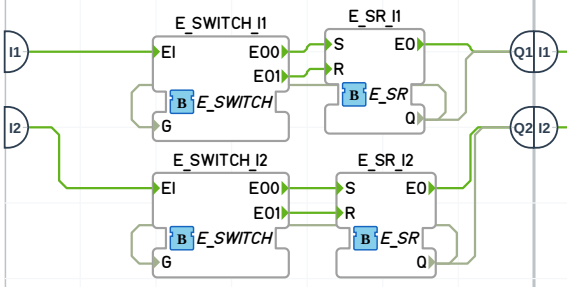
Uebung_004a9

mit E_SPLIT_3



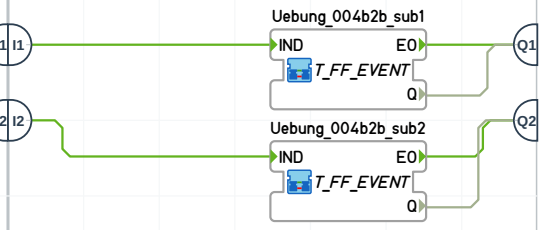
Uebung_004b

Toggle Flip-Flop mit IE / E_SWITCH + E_SR



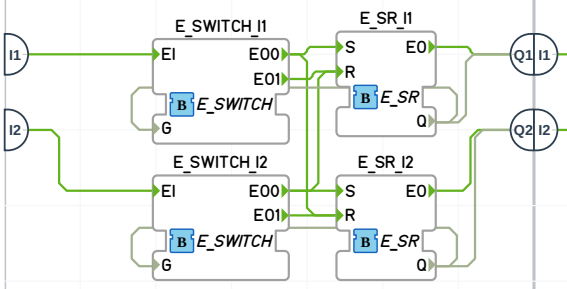
Uebung_004b2

Toggle Flip-Flop mit IE / Split / doppelt



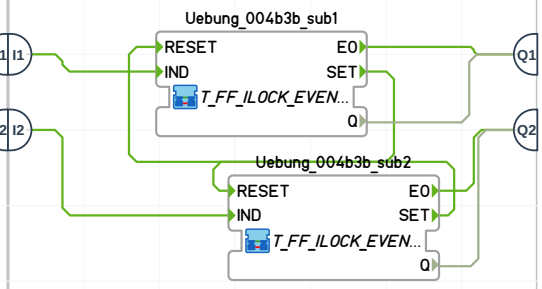
Uebung_004b2b

Zwei unabhängige Toggle-Flip-Flops unter Verwendung von Sub-Applikationen



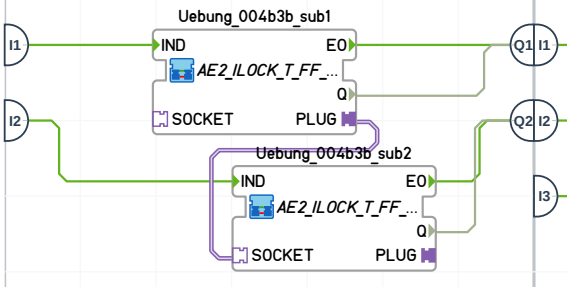
Uebung_004b3

Toggle Flip-Flop mit IE / Split / Verriegelt



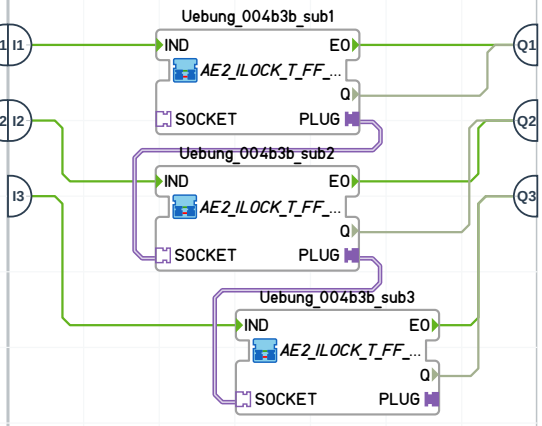
Uebung_004b3b

Zwei gegenseitig verriegelte Toggle-Flip-Flops (Radio-Button-Logik) via Event-Verdrahtung



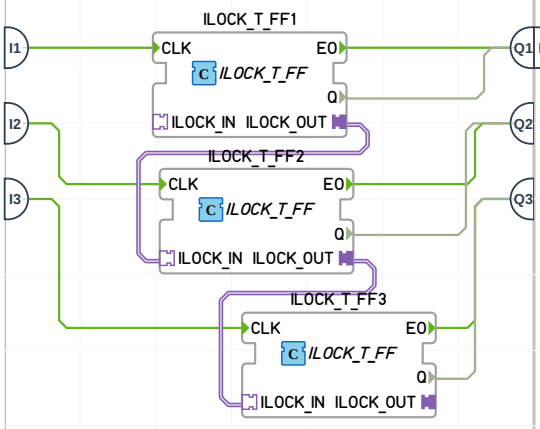
Uebung_004b3c

Zwei gegenseitig verriegelte Toggle-Flip-Flops unter Verwendung von bidirektionalen Adaptern



Uebung_004b4c

Drei gegenseitig verriegelte Toggle-Flip-Flops in einer Kette via AE2-Adapter



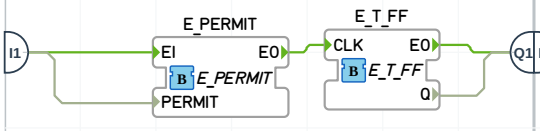
Uebung_004b4d

Drei gegenseitig verriegelte Toggle-Flip-Flops in einer Kette via AE2-Adapter mit ILOCK_T_FF



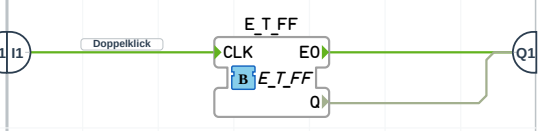
Uebung_004b5

Toggle Flip-Flop mit IX / E_SWITCH + E_T_FF



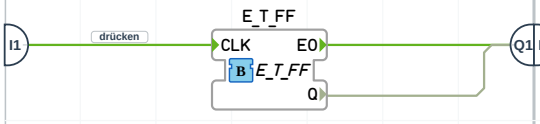
Uebung_004b6

Toggle Flip-Flop mit IX / E_PERMIT + E_T_FF



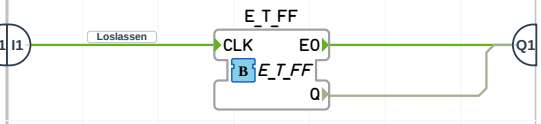
Uebung_004c1

Toggle Flip-Flop mit IE mit BUTTON_DOUBLE_CLICK



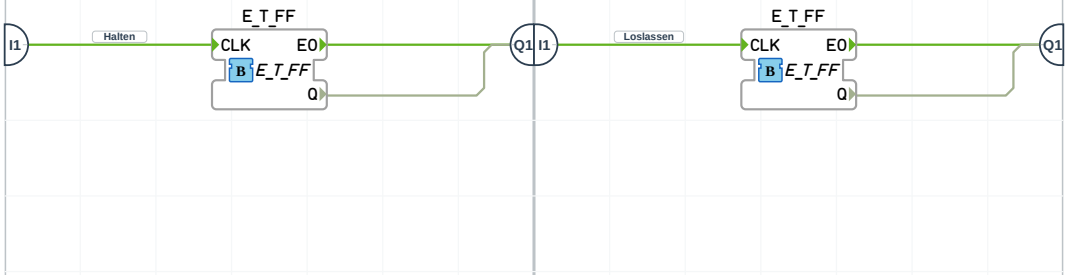
Uebung_004c2

Toggle Flip-Flop mit IE mit BUTTON_LONG_PRESS_START



Uebung_004c3

Toggle Flip-Flop mit IE mit BUTTON_LONG_PRESS_UP

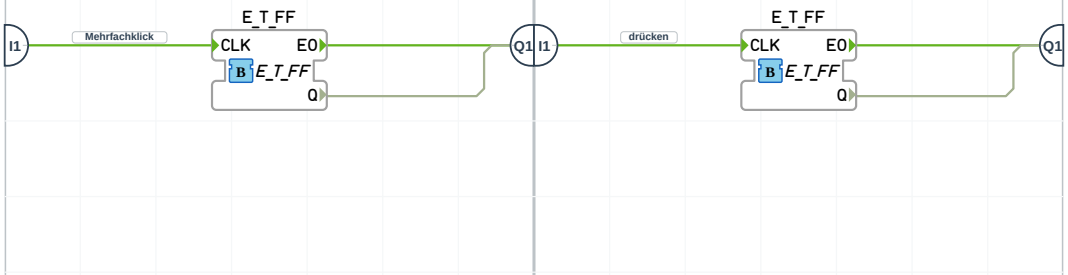


Uebung_004c4

Toggle Flip-Flop mit IE mit BUTTON_LONG_PRESS_HOLD

Uebung_004c5

Toggle Flip-Flop mit IE mit BUTTON_PRESS_END

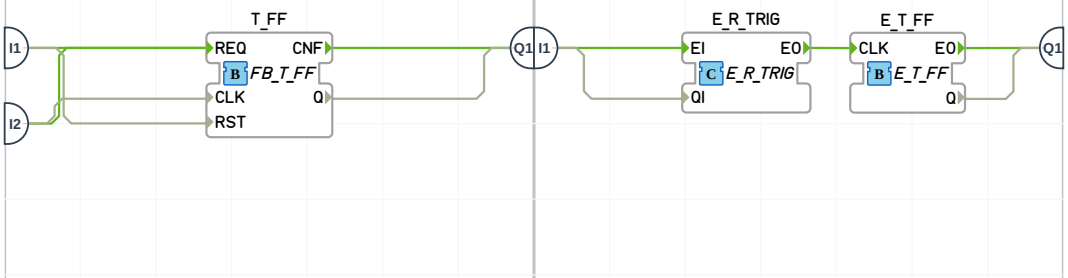


Uebung_004c6

Toggle Flip-Flop mit IE mit BUTTON_MULTIPLE_CLICK
(3-fach Klick)

Uebung_004c7

Toggle Flip-Flop mit IE mit BUTTON_PRESS_END

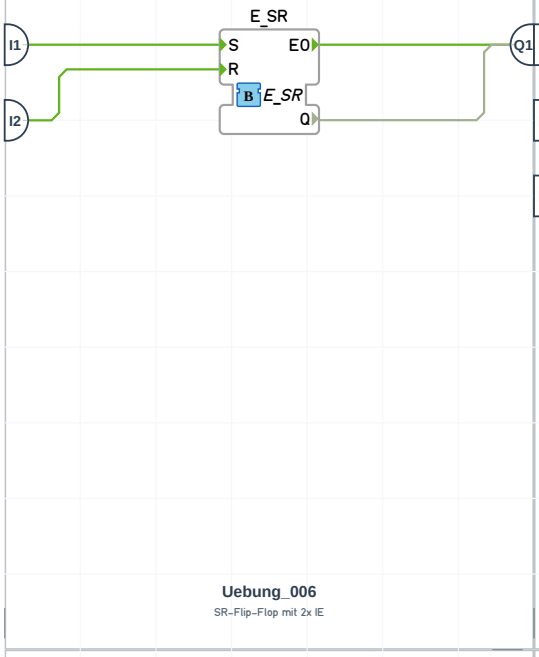


Uebung_004d_T

Exercise for FB_T_FF (Toggle Flip-Flop)

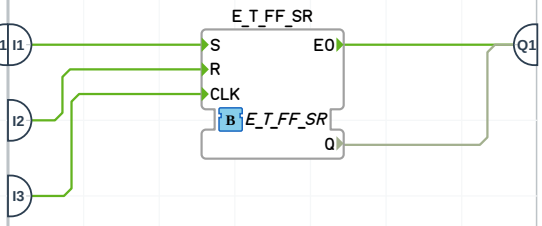
Uebung_005

Toggle Flip-Flop mit IX



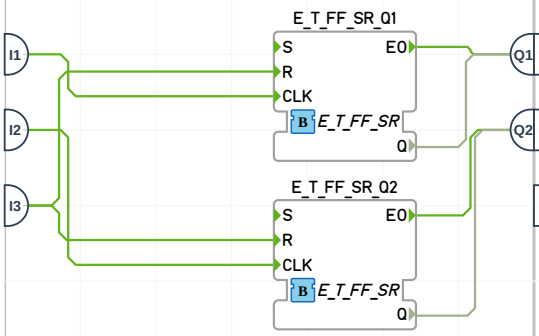
Uebung_006

SR-Flip-Flop mit 2x IE



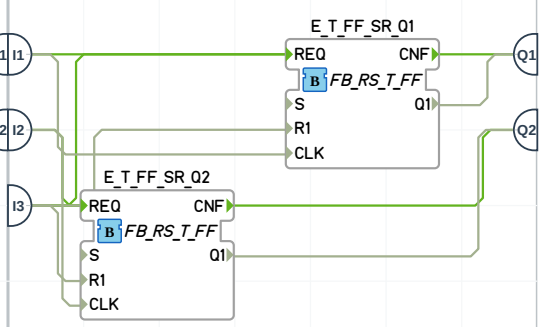
Uebung_006a

SR und T-Flip-Flop mit 3x IE



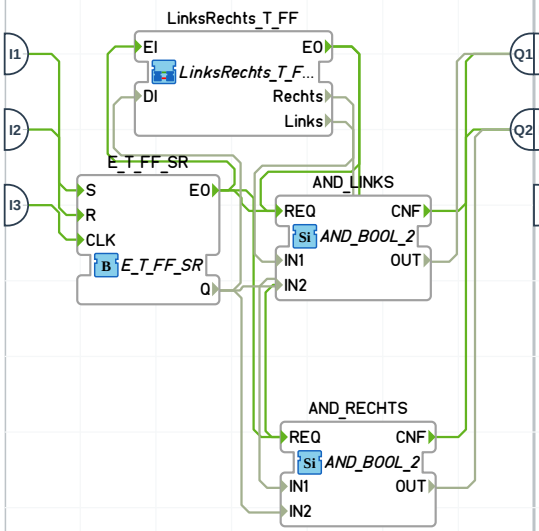
Uebung_006a2

2x SR und T-Flip-Flop mit IE



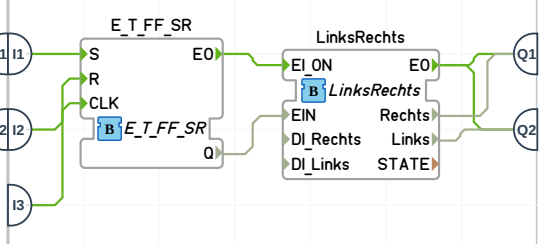
Uebung_006a2b

2x SR und T-Flip-Flop mit IX



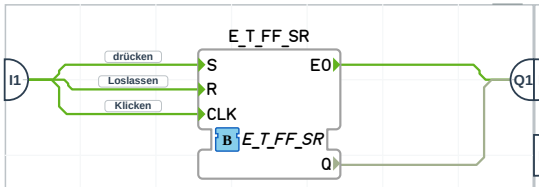
Uebung_006a3

SR und T-Flip-Flop mit 3x IE



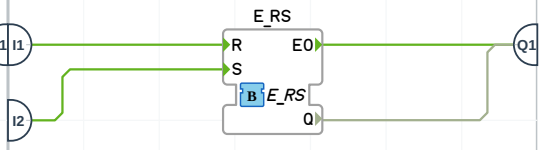
Uebung_006a4

SR und T-Flip-Flop mit 3x IE und LinksRechts Baustein



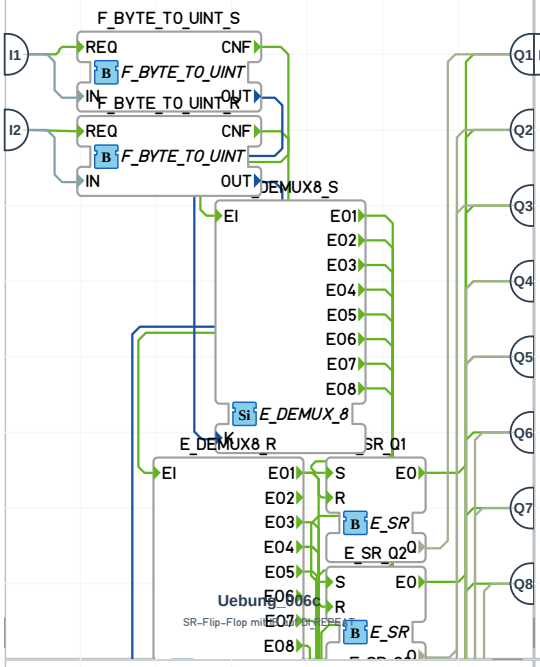
Uebung_006a5

SR und T-Flip-Flop als Rastend/Tastend Implementierung



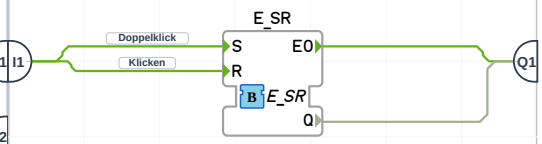
Uebung_006b

RS-Flip-Flop mit 2x IE



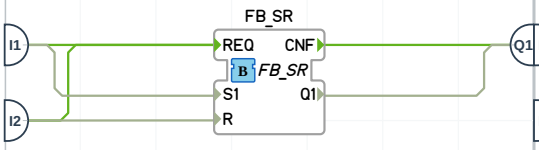
Uebung_006c

SR-Flip-Flop mit [illegible]



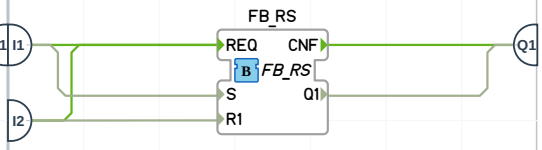
Uebung_006d

SR-Flip-Flop mit 2x IE und Doppelklick.



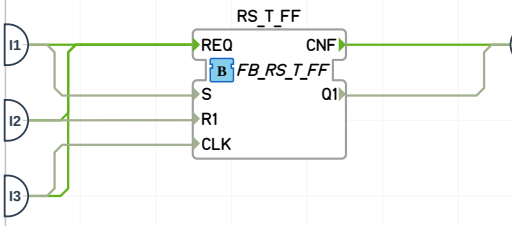
Uebung_006e1

SR-Flip-Flop mit 2x IX



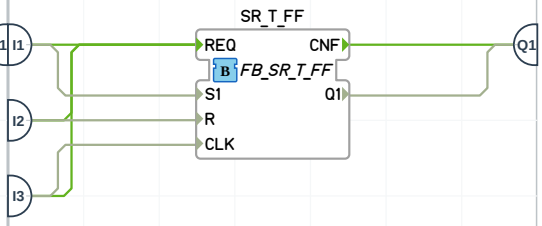
Uebung_006e2

RS-Flip-Flop mit 2x IX



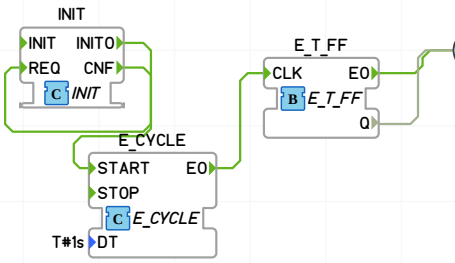
Uebung_006f_RS

Exercise for FB_RS_T_FF (Reset-Dominant Toggle Flip-Flop)



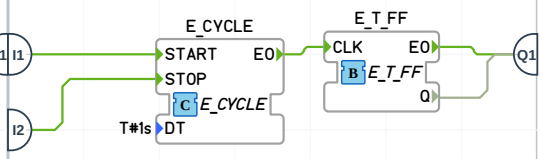
Uebung_006f_SR

Exercise for FB_SR_T_FF (Set-Dominant Toggle Flip-Flop)



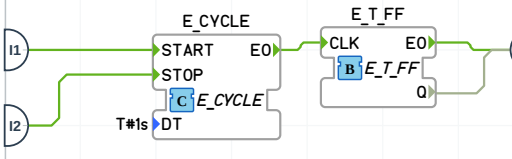
Uebung_007

Blinker mit E_CYCLE und E_T_FF



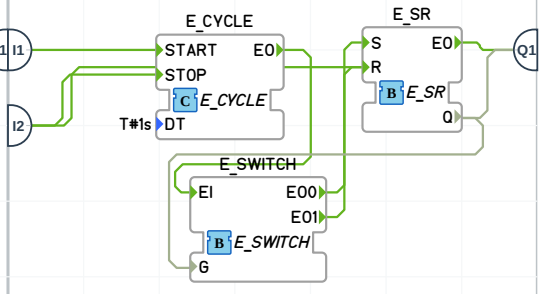
Uebung_007a1

Blinker mit E_CYCLE und E_T_FF



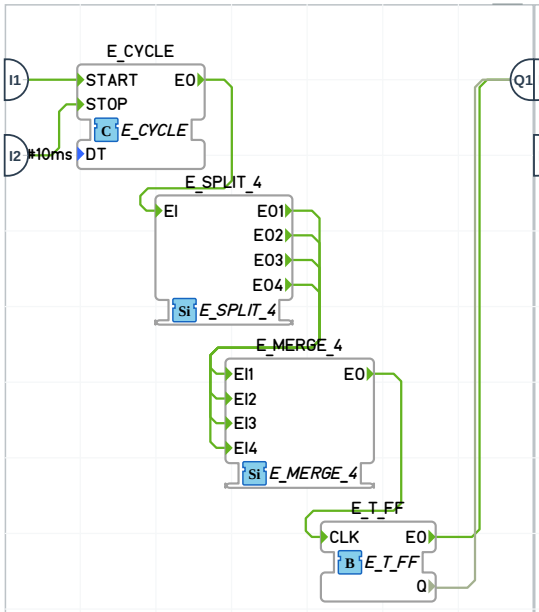
Uebung_007a2

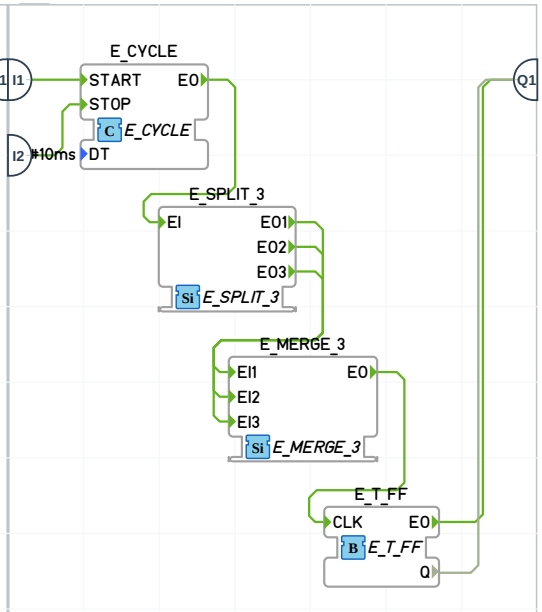
Blinker mit E_CYCLE und E_T_FF

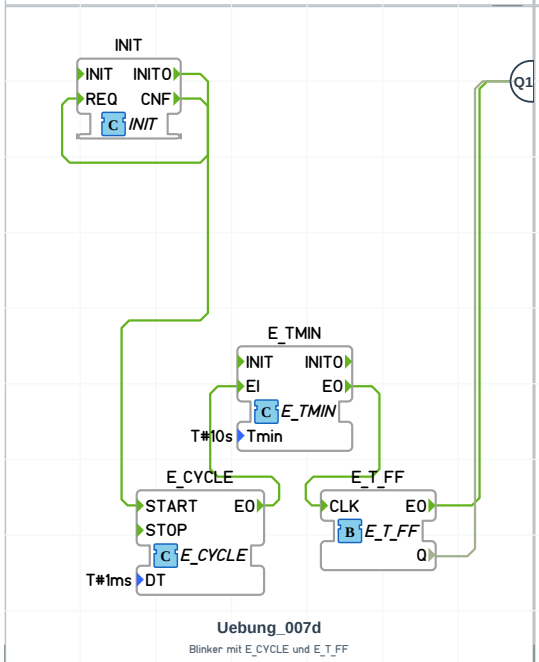


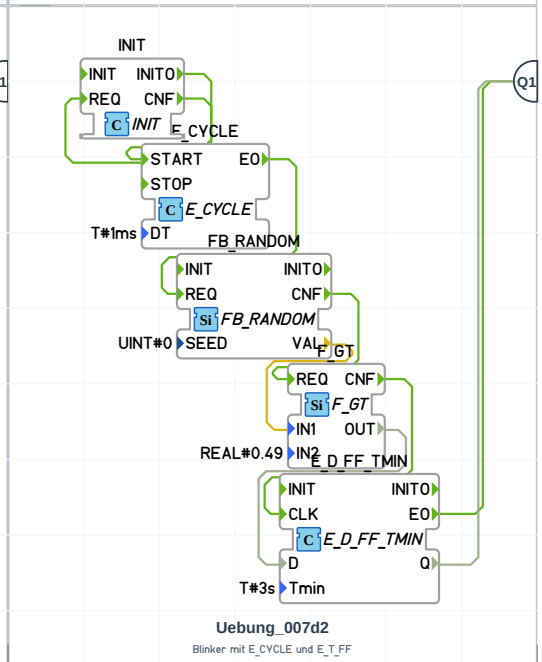
Uebung_007a3

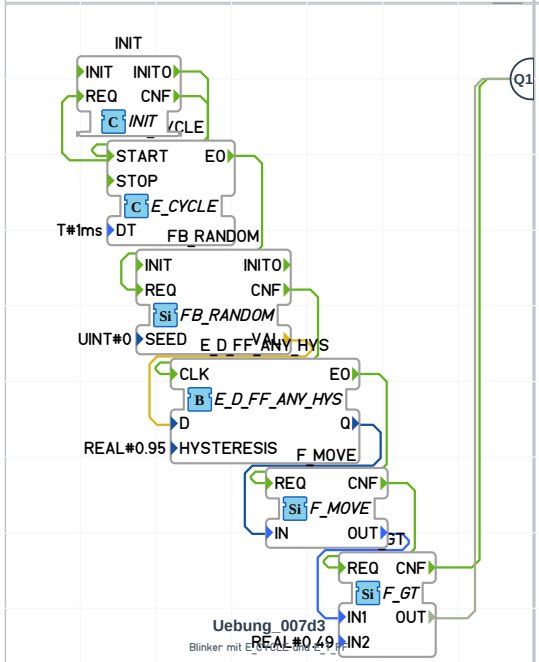
Blinker mit E_CYCLE, E_SWITCH und E_SR

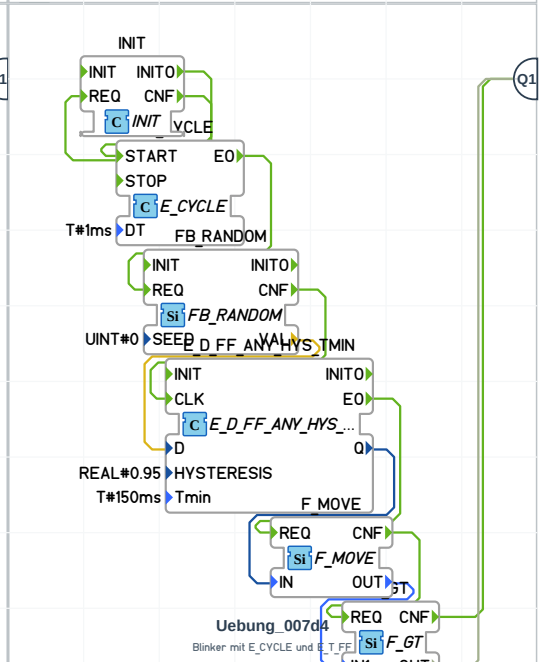

E_CYCLE
I1
START
EO
STOP
C E_CYCLE
I2 #10ms
DT
E_SPLIT_4
EI
EO1
EO2
EO3
EO4
Si E_SPLIT_4
E_MERGE_4
EI1
EO
EI2
EI3
EI4
Si E_MERGE_4
E_T_FF
CLK
EO
B E_T_FF
Q
Q1
Uebung_007b
Blinker mit E_CYCLE und E_T_FF

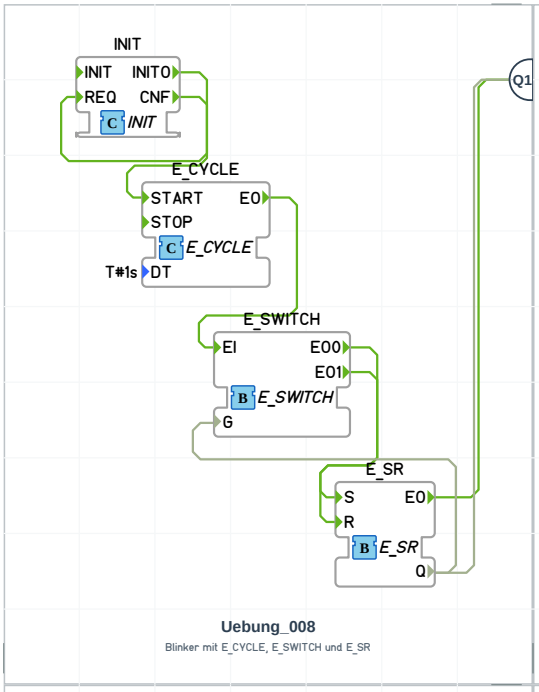

E_CYCLE
I1
START
EO
STOP
C E_CYCLE
I2 #10ms
DT
E_SPLIT_3
EI
EO1
EO2
EO3
Si E_SPLIT_3
E_MERGE_3
EI1
EO
EI2
EI3
Si E_MERGE_3
E_T_FF
CLK
EO
B E_T_FF
Q
Q1
Uebung_007c
Blinker mit E_CYCLE und E_T_FF


INIT
INIT
INITO
REQ
CNF
C INIT
E_TMIN
INIT
INITO
EI
EO
C E_TMIN
T#10s
Tmin
E_CYCLE
START
EO
STOP
C E_CYCLE
T#1ms
DT
E_T_FF
CLK
EO
B E_T_FF
Q
Q1
Uebung_007d
Blinker mit E_CYCLE und E_T_FF


INIT
INIT
INITO
REQ
CNF
C INIT
E_CYCLE
START
EO
STOP
C E_CYCLE
T#1ms
DT
FB_RANDOM
INIT
INITO
REQ
CNF
Si FB_RANDOM
UINT#0
SEED
VAL
F_GT
REQ
CNF
Si F_GT
IN1
OUT
REAL#0.49
IN2
E_D_FF_TMIN
INIT
INITO
CLK
EO
C E_D_FF_TMIN
D
Q
T#3s
Tmin
Q1
Uebung_007d2
Blinker mit E_CYCLE und E_T_FF

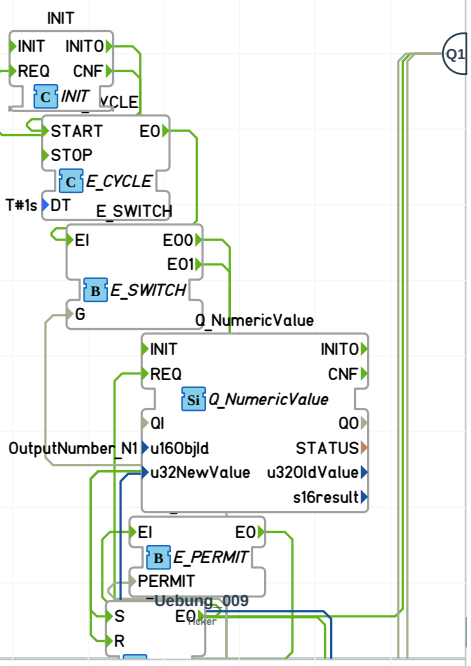

INIT
INIT
INITO
REQ
CNF
C INIT
VCLE
START
EO
STOP
C E_CYCLE
T#1ms
DT
FB_RANDOM
INIT
INITO
REQ
CNF
Si FB_RANDOM
UINT#0
SEED
VAL
E_D_FF_ANY_HYS
CLK
EO
B E_D_FF_ANY_HYS
D
Q
REAL#0.95
HYSTERESIS
F_MOVE
REQ
CNF
Si F_MOVE
IN
OUT
GT
REQ
CNF
Si F_GT
IN1
OUT
REAL#0.49
IN2
Q1
Uebung_007d3
Blinker mit E_CYCLE und E_T_FF


INIT
INIT
INITO
REQ
CNF
C INIT
VCLE
START
EO
STOP
C E_CYCLE
T#1ms
DT
FB_RANDOM
INIT
INITO
REQ
CNF
Si FB_RANDOM
UINT#0
SEED
VAL
E_D_FF_ANY_HYS_TMIN
INIT
INITO
CLK
EO
C E_D_FF_ANY_HYS_...
D
Q
REAL#0.95
HYSTERESIS
T#150ms
Tmin
F_MOVE
REQ
CNF
Si F_MOVE
IN
OUT
GT
REQ
CNF
Si F_GT
IN1
OUT
REAL#0.49
IN2
Q1
Uebung_007d4
Blinker mit E_CYCLE und E_T_FF



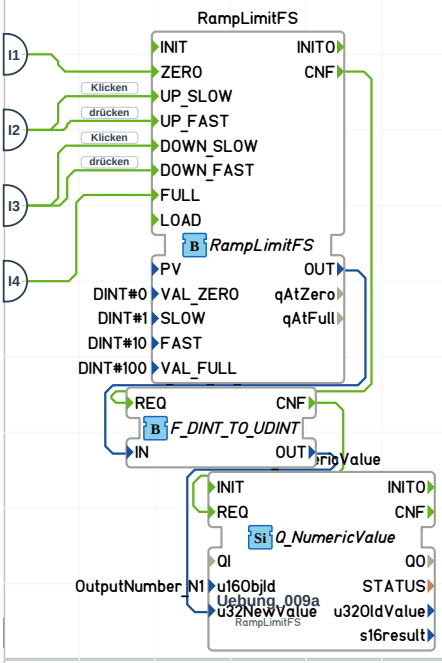
Uebung_008

Blinker mit E_CYCLE, E_SWITCH und E_SR



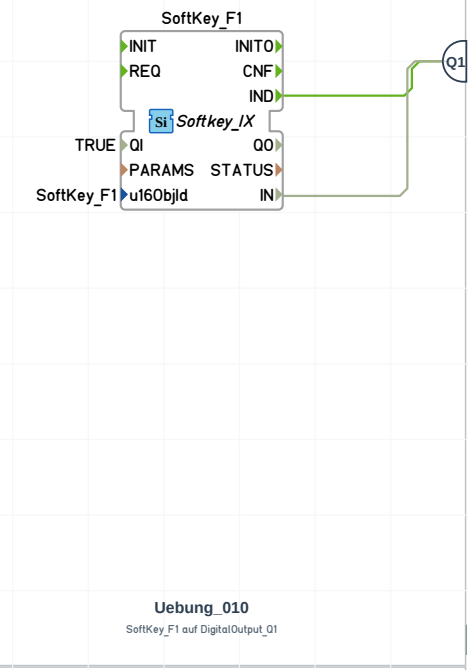
Uebung_009

Taster



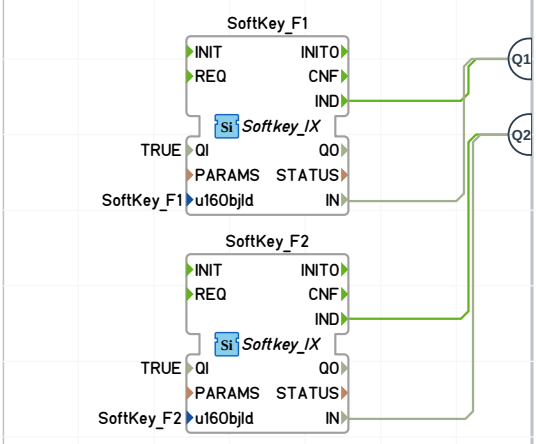
Uebung_009a

RampLimitFS



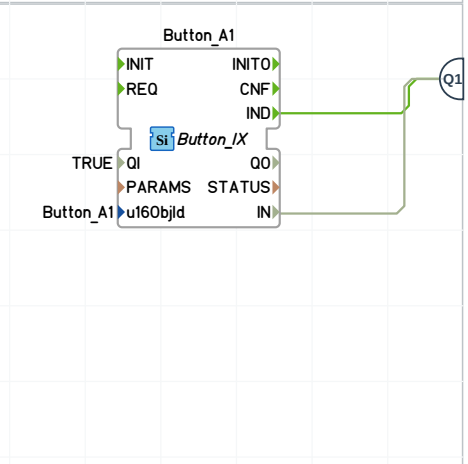
Uebung_010

SoftKey_F1 auf DigitalOutput_Q1



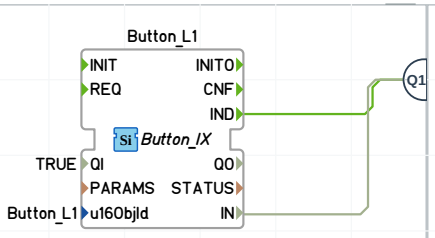
Uebung_010a

SoftKey_F1/_F2 auf DigitalOutput_Q1/_Q2



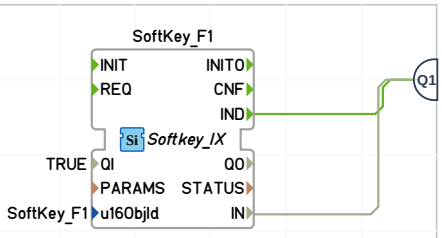
Uebung_010a2

Button_A1 auf DigitalOutput_Q1



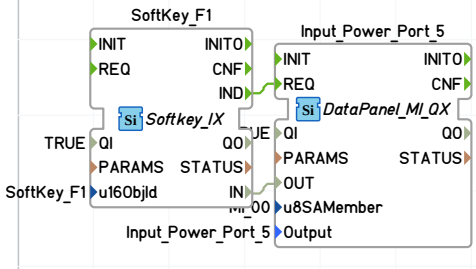
Uebung_010a3

Button_L1 (Latching) auf DigitalOutput_Q1



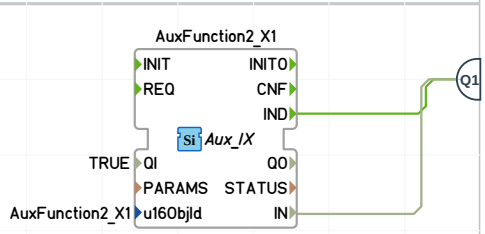
Uebung_010a4

SoftKey_F1 auf DigitalOutput_Q1 (Datapanel)



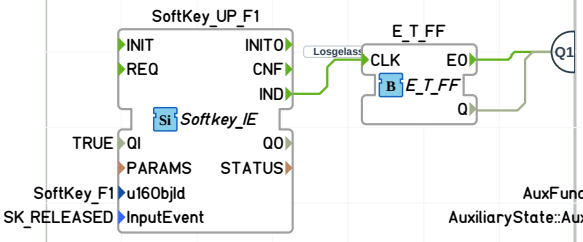
Uebung_010a5

SoftKey_F1 auf DigitalOutput_Q1 (Datapanel)



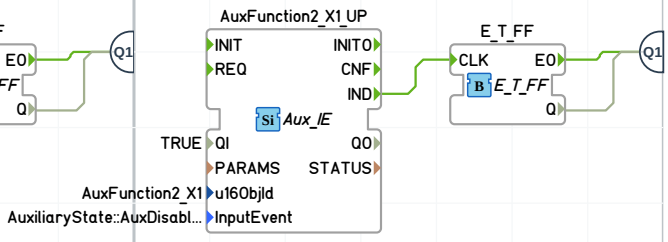
Uebung_010b1

AuxFunction2_X1 auf DigitalOutput_Q1



Uebung_010b2

Toggle Flip-Flop mit IE SoftKey_F1 SK_RELEASED



Uebung_010b3

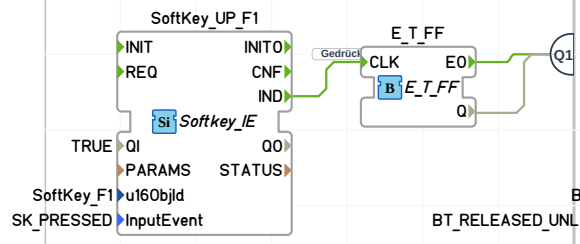
Toggle Flip-Flop mit IE AuxFunction2_X1

Uebung_010b4

SoftKey_F1–8 auf DigitalOutput_Q1–8

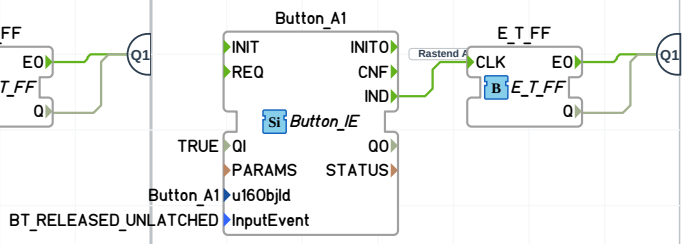
Uebung_010b5

SoftKey_F1–10 auf DigitalOutput_Q1–10



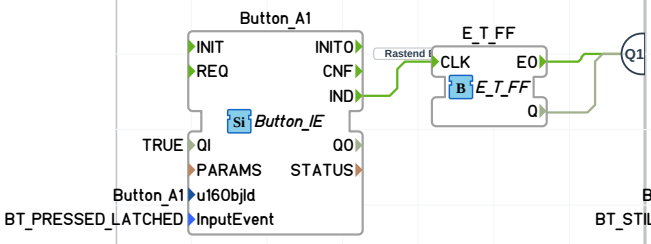
Uebung_010b6

Toggle Flip-Flop mit IE SoftKey_F1 SK_PRESSED



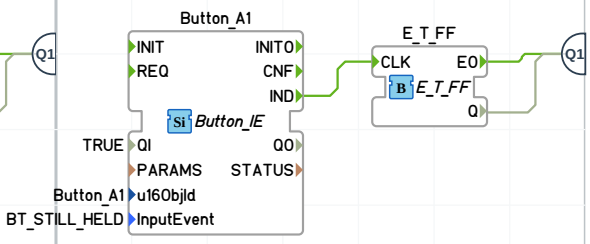
Uebung_010b7

Toggle Flip-Flop mit IE Button_A1 BT_RELEASED_UNLATCHED



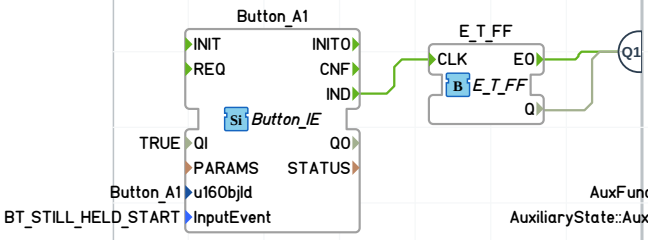
Uebung_010b8

Toggle Flip-Flop mit IE Button_A1 BT_PRESSED_LATCHED



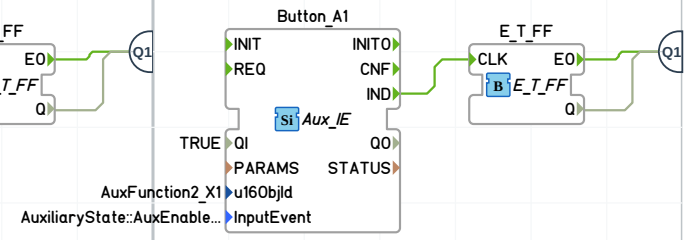
Uebung_010b9

Toggle Flip-Flop mit IE Button_A1 BT_STILL_HELD



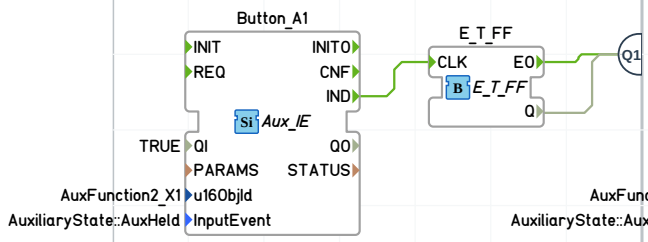
Uebung_010bA

Toggle Flip-Flop mit IE Button_A1 BT_STILL_HELD



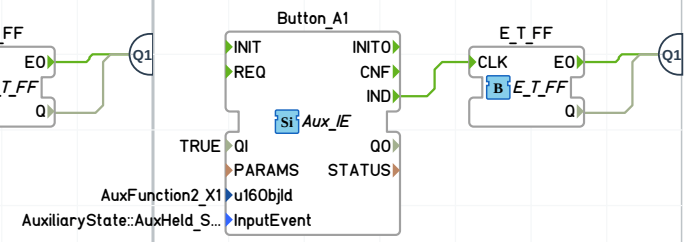
Uebung_010bA2

Toggle Flip-Flop mit IE AuxFunction2_X1 AuxEnabled



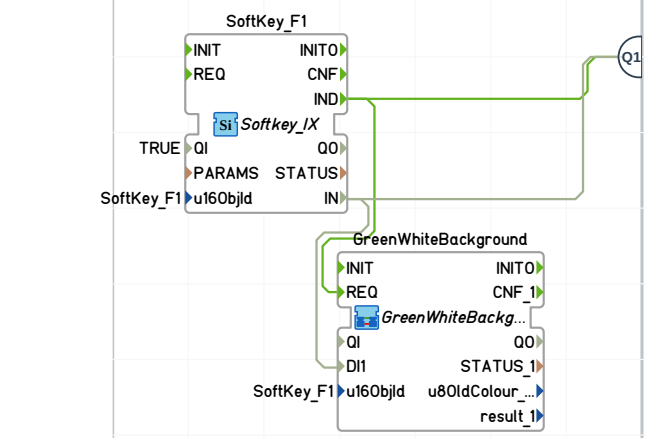
Uebung_010bA3

Toggle Flip-Flop mit IE AuxFunction2_X1 AuxHeld



Uebung_010bA4

Toggle Flip-Flop mit IE AuxFunction2_X1 AuxHeld_START



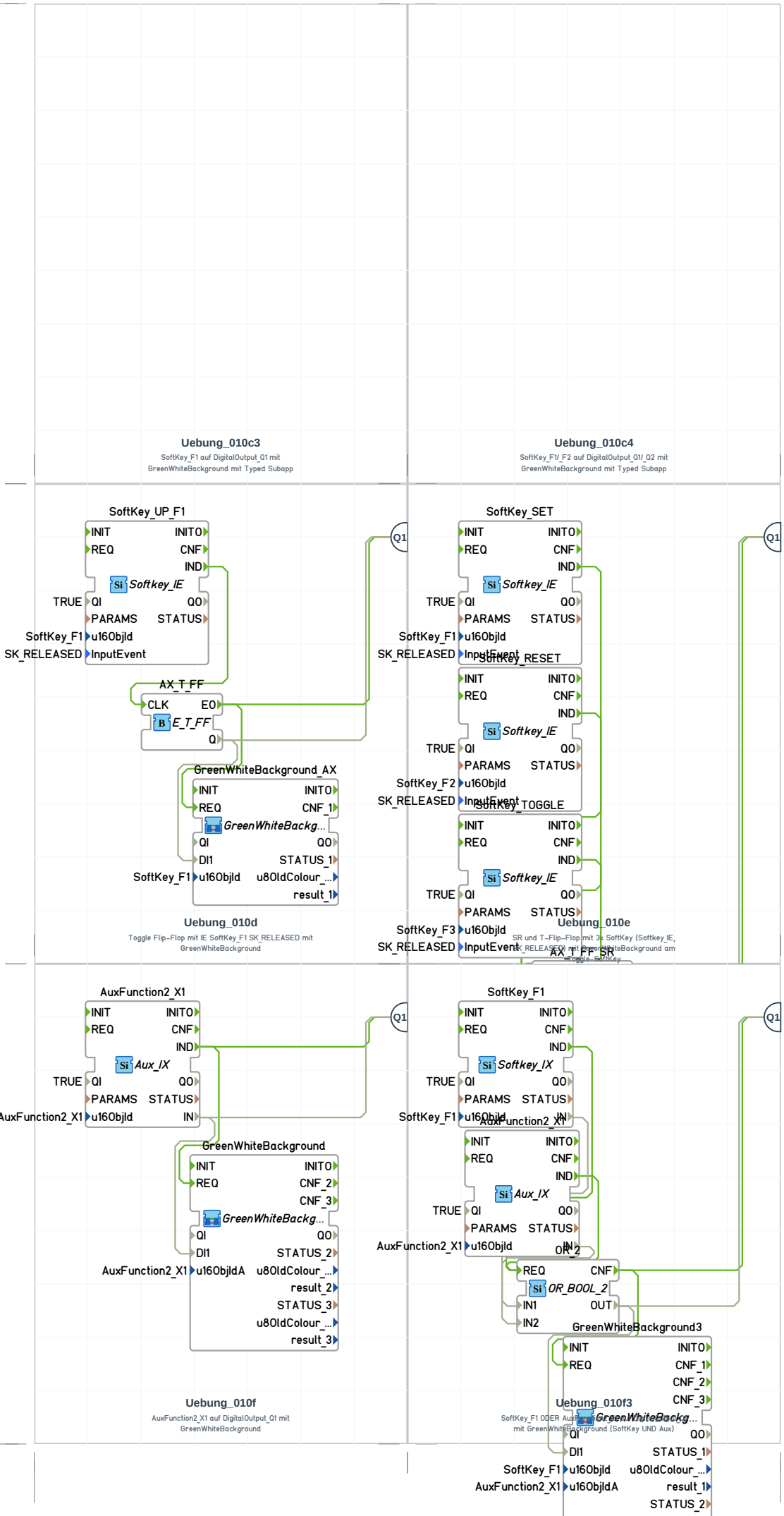
Uebung_010c

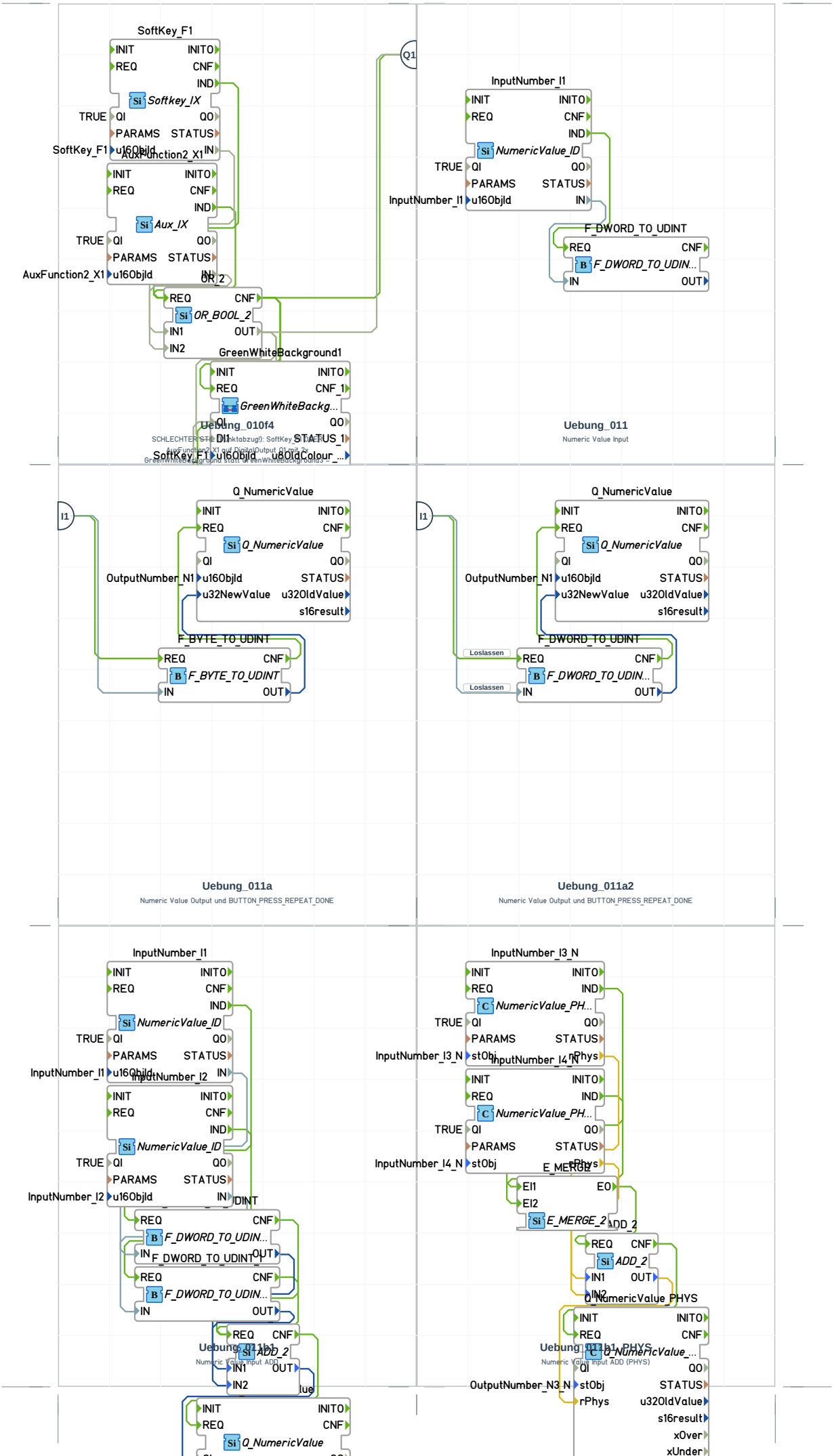
SoftKey_F1 auf DigitalOutput_Q1 mit GreenWhiteBackground

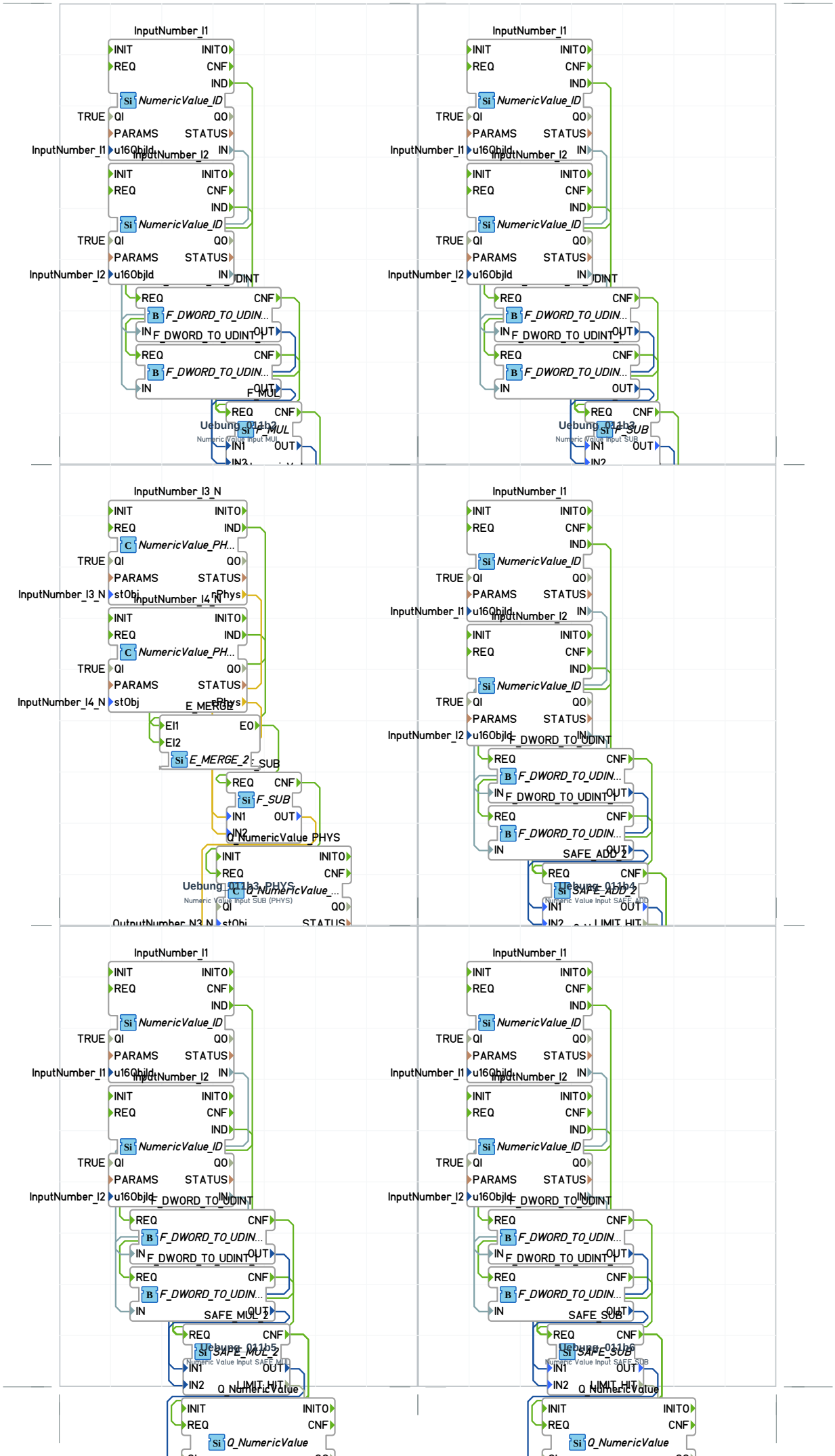


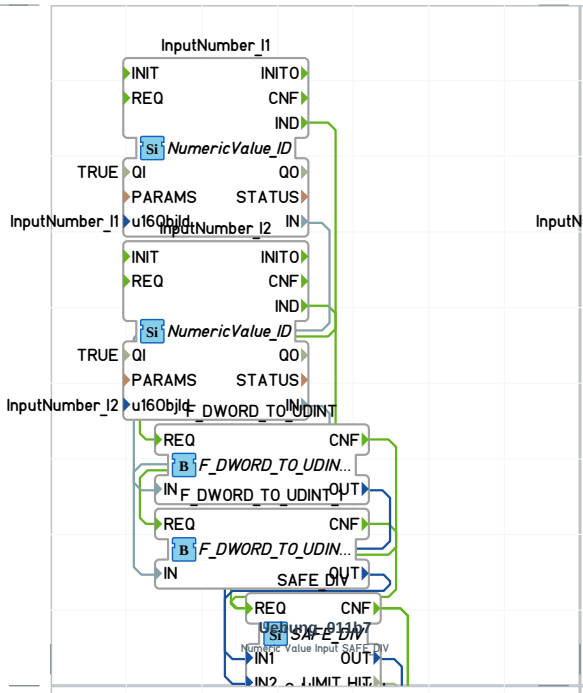
Uebung_010c2

SoftKey_F1 auf DigitalOutput_Q1 mit GreenWhiteBackground mit Subapp


Uebung_010c3
SoftKey_F1 auf DigitalOutput_Q1 mit GreenWhiteBackground mit Typed Subapp
Uebung_010c4
SoftKey_F1/_F2 auf DigitalOutput_Q1/_Q2 mit GreenWhiteBackground mit Typed Subapp
SoftKey_UP_F1
INIT
INITO
REQ
CNF
IND
Softkey_IE
TRUE
QI
QO
PARAMS
STATUS
SoftKey_F1
u16ObjId
SK_RELEASED
InputEvent
Q1
AX_T_FF
CLK
EO
E_T_FF
Q
GreenWhiteBackground_AX
CNF_1
GreenWhiteBackg...
DI1
STATUS_1
u8OldColour_...
result_1
Uebung_010d
Toggle Flip-Flop mit IE SoftKey_F1 SK_RELEASED mit GreenWhiteBackground
SoftKey_SET
SoftKey_RESET
SoftKey_TOGGLE
SoftKey_F2
SoftKey_F3
AX_T_FF_SR
Uebung_010e
SR und T-Flip-Flop mit 3x SoftKey (Softkey_IE, SK_RELEASED) mit GreenWhiteBackground am
Toggle_SoftKey
AuxFunction2_X1
Aux_IX
IN
GreenWhiteBackground
CNF_2
CNF_3
STATUS_2
u8OldColour_...
result_2
STATUS_3
result_3
Uebung_010f
AuxFunction2_X1 auf DigitalOutput_Q1 mit GreenWhiteBackground
SoftKey_F1
Softkey_IX
OR_2
OR_BOOL_2
IN1
IN2
OUT
GreenWhiteBackground3
CNF_1
Uebung_010f3
SoftKey_F1 ODER AuxFunction2_X1 mit GreenWhiteBackground (SoftKey UND Aux)

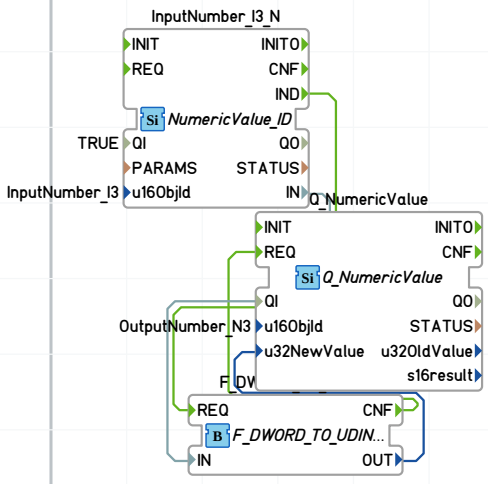

SoftKey_F1
INIT
INITO
REQ
CNF
IND
Softkey_IX
TRUE
QI
QO
PARAMS
STATUS
SoftKey_F1
u16ObjId
IN
AuxFunction2_X1
Aux_IX
AuxFunction2_X1
OR_2
OR_BOOL_2
IN1
IN2
OUT
Q1
GreenWhiteBackground1
CNF_1
GreenWhiteBackg...
STATUS_1
u8ObjColour_...
Uebung_010f4
InputNumber_I1
NumericValue_ID
F_DWORD_TO_UDINT
F_DWORD_TO_UDIN...
Uebung_011
Numeric Value Input
Q_NumericValue
I1
OutputNumber_N1
u32NewValue
u32OldValue
s16result
F_BYTE_TO_UDINT
Uebung_011a
Numeric Value Output und BUTTON_PRESS_REPEAT_DONE
Loslassen
Uebung_011a2
InputNumber_I2
InputNumber_I3_N
InputNumber_I4_N
NumericValue_PH...
stObj
rPhys
E_MERGE
EI1
EI2
EO
E_MERGE_2
ADD_2
IN1
IN2
Uebung_011b1
Numeric Value Input ADD
Uebung_011b1_PHYS
Numeric Value Input ADD (PHYS)
Q_NumericValue_...
OutputNumber_N3_N
xOver
xUnder


InputNumber_I1
InputNumber_I2
NumericValue_ID
F_DWORD_TO_UDIN...
F_DWORD_TO_UDINT
F_MUL
F_SUB
InputNumber_I3_N
InputNumber_I4_N
NumericValue_PH...
E_MERGE
E_MERGE_2
Q_NumericValue_PHYS
SAFE_ADD_2
SAFE_MUL_2
SAFE_SUB
Q_NumericValue
INIT
INITO
REQ
CNF
IND
QI
QO
PARAMS
STATUS
IN
OUT
TRUE
u16ObjId
stObj
Phys
EI1
EI2
EO
IN1
IN2
LIMIT_HIT



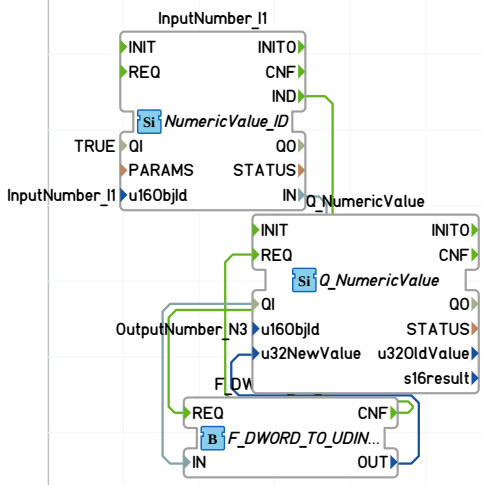
Uebung_011b7

Numeric Value Input SAFE_DIV



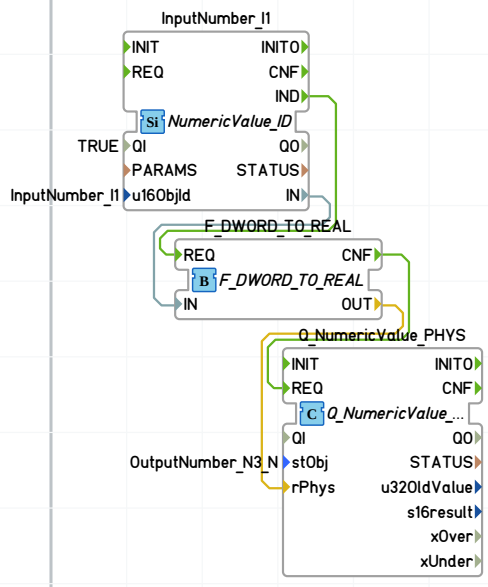
Uebung_011c

Numeric Value Input I3 Durchschleifen auf N3



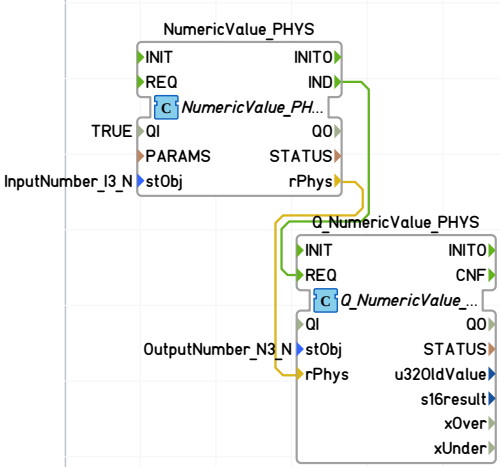
Uebung_011d

Numeric Value Input I1 Durchschleifen auf N3
(Offset/Scale Effekt)



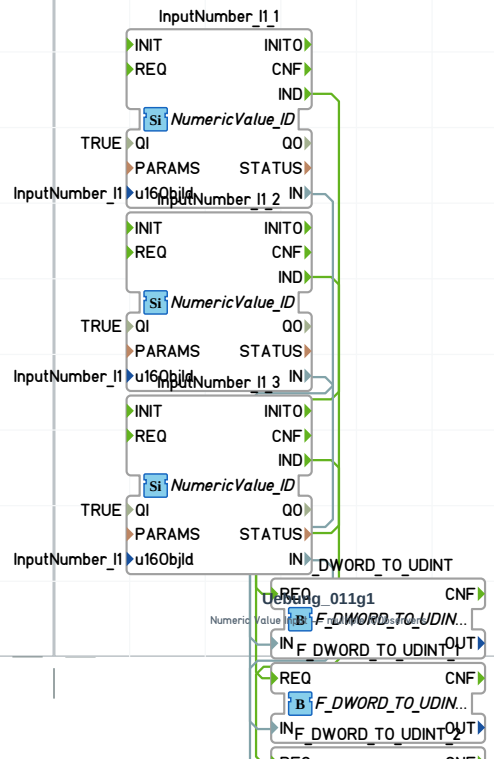
Uebung_011e_MIX

Numeric Value Input I1 Durchschleifen auf N3
(Software Scale via NumericObjectPool_S) falsch gemischt!



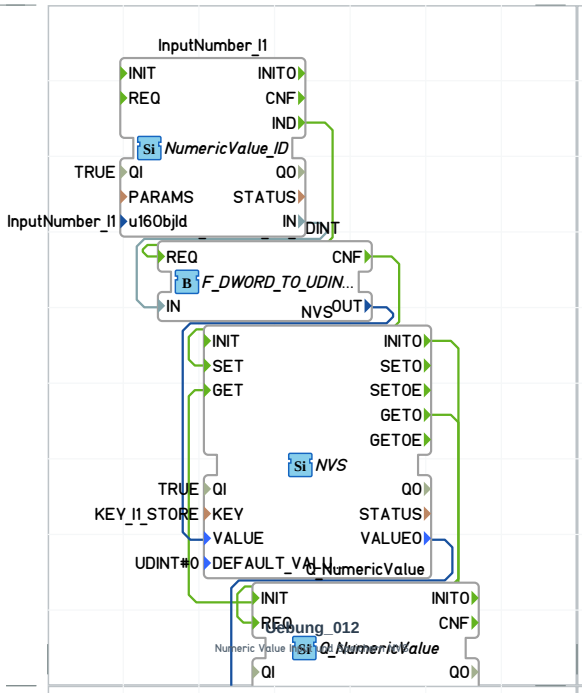
Uebung_011f

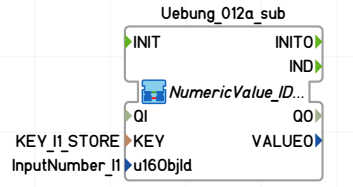
Numeric Value Input I3 Durchschleifen auf N3 (Input und Output PHYS via NumericObjectPool_S)



Uebung_011g1

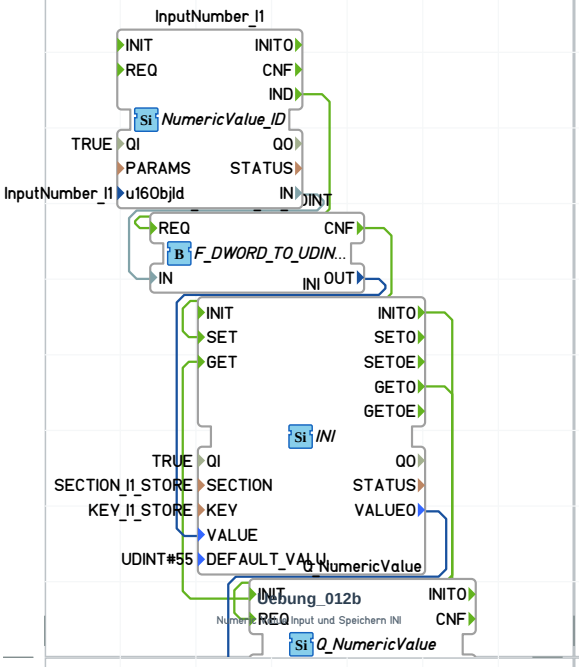
Numeric Value Input I1 Durchschleifen auf N3

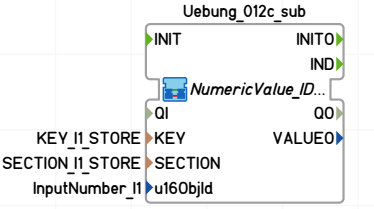




Uebung_012a

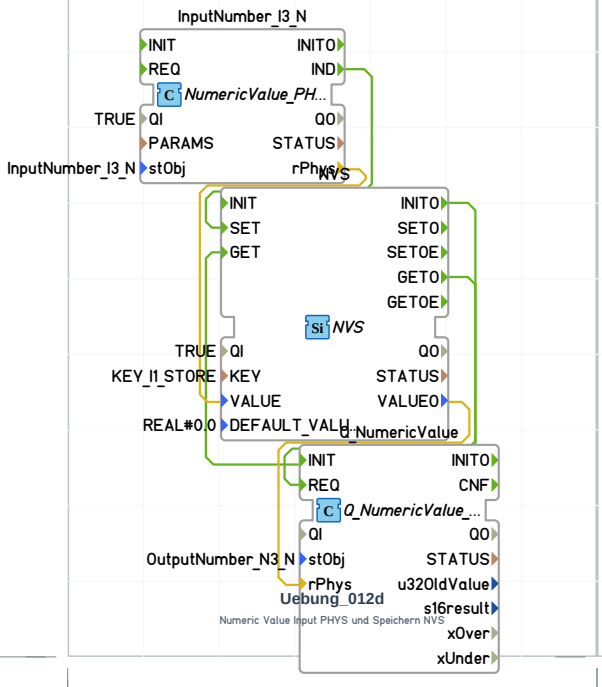
Numeric Value Input und Speichern NVS mit Subapp

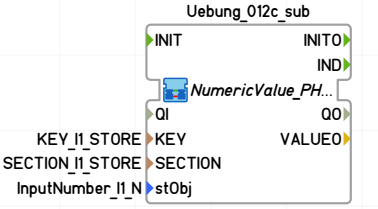




Uebung_012c

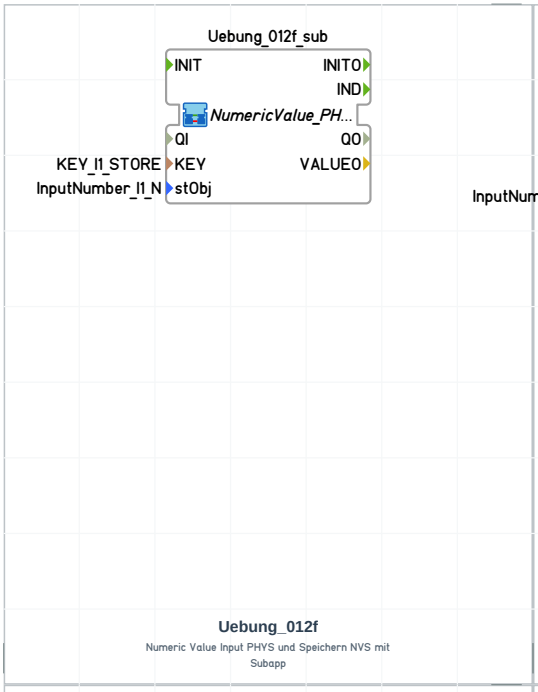
Numeric Value Input und Speichern INI mit Subapp

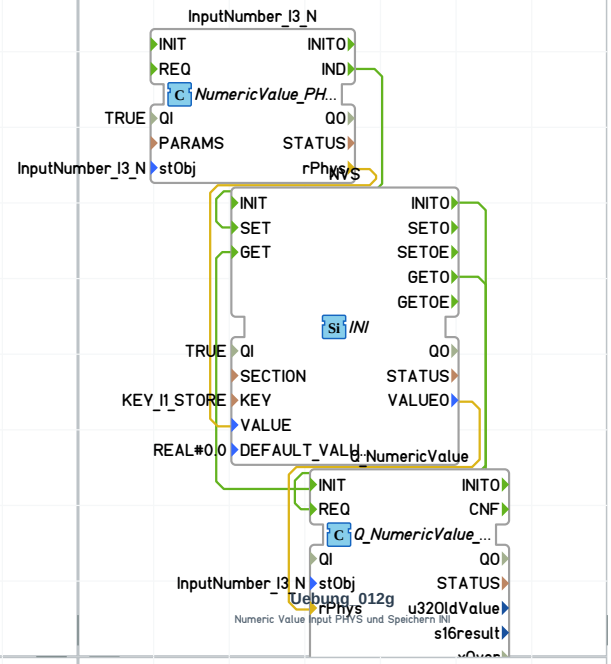


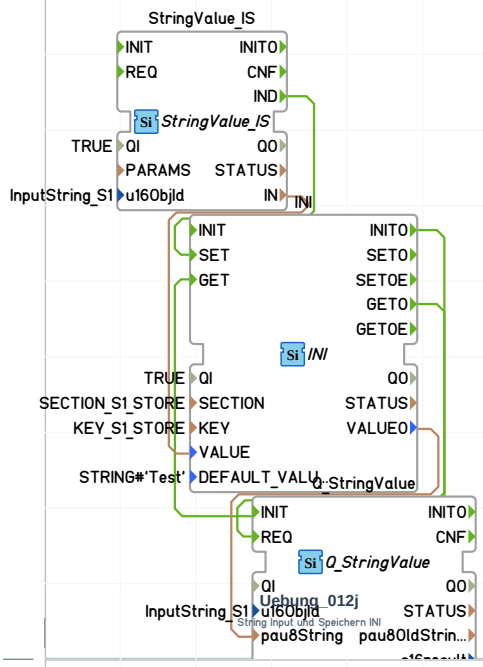


Uebung_012e

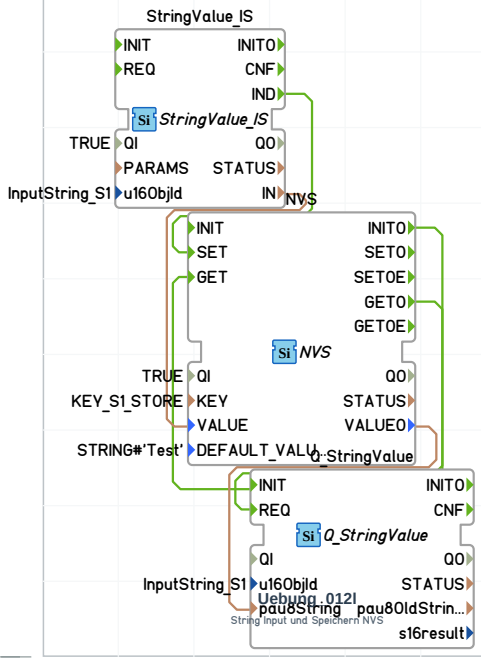
Numeric Value Input PHYS und Speichern INI mit Subapp

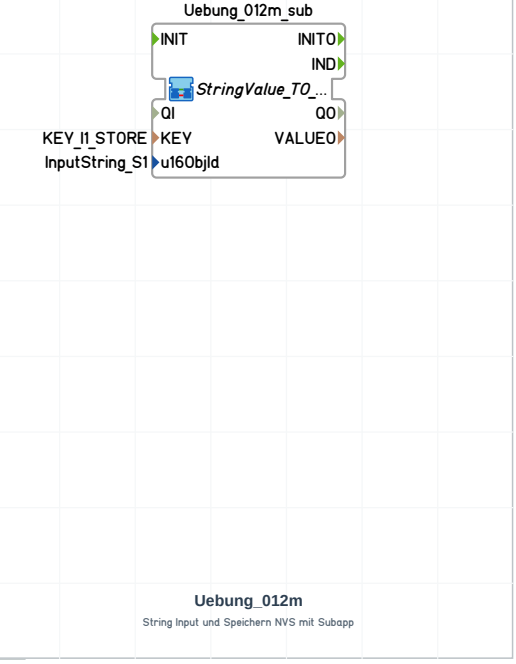

Uebung_012f_sub
INIT
INITO
IND
NumericValue_PH...
QI
QO
KEY_I1_STORE
KEY
VALUEO
InputNumber_I1_N
stObj
Uebung_012f
Numeric Value Input PHYS und Speichern NVS mit Subapp

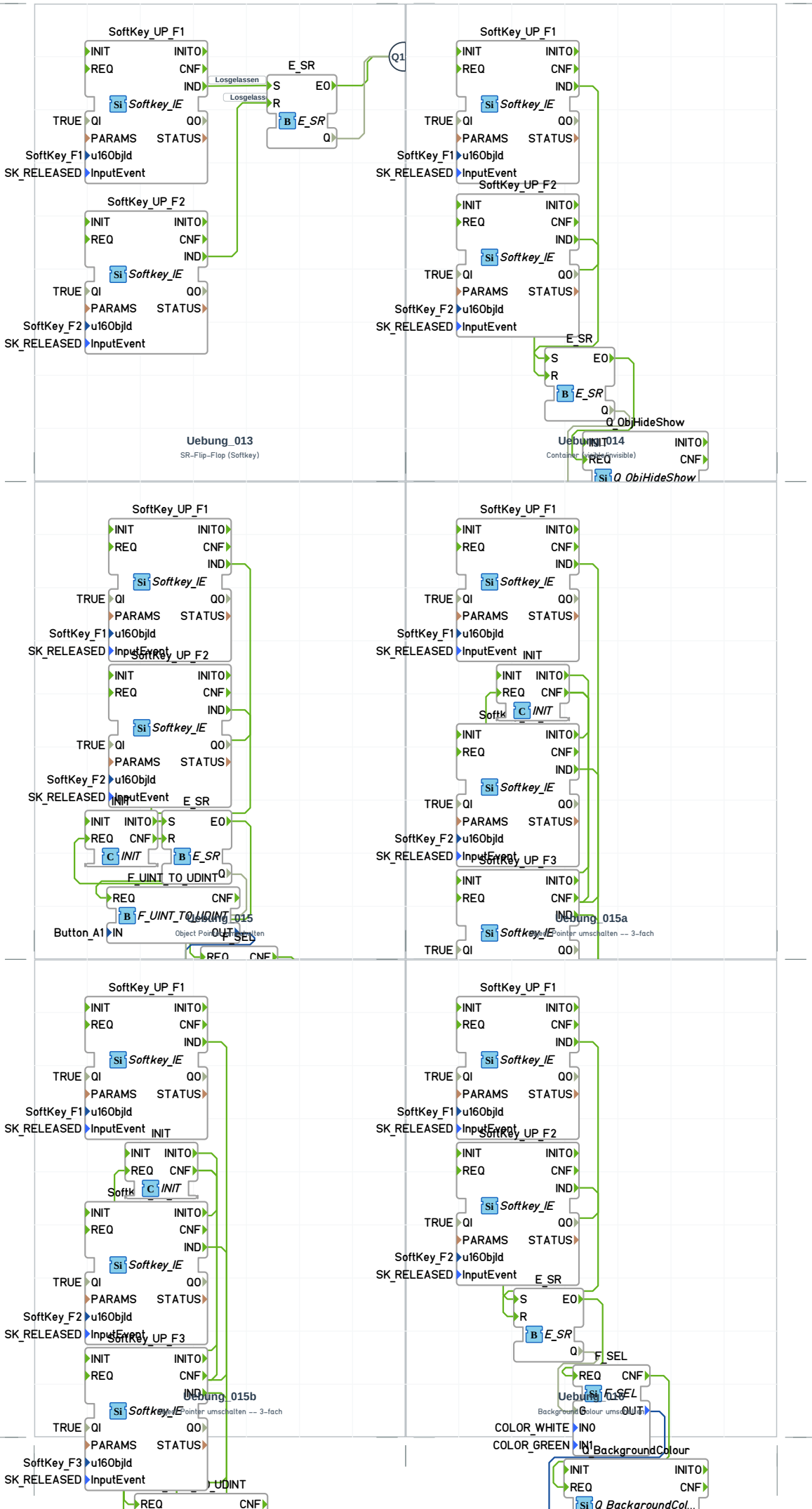

InputNumber_I3_N
INIT
INITO
REQ
IND
NumericValue_PH...
TRUE
QI
QO
PARAMS
STATUS
InputNumber_I3_N
stObj
rPhys
NVS
INIT
INITO
SET
SETO
GET
SETOE
GETO
GETOE
INI
TRUE
QI
QO
SECTION
STATUS
KEY_I1_STORE
KEY
VALUEO
VALUE
REAL#0.0
DEFAULT_VALU
Q_NumericValue
INIT
INITO
REQ
CNF
Q_NumericValue_...
QI
QO
InputNumber_I3_N
stObj
STATUS
rPhys
u32OldValue
s16result
Uebung_012g
Numeric Value Input PHYS und Speichern INI

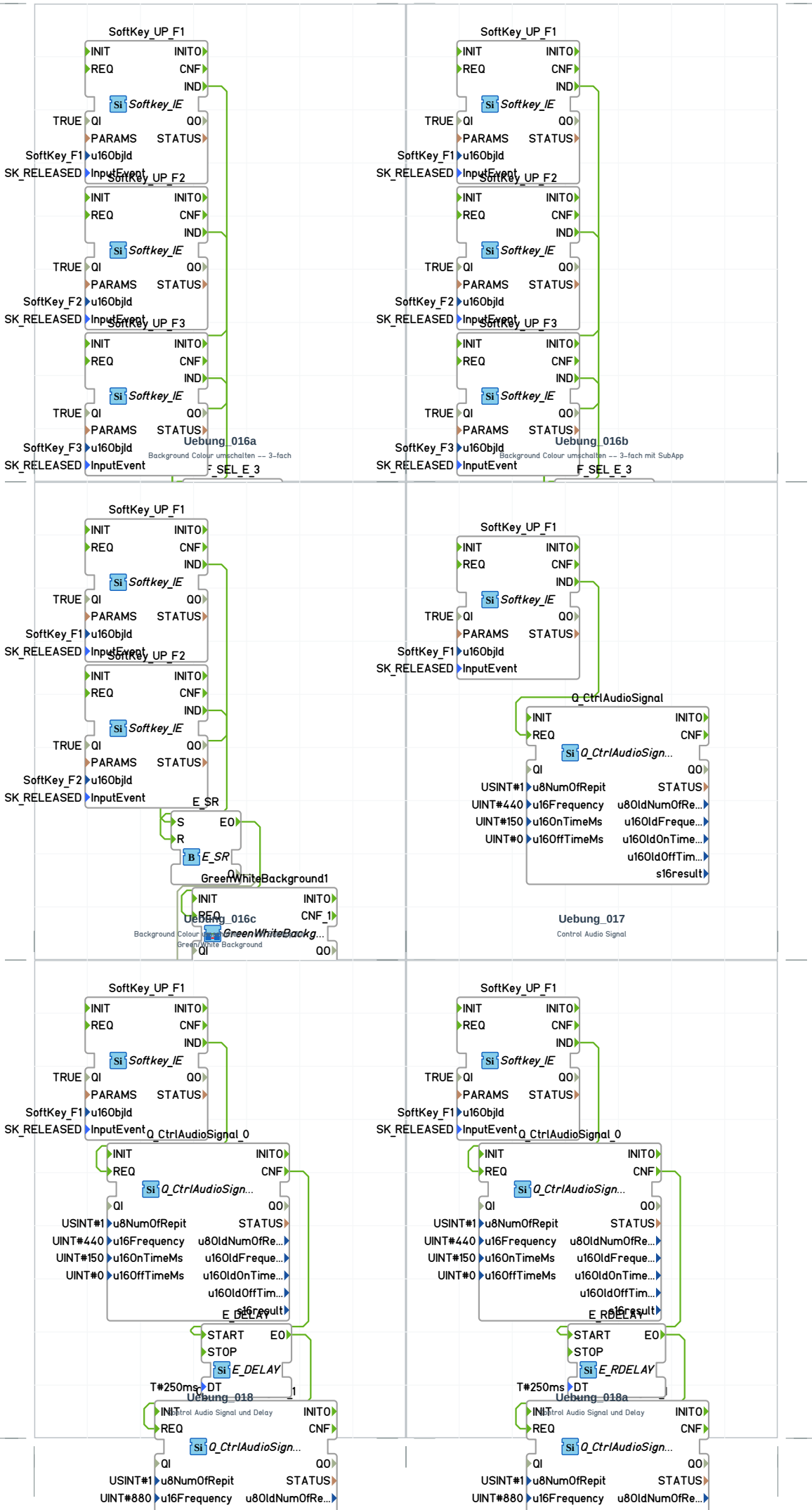

StringValue_IS
INIT
INITO
REQ
CNF
IND
StringValue_IS
TRUE
QI
QO
PARAMS
STATUS
InputString_S1
u16ObjId
IN
INI
INIT
INITO
SET
SETO
GET
SETOE
GETO
GETOE
INI
TRUE
QI
QO
SECTION_S1_STORE
SECTION
STATUS
KEY_S1_STORE
KEY
VALUEO
VALUE
STRING#'Test'
DEFAULT_VALU
Q_StringValue
INIT
INITO
REQ
CNF
Q_StringValue
QI
QO
InputString_S1
u16ObjId
STATUS
pau8String
pau8OldStrin...
Uebung_012j
String Input und Speichern INI

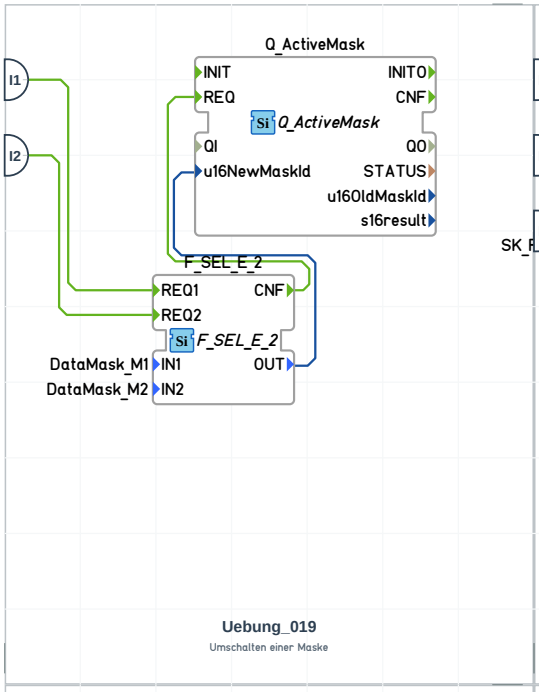

Uebung_012c_sub
INIT
INITO
IND
StringValue_TO_...
QI
QO
KEY_I1_STORE
KEY
VALUEO
SECTION_I1_STORE
SECTION
InputString_S1
u16ObjId
Uebung_012k
String Input und Speichern INI mit Subapp

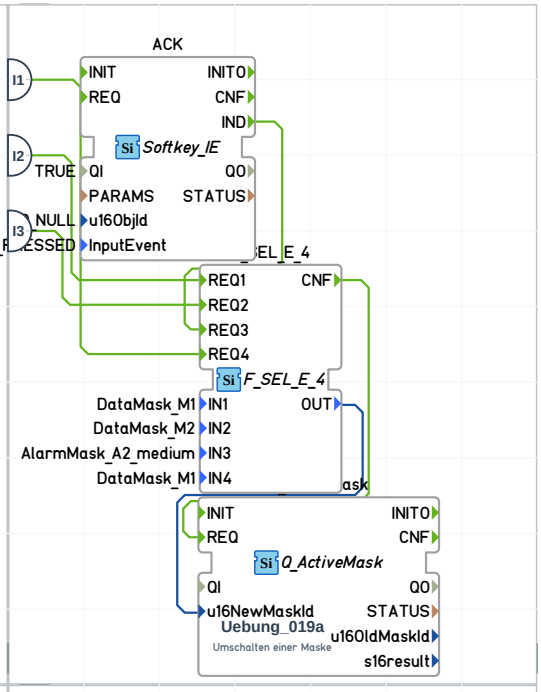

StringValue_IS
INIT
INITO
REQ
CNF
IND
StringValue_IS
TRUE
QI
QO
PARAMS
STATUS
InputString_S1
u16ObjId
IN
NVS
INIT
INITO
SET
SETO
GET
SETOE
GETO
GETOE
NVS
TRUE
QI
QO
KEY_S1_STORE
KEY
STATUS
VALUE
VALUEO
STRING#'Test'
DEFAULT_VALU
Q_StringValue
INIT
INITO
REQ
CNF
Q_StringValue
QI
QO
InputString_S1
u16ObjId
STATUS
pau8String
pau8OldStrin...
s16result
Uebung_012l
String Input und Speichern NVS

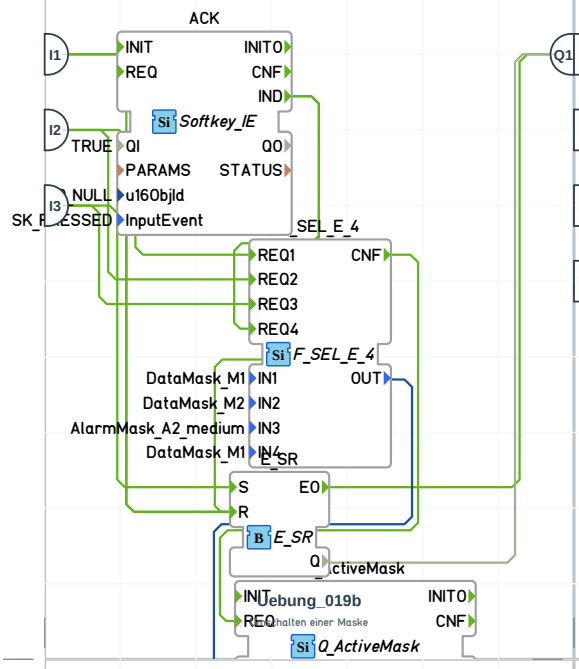

Uebung_012m_sub
INIT
INITO
IND
StringValue_TO_...
QI
QO
KEY_I1_STORE
KEY
VALUEO
InputString_S1
u16ObjId
Uebung_012m
String Input und Speichern NVS mit Subapp

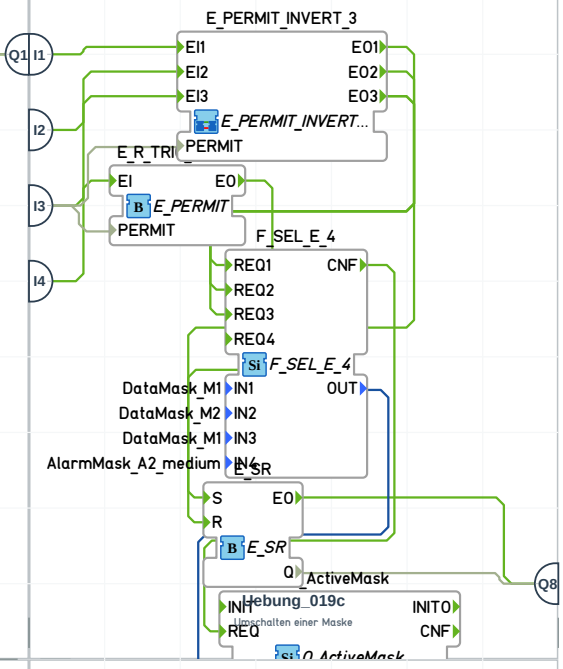

SoftKey_UP_F1
INIT
REQ
INITO
CNF
IND
Si Softkey_IE
TRUE QI
QO
PARAMS
STATUS
SoftKey_F1 u16ObjId
SK_RELEASED InputEvent
SoftKey_UP_F2
SoftKey_F2 u16ObjId
E_SR
Losgelassen
Losgelassen
S
EO
R
B E_SR
Q
Q1
Uebung_013
SR-Flip-Flop (Softkey)
SoftKey_UP_F3
SoftKey_F3 u16ObjId
C INIT
F_UINT_TO_UDINT
B F_UINT_TO_UDINT
Uebung_015
Object Pointer umschalten
Button_A1 IN
OUT
F_SEL
Uebung_015a
Object Pointer umschalten -- 3-fach
Uebung_015b
Uebung_016
Background Colour umschalten
COLOR_WHITE IN0
COLOR_GREEN IN1
Q_BackgroundColour
Uebung_014
Container (Unsichtbar/Invisible)
Q_ObjHideShow

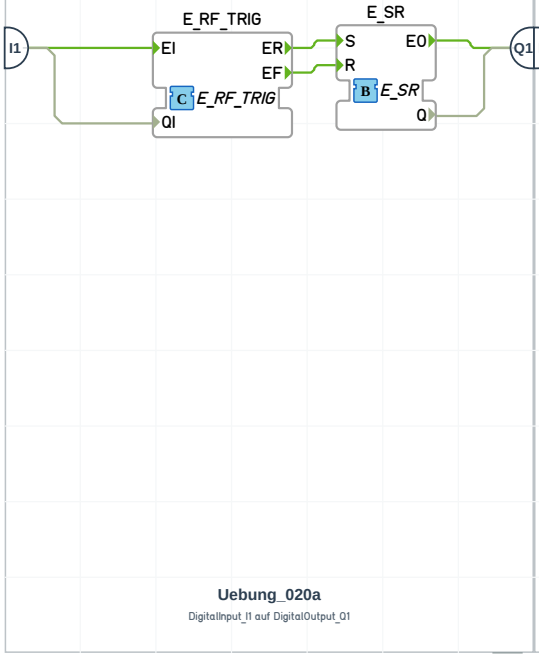

SoftKey_UP_F1
INIT
INITO
REQ
CNF
IND
Softkey_IE
TRUE
QI
QO
PARAMS
STATUS
SoftKey_F1
u16ObjId
SK_RELEASED
InputEvent
SoftKey_UP_F2
SoftKey_F2
SoftKey_UP_F3
SoftKey_F3
Uebung_016a
Background Colour umschalten -- 3-fach
F_SEL_E_3
Uebung_016b
Background Colour umschalten -- 3-fach mit SubApp
E_SR
S
EO
R
E_SR
GreenWhiteBackground1
CNF_1
Uebung_016c
Q_CtrlAudioSignal
Q_CtrlAudioSign...
USINT#1
u8NumOfRepit
UINT#440
u16Frequency
UINT#150
u16OnTimeMs
UINT#0
u16OffTimeMs
u80ldNumOfRe...
u160ldFreque...
u160ldOnTime...
u160ldOffTim...
s16result
Uebung_017
Control Audio Signal
Q_CtrlAudioSignal_0
E_DELAY
START
STOP
E_DELAY
T#250ms
DT
Uebung_018
Control Audio Signal und Delay
E_RDELAY
E_RDELAY
Uebung_018a
UINT#880

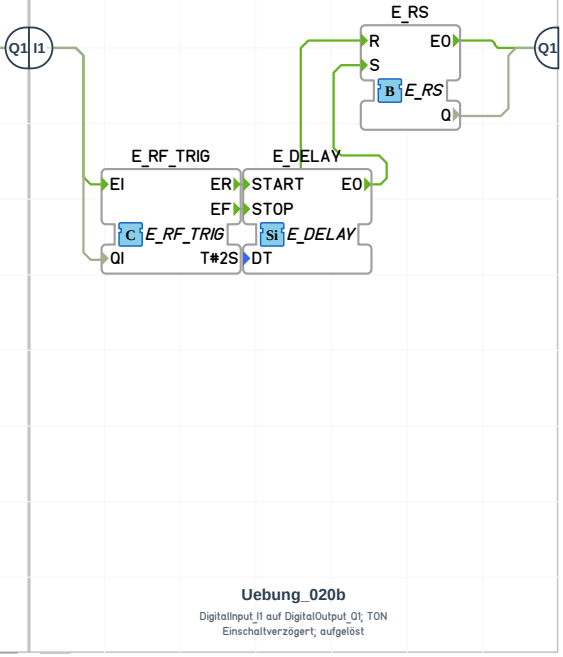

Q_ActiveMask
INIT
REQ
QI
u16NewMaskId
INITO
CNF
Q_ActiveMask
QO
STATUS
u16OldMaskId
s16result
I1
I2
F_SEL_E_2
REQ1
REQ2
F_SEL_E_2
DataMask_M1
DataMask_M2
IN1
IN2
OUT
CNF
Uebung_019
Umschalten einer Maske

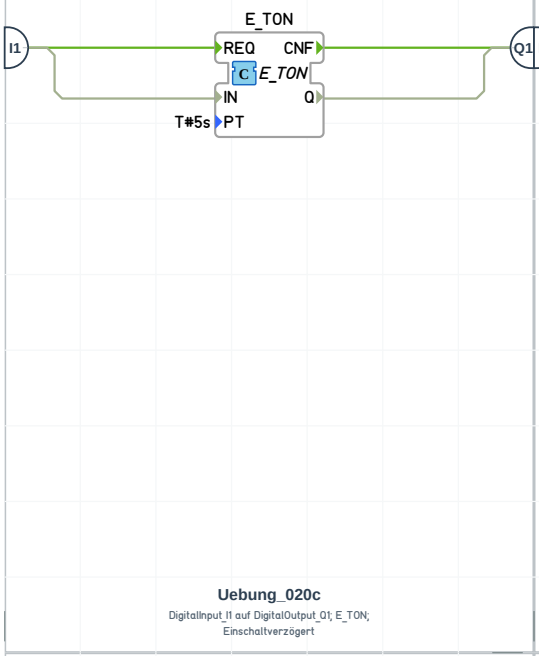

ACK
INIT
REQ
TRUE
QI
PARAMS
NULL
u16ObjId
InputEvent
INITO
CNF
IND
Softkey_IE
QO
STATUS
I1
I2
I3
REQ1
REQ2
REQ3
REQ4
F_SEL_E_4
DataMask_M1
DataMask_M2
AlarmMask_A2_medium
IN1
IN2
IN3
IN4
OUT
Q_ActiveMask
u16NewMaskId
u16OldMaskId
s16result
Uebung_019a
Umschalten einer Maske


ACK
INIT
REQ
TRUE
QI
PARAMS
NULL
u16ObjId
InputEvent
INITO
CNF
IND
Softkey_IE
QO
STATUS
I1
I2
I3
REQ1
REQ2
REQ3
REQ4
F_SEL_E_4
DataMask_M1
DataMask_M2
AlarmMask_A2_medium
IN1
IN2
IN3
IN4
OUT
E_SR
S
R
EO
Q
Uebung_019b
Umschalten einer Maske


E_PERMIT_INVERT_3
EI1
EI2
EI3
EO1
EO2
EO3
E_PERMIT_INVERT...
PERMIT
I1
I2
I3
I4
Q1
E_R_TRIG
EI
EO
E_PERMIT
F_SEL_E_4
REQ1
REQ2
REQ3
REQ4
F_SEL_E_4
DataMask_M1
DataMask_M2
AlarmMask_A2_medium
IN1
IN2
IN3
IN4
OUT
E_SR
S
R
EO
Q
Q8
Uebung_019c
Umschalten einer Maske

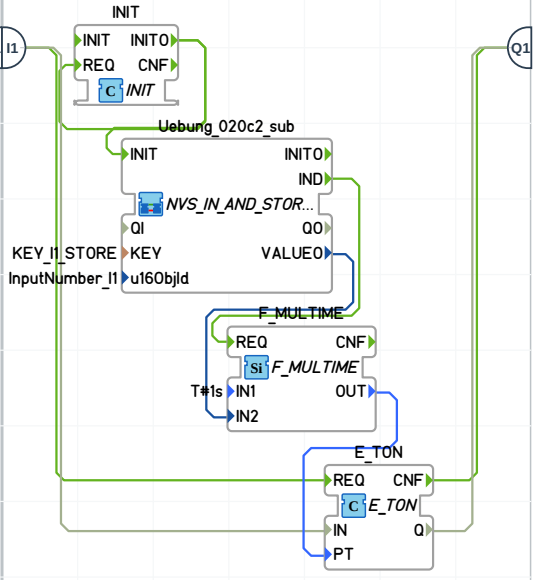

E_RF_TRIG
E_SR
I1
EI
ER
EF
QI
E_RF_TRIG
S
R
EO
Q
E_SR
Q1
Uebung_020a
DigitalInput_I1 auf DigitalOutput_Q1


E_RS
R
S
EO
Q
E_RS
Q1
I1
E_RF_TRIG
E_DELAY
EI
ER
EF
QI
T#2S
START
STOP
DT
E_RF_TRIG
E_DELAY
Uebung_020b
DigitalInput_I1 auf DigitalOutput_Q1; TON
Einschaltverzögert; aufgelöst



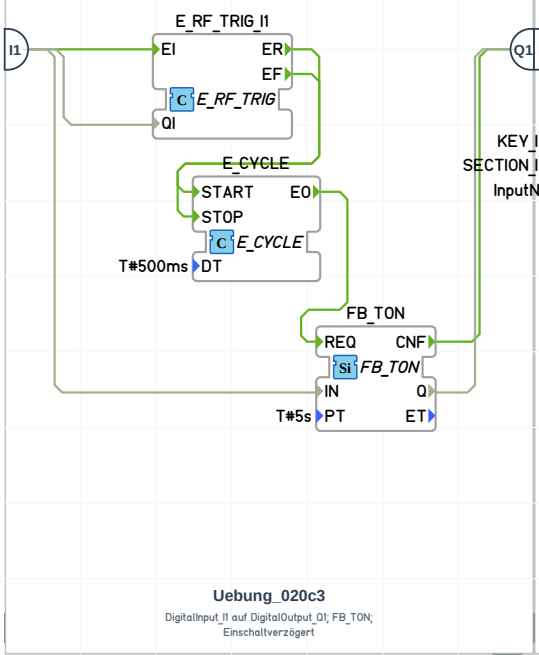
Uebung_020c

DigitalInput_I1 auf DigitalOutput_Q1; E_TON;
Einschaltverzögert



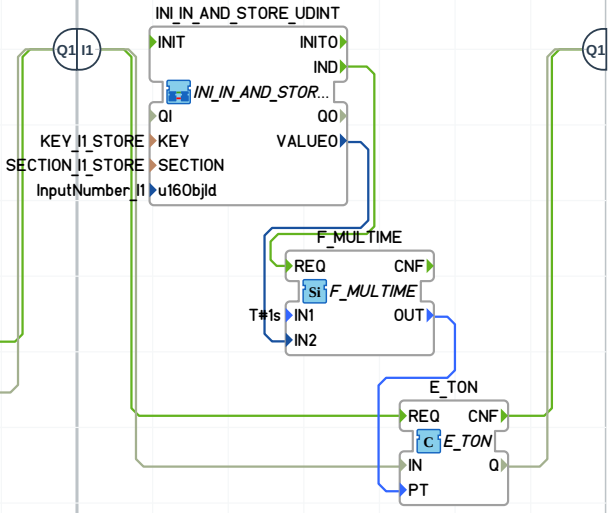
Uebung_020c2

DigitalInput_I1 auf DigitalOutput_Q1; E_TON
Einschaltverzögert; Zeit eingeben und speichern, NVS



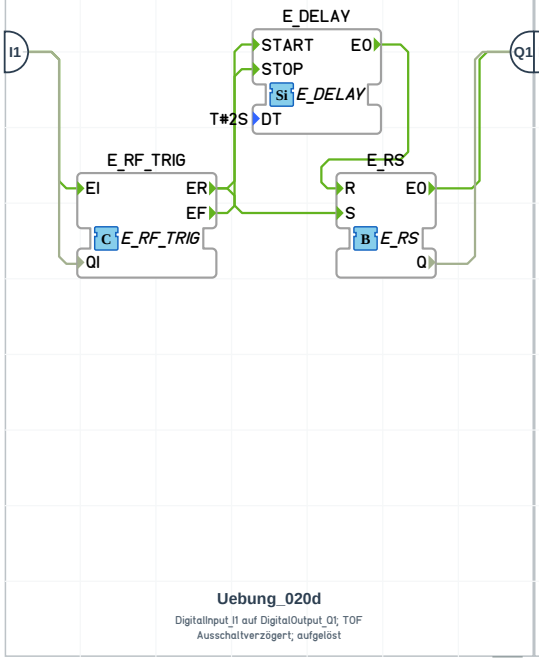
Uebung_020c3

DigitalInput_I1 auf DigitalOutput_Q1; FB_TON;
Einschaltverzögert



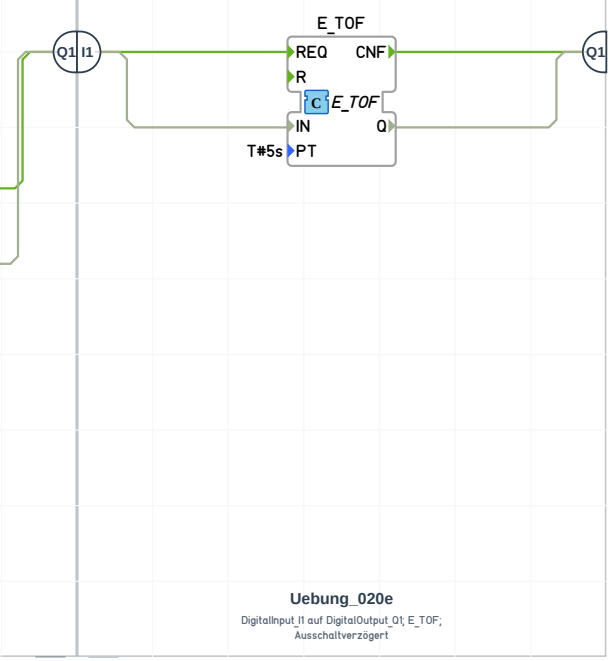
Uebung_020c4

DigitalInput_I1 auf DigitalOutput_Q1; E_TON
Einschaltverzögert; Zeit eingeben und speichern, INI



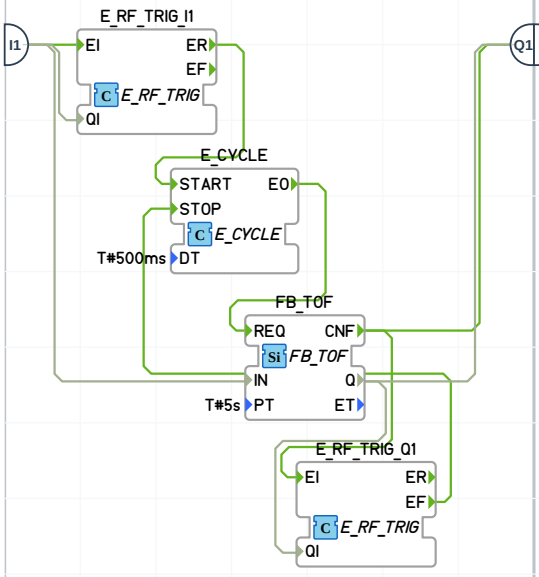
Uebung_020d

DigitalInput_I1 auf DigitalOutput_Q1; TOF
Ausschaltverzögert; aufgelöst



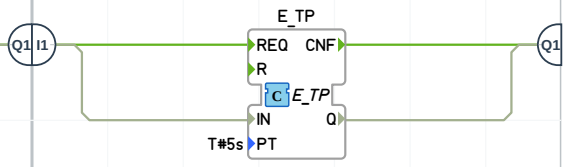
Uebung_020e

DigitalInput_I1 auf DigitalOutput_Q1; E_TOF;
Ausschaltverzögert



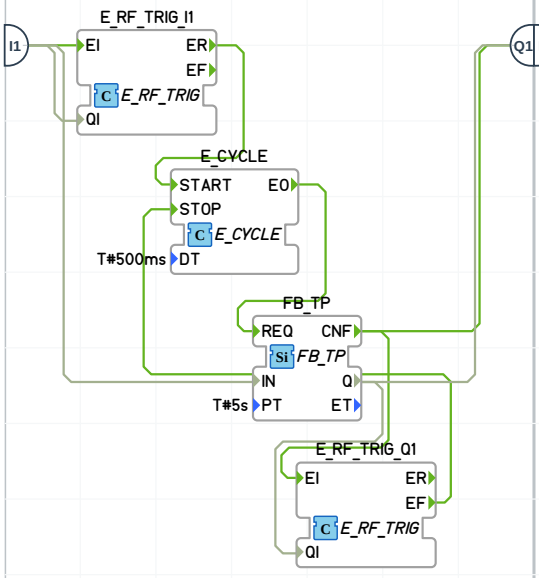
Uebung_020e2

DigitalInput_I1 auf DigitalOutput_Q1; FB_TOF; Ausschaltverzögert



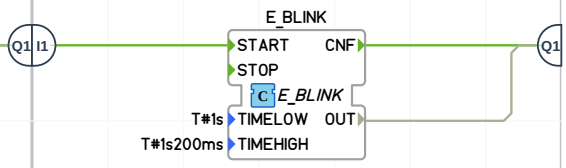
Uebung_020f

DigitalInput_I1 auf DigitalOutput_Q1; E_TP; Impulsformend



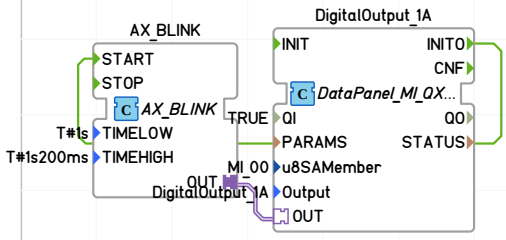
Uebung_020f2

DigitalInput_I1 auf DigitalOutput_Q1; FB_TP; Impulsformend



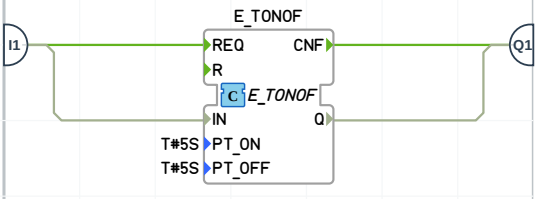
Uebung_020f3

DigitalInput_I1 auf DigitalOutput_Q1; BLINKER



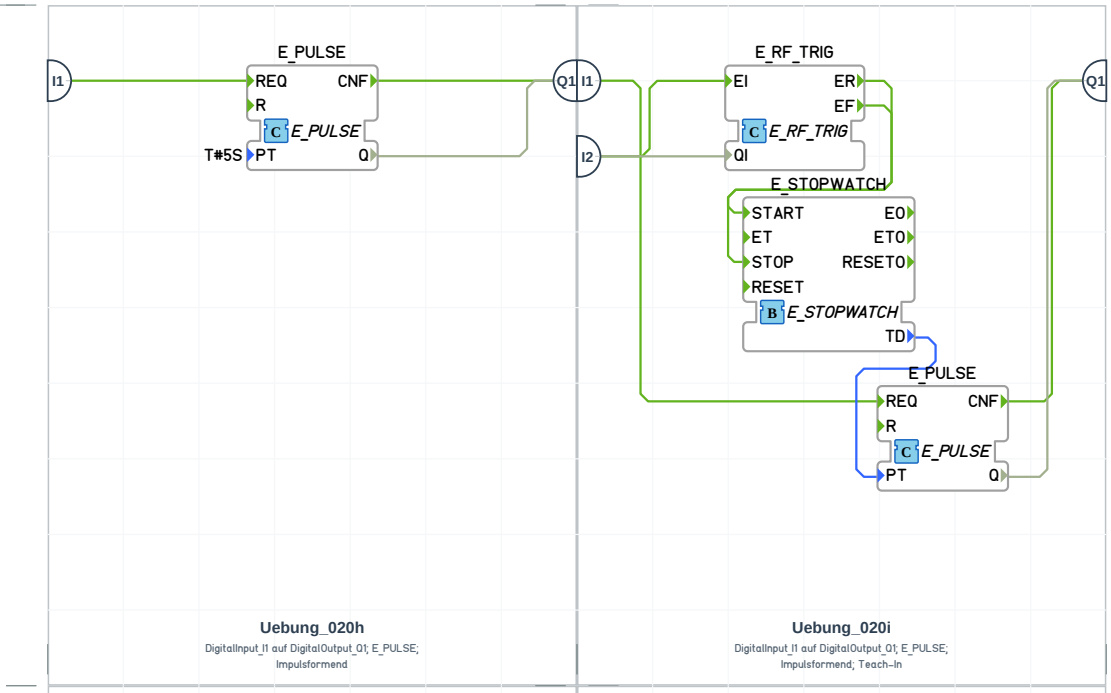
Uebung_020f4

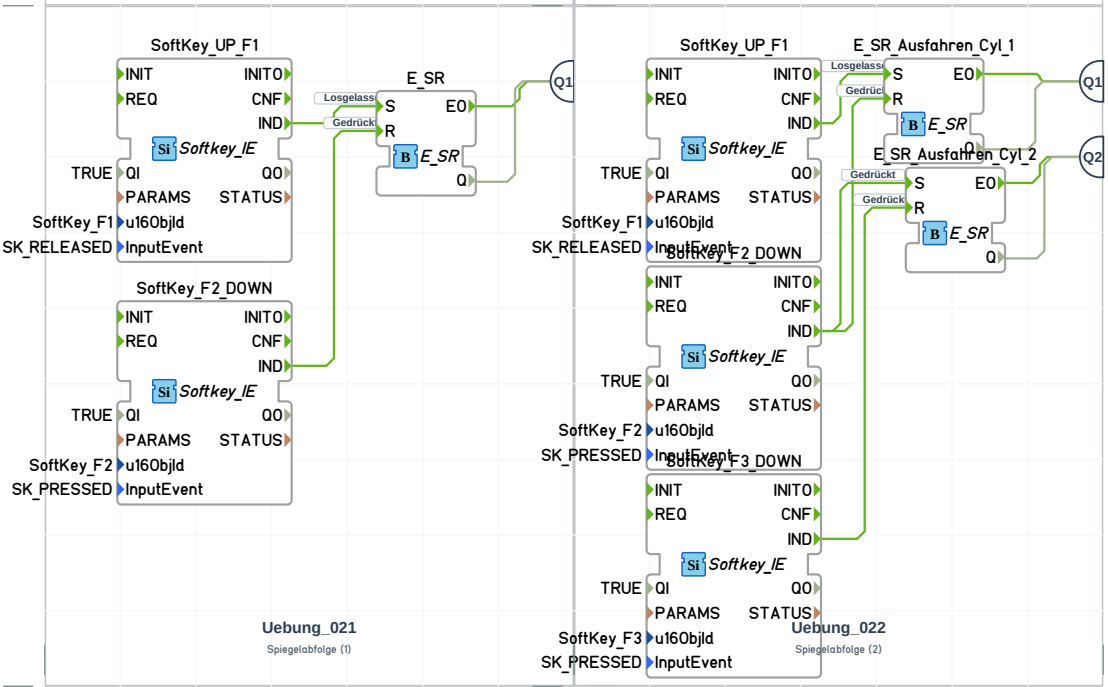
DigitalInput_I1 auf DataPanel_1A; BLINKER

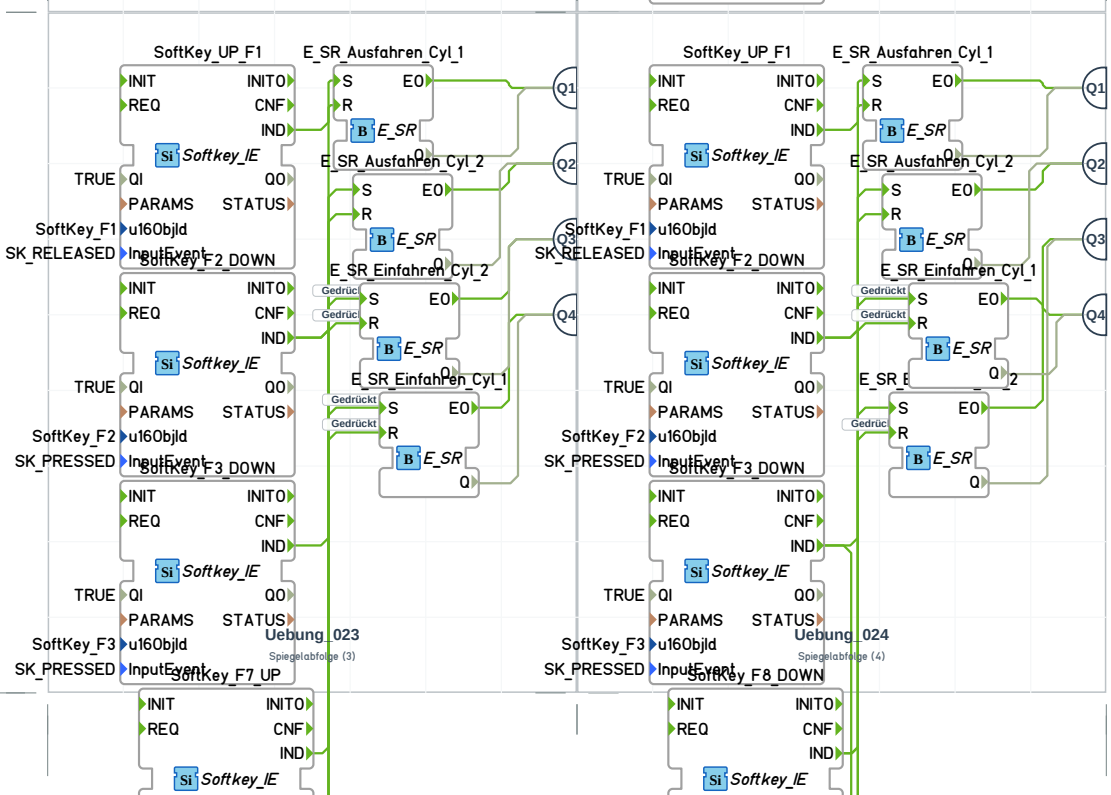


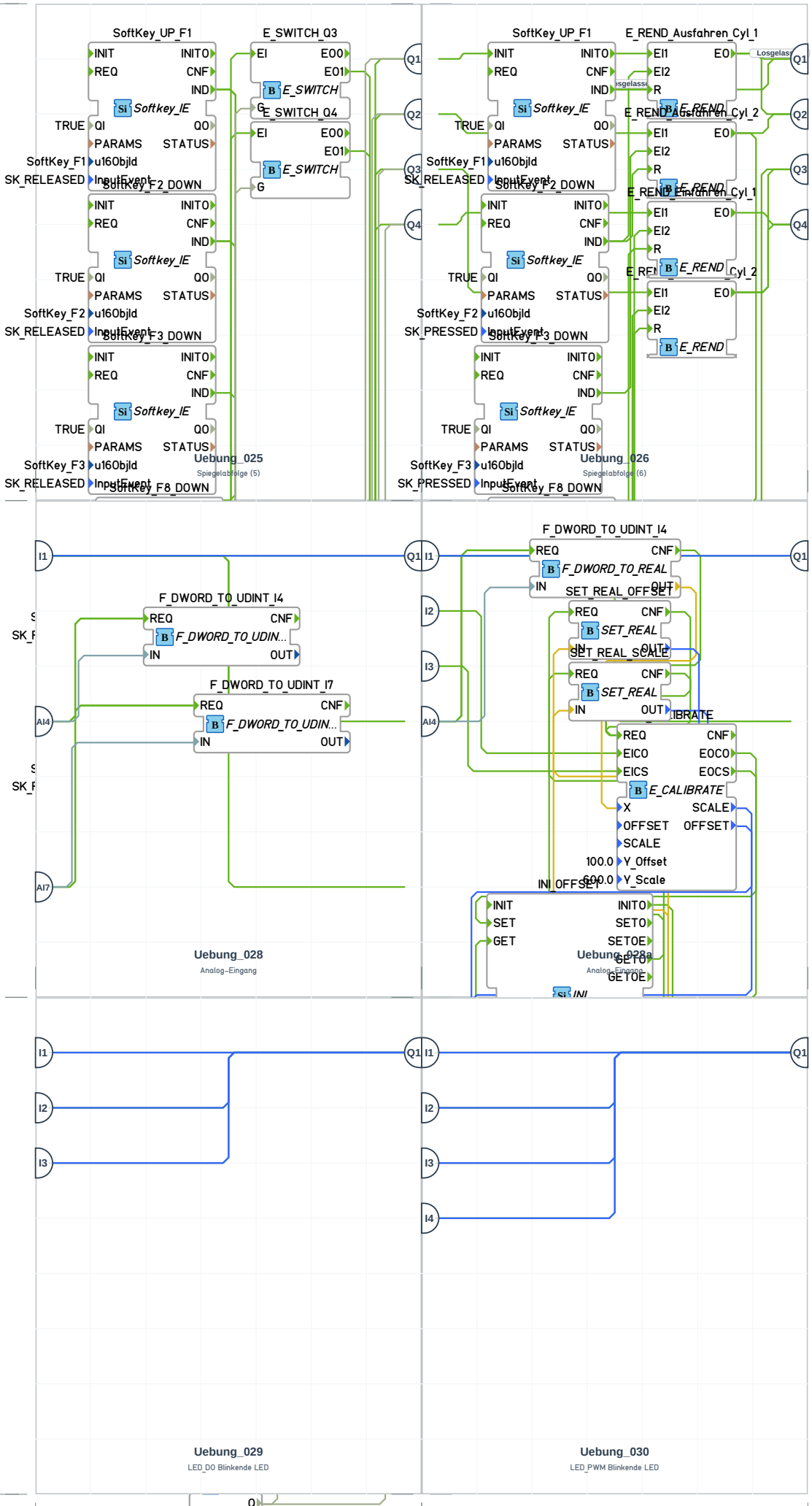
Uebung_020g

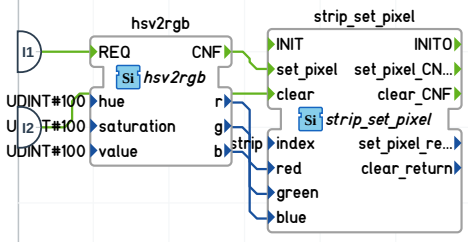
DigitalInput_I1 auf DigitalOutput_Q1; E_TONOF; Ein-/Ausschaltverzögert


E_PULSE
I1
REQ
CNF
R
C E_PULSE
T#5S PT
Q
Q1
E_RF_TRIG
EI
ER
EF
C E_RF_TRIG
QI
I2
E_STOPWATCH
START
EO
ET
ETO
STOP
RESETO
RESET
B E_STOPWATCH
TD
Uebung_020h
DigitalInput_I1 auf DigitalOutput_Q1; E_PULSE; Impulsformend
Uebung_020i
DigitalInput_I1 auf DigitalOutput_Q1; E_PULSE; Impulsformend; Teach-In


SoftKey_UP_F1
INIT
INITO
REQ
CNF
IND
Si Softkey_IE
TRUE
QI
QO
PARAMS
STATUS
SoftKey_F1
u16ObjId
SK_RELEASED
InputEvent
E_SR
Losgelass
Gedrück
S
EO
R
B E_SR
Q
SoftKey_F2_DOWN
SoftKey_F2
SK_PRESSED
SoftKey_F3_DOWN
SoftKey_F3
E_SR_Ausfahren_Cyl_1
E_SR_Ausfahren_Cyl_2
Gedrückt
Uebung_021
Spiegelabfolge (1)
Uebung_022
Spiegelabfolge (2)

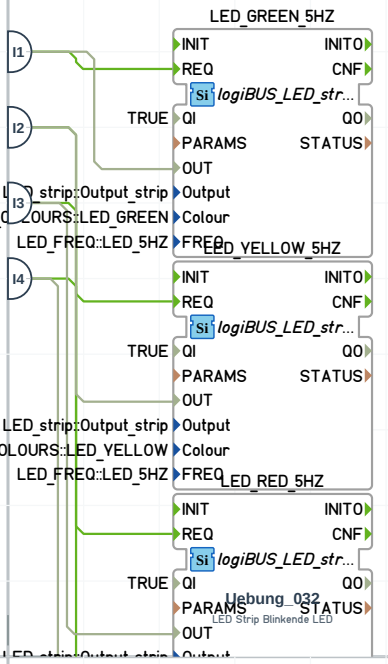

SoftKey_UP_F1
E_SR_Ausfahren_Cyl_1
E_SR_Ausfahren_Cyl_2
E_SR_Einfahren_Cyl_2
E_SR_Einfahren_Cyl_1
SoftKey_F2_DOWN
SoftKey_F3_DOWN
SoftKey_F7_UP
SoftKey_F8_DOWN
Uebung_023
Spiegelabfolge (3)
Uebung_024
Spiegelabfolge (4)


SoftKey_UP_F1
E_SWITCH_Q3
INIT
REQ
INITO
CNF
IND
Softkey_IE
TRUE
QI
QO
PARAMS
STATUS
SoftKey_F1
u16ObjId
SK_RELEASED
InputEvent
Softkey_F2_DOWN
SoftKey_F2
Softkey_F3_DOWN
SoftKey_F3
Softkey_F8_DOWN
EI
EO00
EO01
E_SWITCH
G
E_SWITCH_Q4
Q1
Q2
Q3
Q4
Uebung_025
Spiegelabfolge (5)
E_REND_Ausfahren_Cyl_1
E_REND_Ausfahren_Cyl_2
E_REND_Einfahren_Cyl_1
E_REND_Einfahren_Cyl_2
EI1
EI2
EO
R
E_REND
SK_PRESSED
Uebung_026
Spiegelabfolge (6)
F_DWORD_TO_UDINT_I4
F_DWORD_TO_UDINT_I7
F_DWORD_TO_UDIN...
REQ
CNF
IN
OUT
I1
I2
I3
AI4
AI7
Uebung_028
Analog-Eingang
F_DWORD_TO_REAL
SET_REAL_OFFSET
SET_REAL
SET_REAL_SCALE
E_CALIBRATE
EICO
EICS
EOCO
EOCS
X
SCALE
OFFSET
100.0
Y_Offset
600.0
Y_Scale
INI_OFFSET
INIT
SET
GET
INITO
SETO
SETOE
GETOE
Uebung_028a
Analog-Eingang
I4
Uebung_029
LED_DO Blinkende LED
Uebung_030
LED_PWM Blinkende LED
Q



Uebung_031

LED Strip

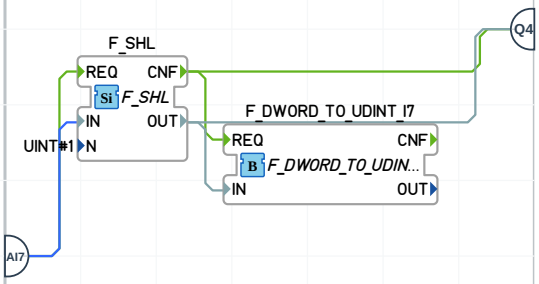


Uebung_032

LED Strip Blinkende LED

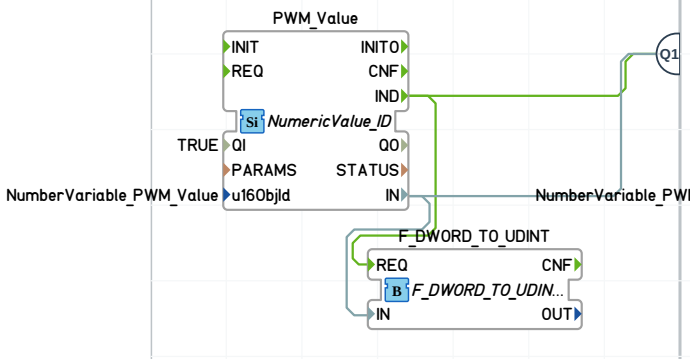
Uebung_033

LED Strip Blinkende LED



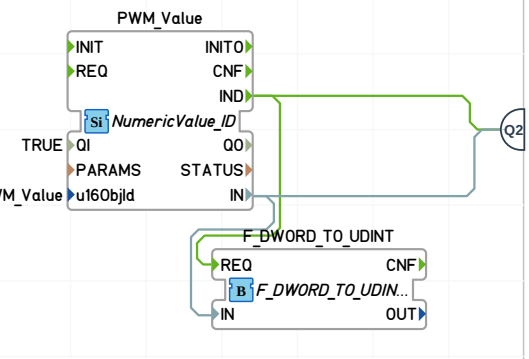
Uebung_034

Analog-Eingang auf PWM Ausgang



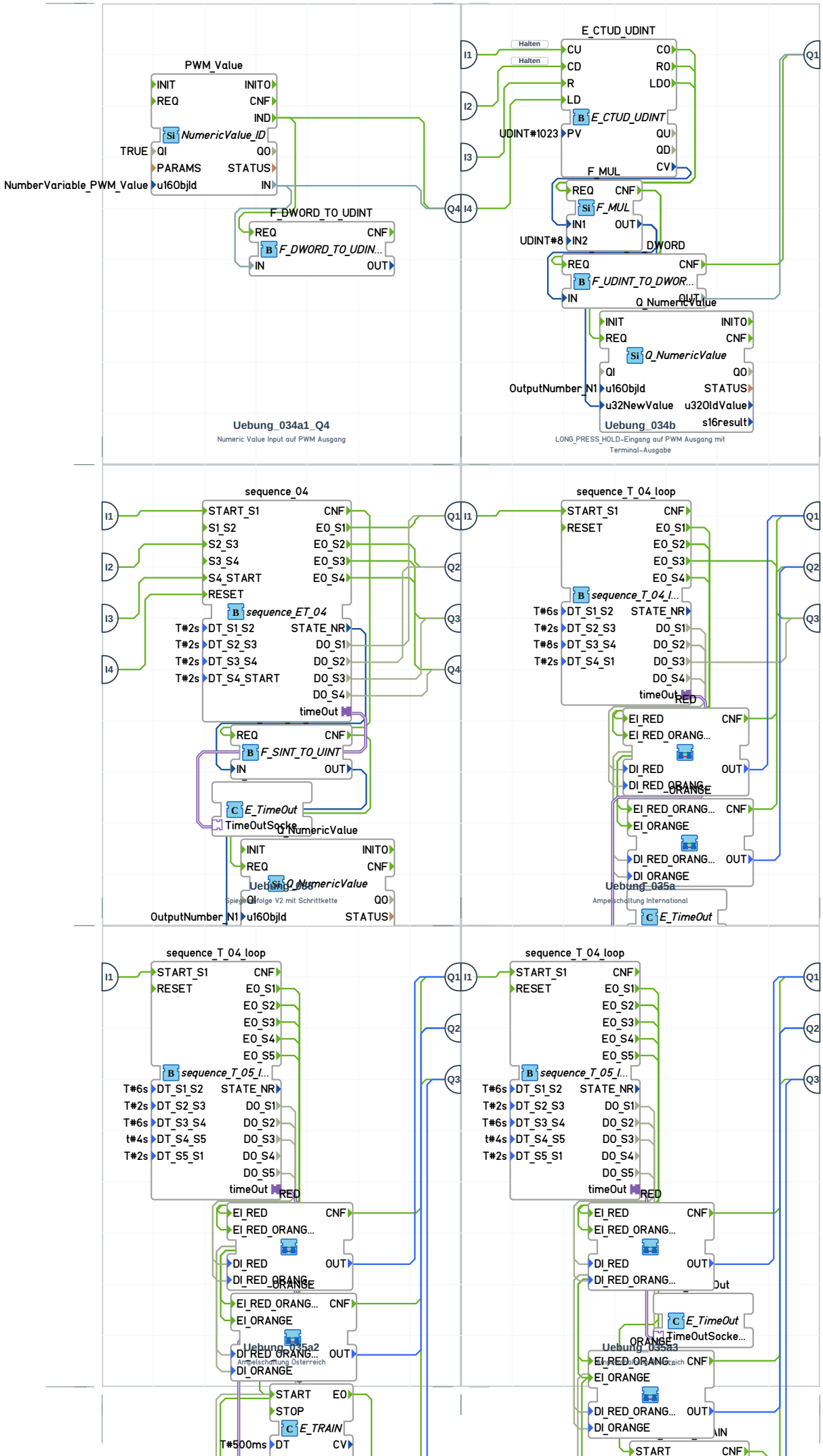
Uebung_034a1_Q1

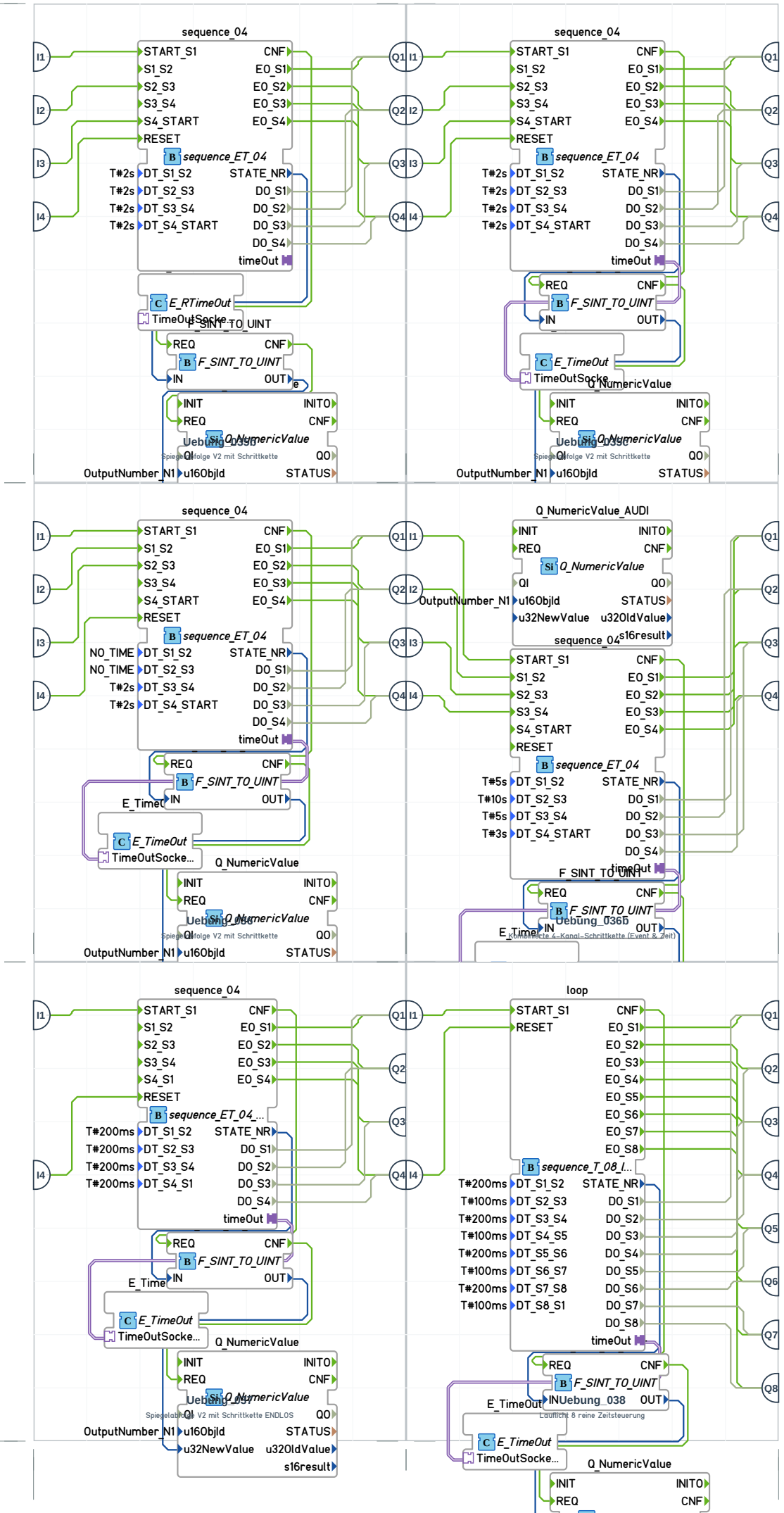
Numeric Value Input auf PWM Ausgang

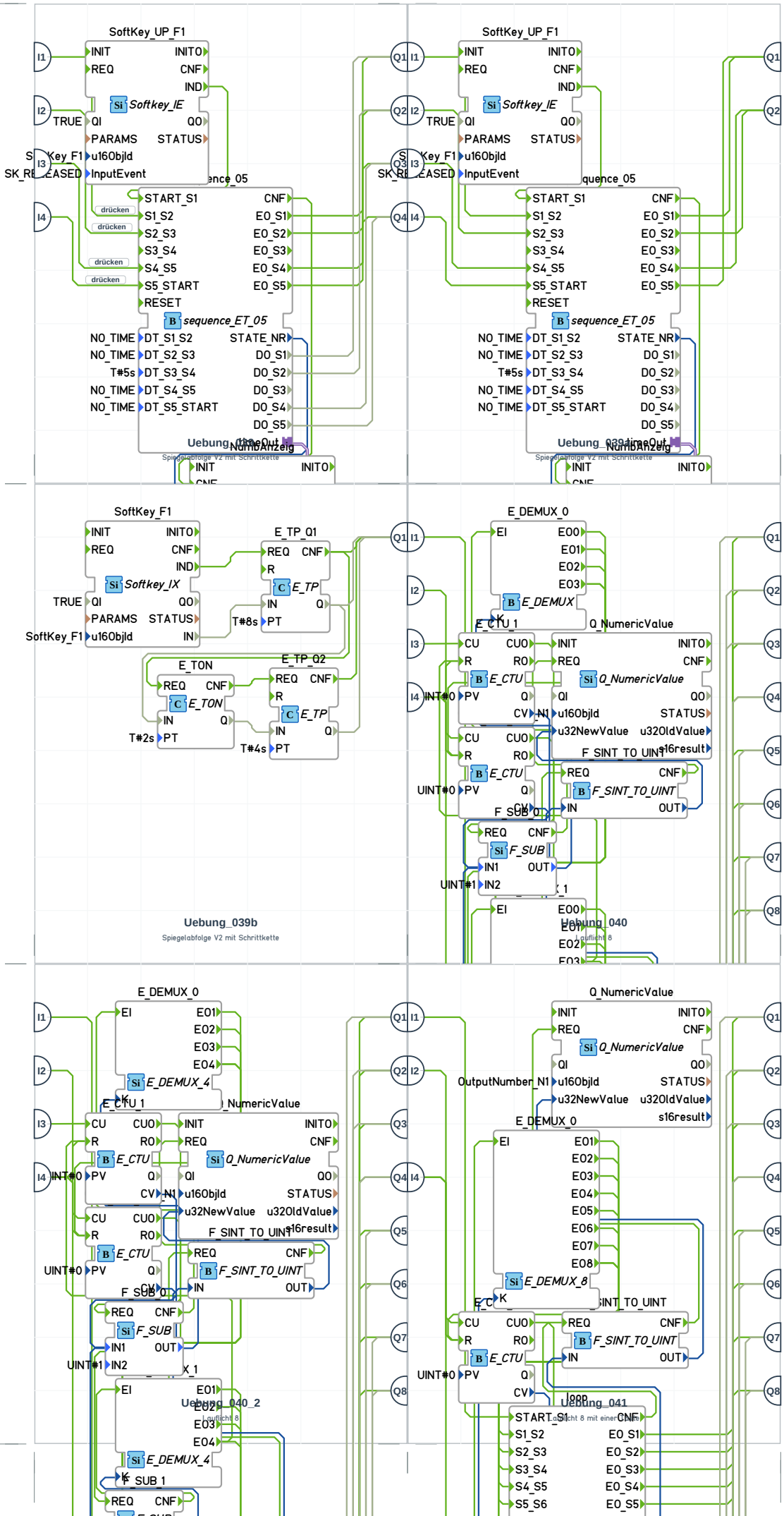


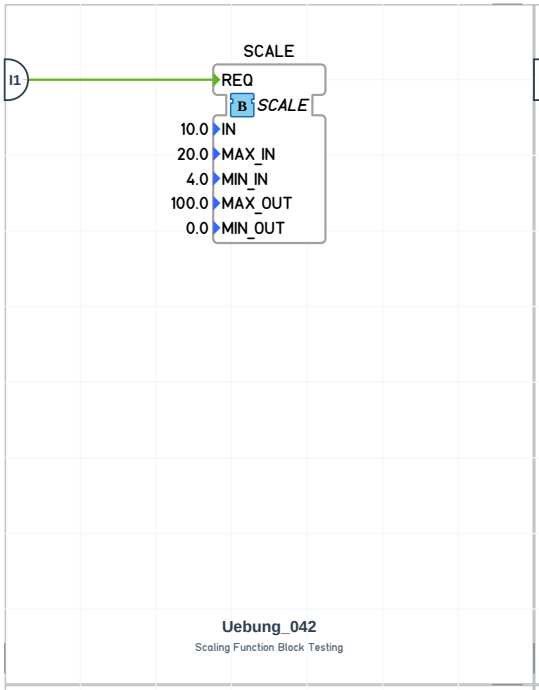
Uebung_034a1_Q2

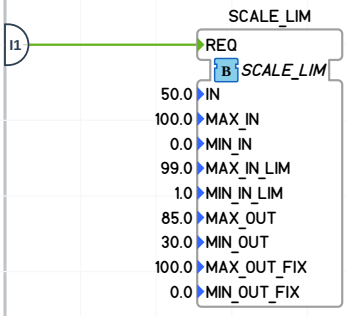
Numeric Value Input auf PWM Ausgang


PWM_Value
INIT
REQ
NumericValue_ID
QI
PARAMS
u16ObjId
INITO
CNF
IND
QO
STATUS
IN
TRUE
NumberVariable_PWM_Value
F_DWORD_TO_UDINT
F_DWORD_TO_UDIN...
Uebung_034a1_Q4
Numeric Value Input auf PWM Ausgang
E_CTUD_UDINT
CU
CD
R
LD
E_CTUD_UDINT
PV
CO
RO
LDO
QU
QD
CV
Halten
Halten
I1
I2
I3
I4
Q1
Q4
UDINT#1023
F_MUL
IN1
IN2
OUT
UDINT#8
DWORD
F_UDINT_TO_DWOR...
Q_NumericValue
OutputNumber_N1
u32NewValue
u32OldValue
s16result
Uebung_034b
LONG_PRESS_HOLD-Eingang auf PWM Ausgang mit Terminal-Ausgabe
sequence_04
START_S1
S1_S2
S2_S3
S3_S4
S4_START
RESET
sequence_ET_04
DT_S1_S2
DT_S2_S3
DT_S3_S4
DT_S4_START
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
timeOut
T#2s
F_SINT_TO_UINT
E_TimeOut
TimeOutSocke...
Uebung_035a
Ampelschaltung International
sequence_T_04_loop
sequence_T_04_l...
T#6s
T#8s
T#4s
EO_S1
EO_S2
EO_S3
EO_S4
EO_S5
DO_S5
RED
EI_RED
EI_RED_ORANG...
DI_RED
DI_RED_ORANGE
EI_ORANGE
DI_ORANGE
sequence_T_05_l...
Uebung_035a2
Ampelschaltung Österreich
START
STOP
E_TRAIN
T#500ms
DT
Uebung_035a3


sequence_04
START_S1
S1_S2
S2_S3
S3_S4
S4_START
RESET
sequence_ET_04
DT_S1_S2
DT_S2_S3
DT_S3_S4
DT_S4_START
CNF
EO_S1
EO_S2
EO_S3
EO_S4
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
timeOut
E_RTimeOut
TimeOutSocke...
F_SINT_TO_UINT
REQ
IN
OUT
INIT
INITO
QI
QO
STATUS
Q_NumericValue
OutputNumber_N1
u16ObjId
Uebung_035b
Spiegelabfolge V2 mit Schrittkette
sequence_04
E_TimeOut
Q_NumericValue_AUDI
u32NewValue
u32OldValue
s16result
Uebung_036b
Kombinierte 4-Kanal-Schrittkette (Event & Zeit)
sequence_04
sequence_ET_04_...
E_TimeOut
Uebung_037
Spiegelabfolge V2 mit Schrittkette ENDLOS
loop
sequence_T_08_I...
Uebung_038
Lauflicht 8 reine Zeitsteuerung


SoftKey_UP_F1
INIT
REQ
INITO
CNF
IND
Softkey_IE
TRUE
QI
QO
PARAMS
STATUS
u16ObjId
InputEvent
START_S1
S1_S2
S2_S3
S3_S4
S4_S5
S5_START
RESET
CNF
EO_S1
EO_S2
EO_S3
EO_S4
EO_S5
sequence_ET_05
NO_TIME
T#5s
DT_S1_S2
DT_S2_S3
DT_S3_S4
DT_S4_S5
DT_S5_START
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
DO_S5
NumbAnzeig
Uebung_039a
Spiegelabfolge V2 mit Schrittkette
SoftKey_F1
Softkey_IX
E_TP_Q1
E_TP
T#8s
E_TON
T#2s
T#4s
Uebung_039b
E_DEMUX_0
E_DEMUX
E_CTU_1
E_CTU
INT#0
Q_NumericValue
F_SINT_TO_UINT
F_SUB_0
F_SUB
UINT#0
UINT#1
Uebung_040
E_DEMUX_4
E_CTU_2
F_SUB_1
Uebung_040_2
E_DEMUX_8
OutputNumber_N1
Uebung_041
Loop
S6



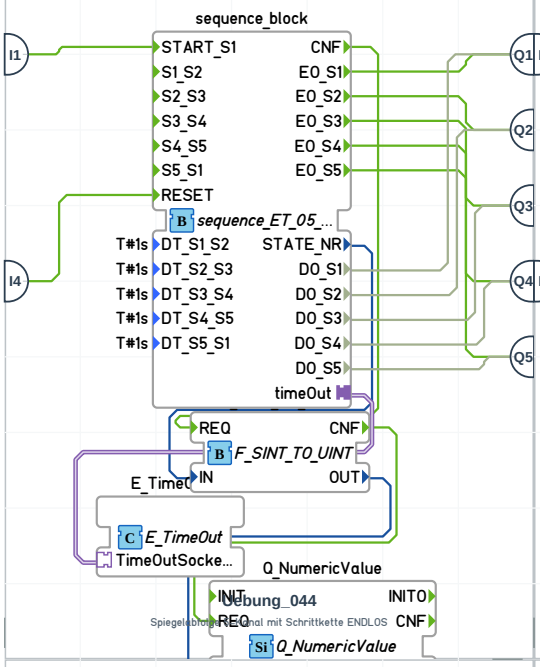


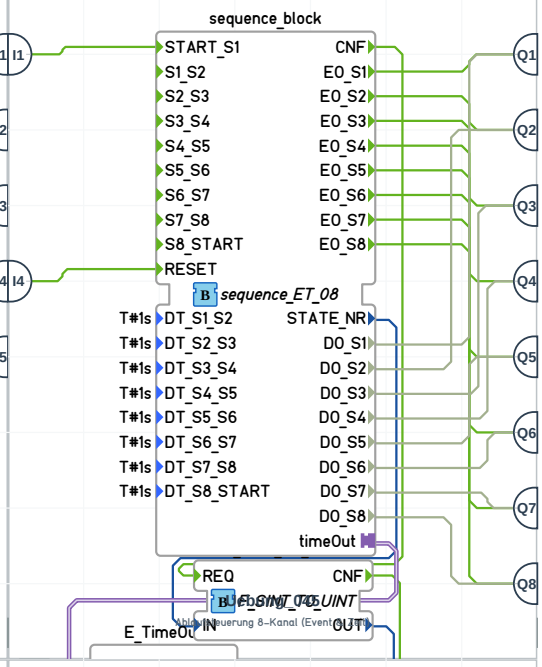
Uebung_042

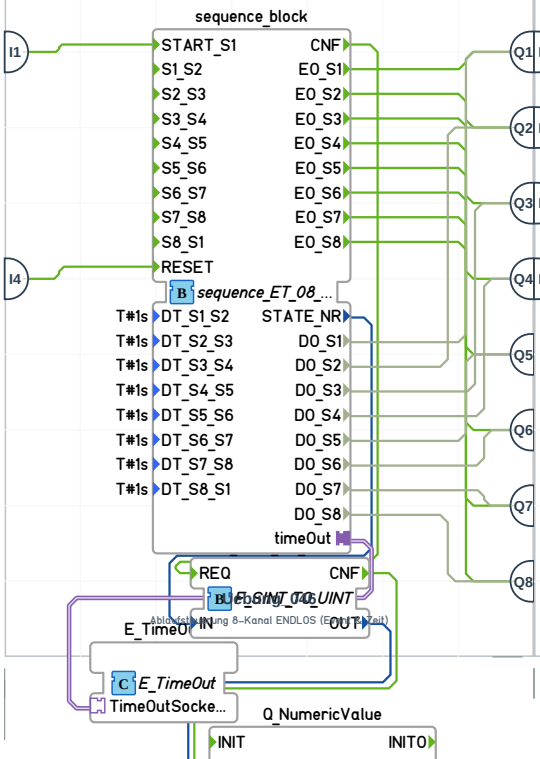
Scaling Function Block Testing

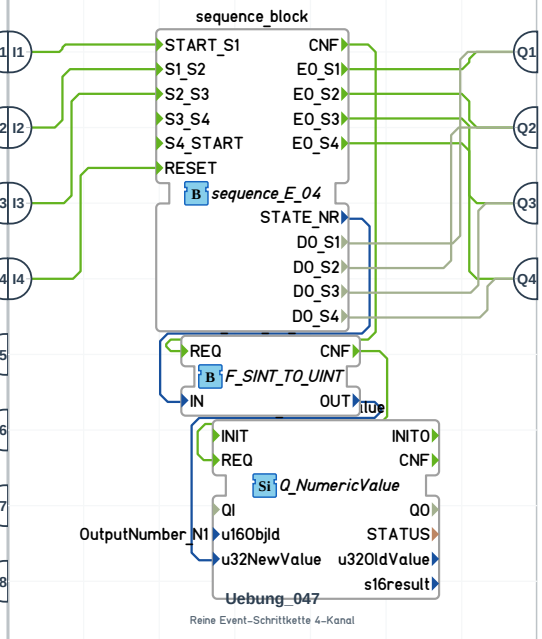
Uebung_043

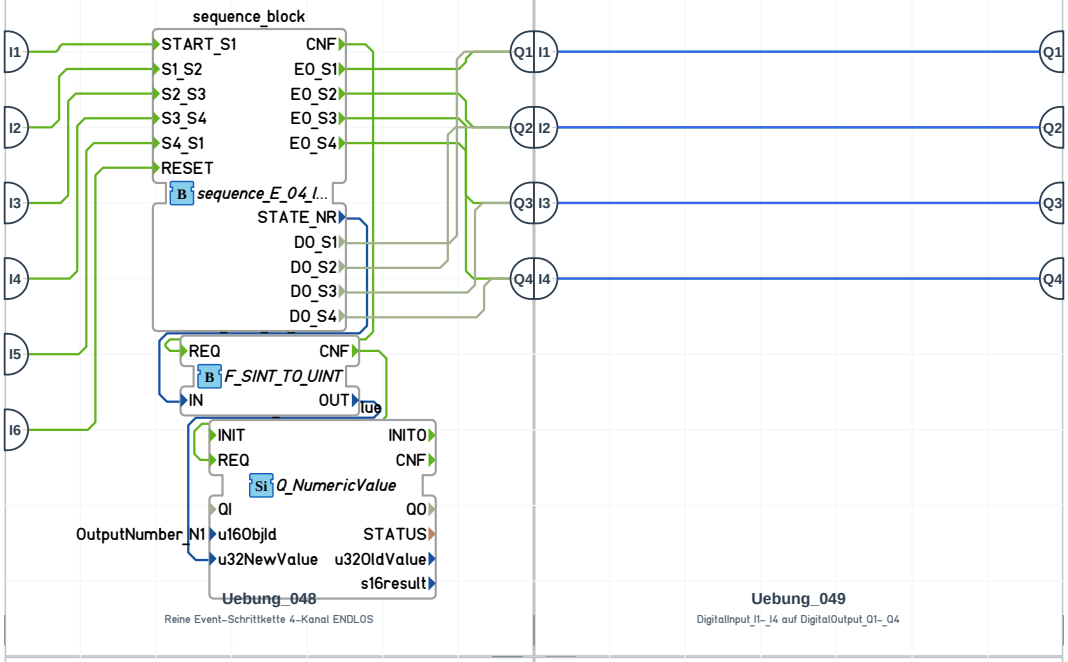
Scaling Function Block with limits Testing









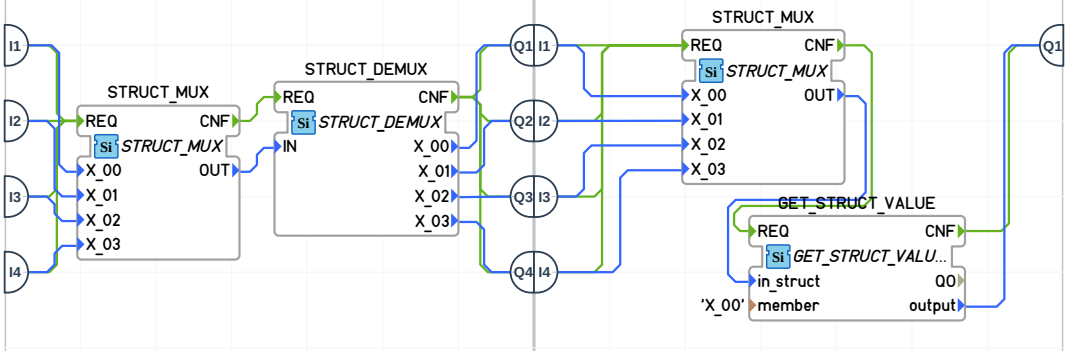


Uebung_048

Reine Event-Schrittkette 4-Kanal ENDLOS

Uebung_049

DigitalInput_I1-_I4 auf DigitalOutput_Q1-_Q4

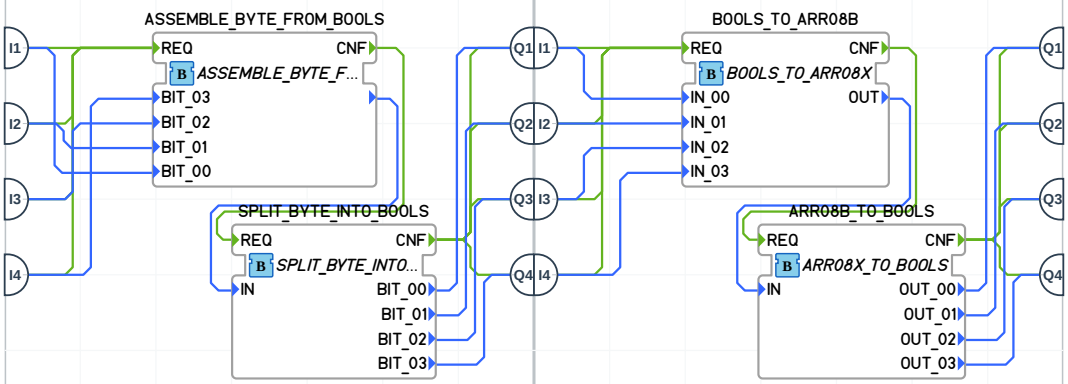


Uebung_051

DigitalInput_I1-_I4 auf DigitalOutput_Q1-_Q4

Uebung_052

DigitalInput_I1-_I4 auf DigitalOutput_Q1-_Q4

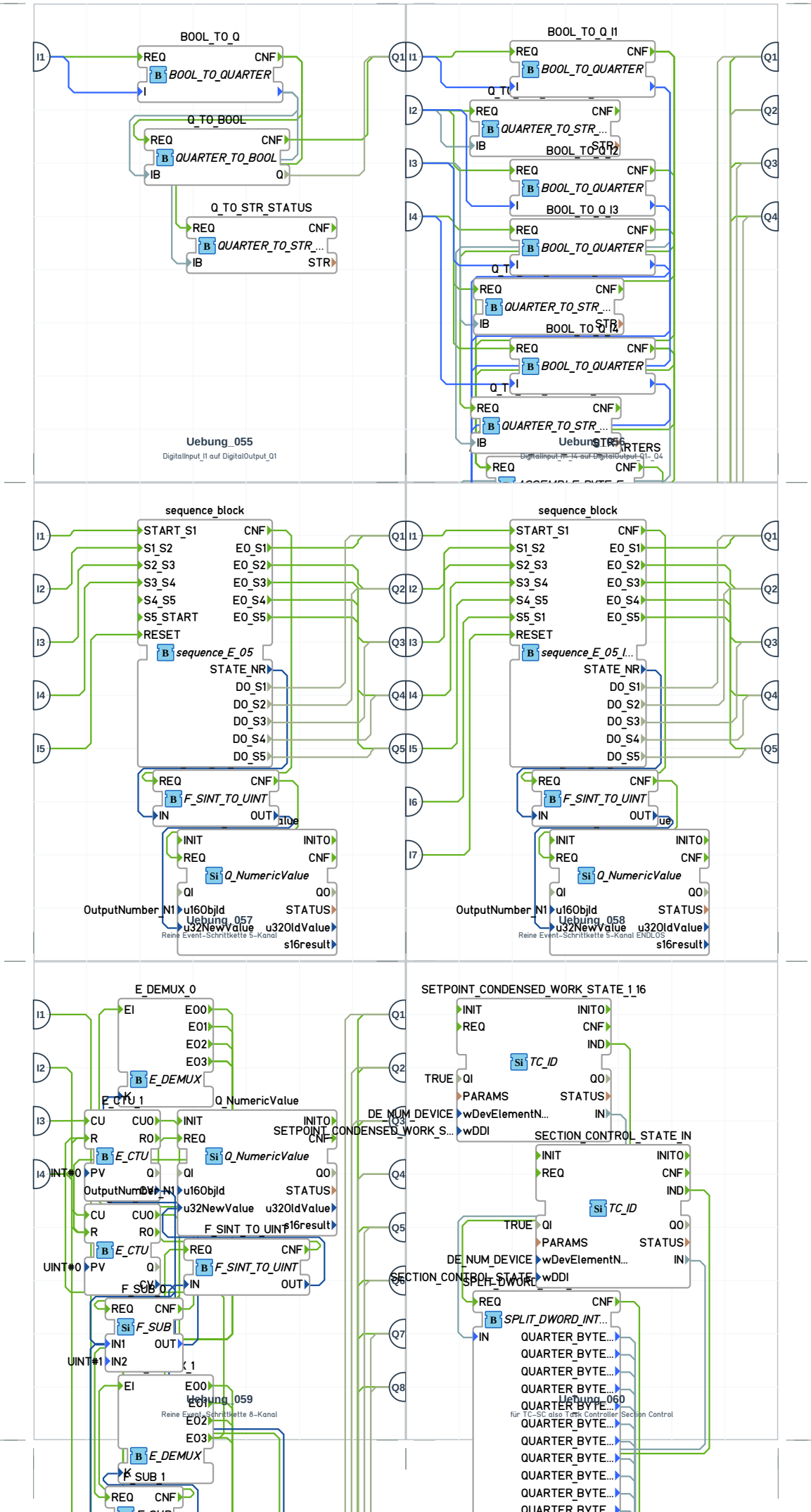


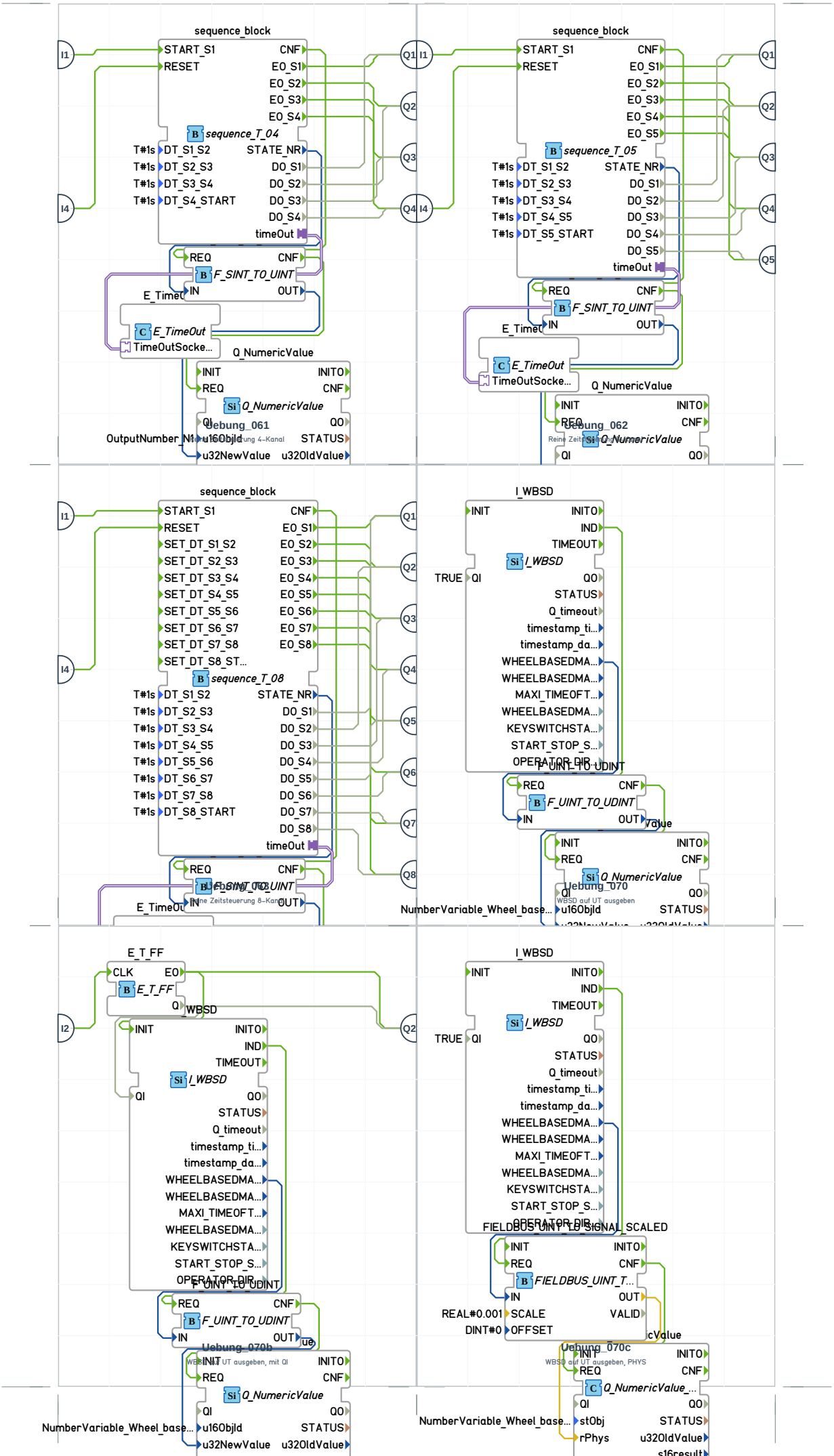
Uebung_053

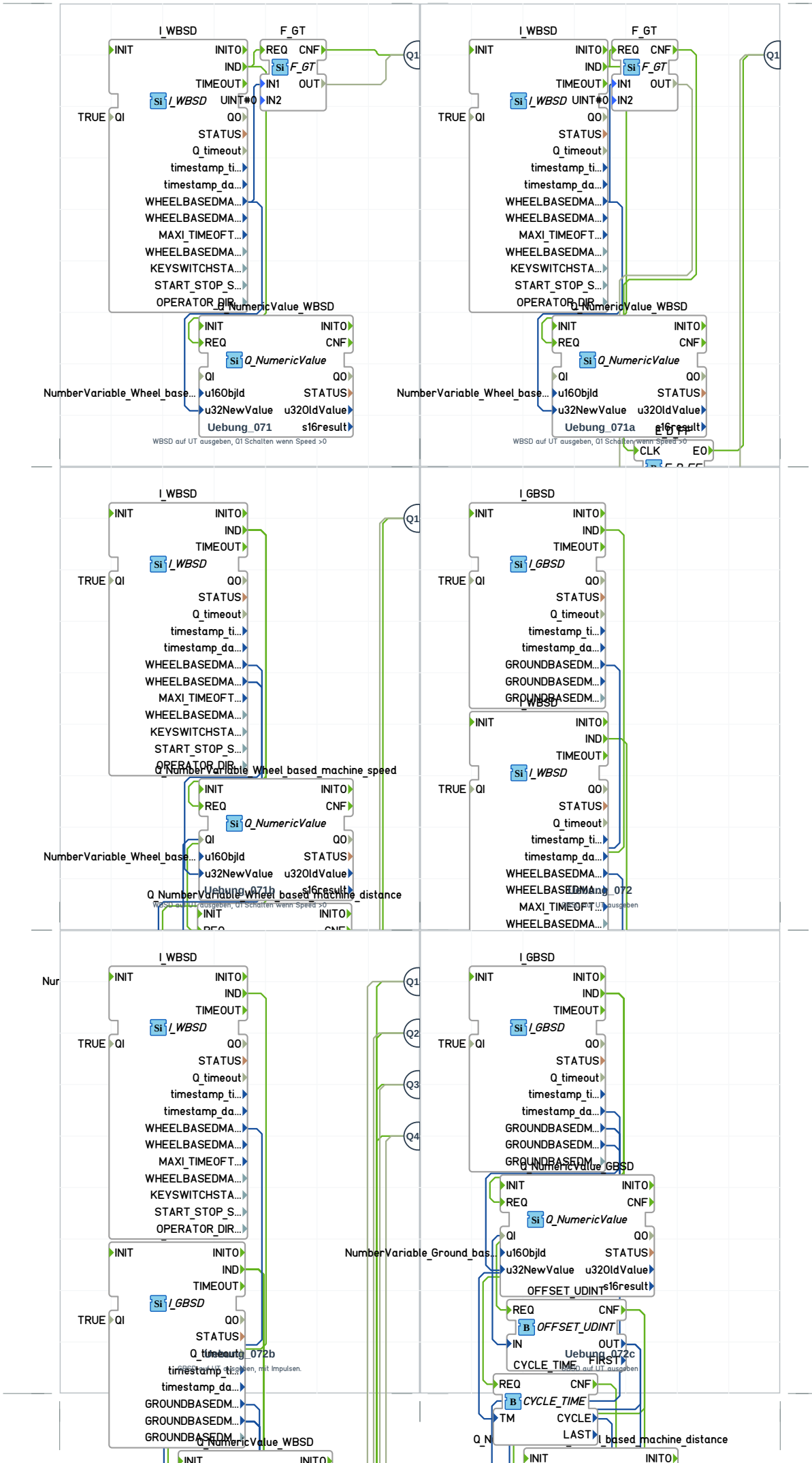
DigitalInput_I1-_I4 auf DigitalOutput_Q1-_Q4

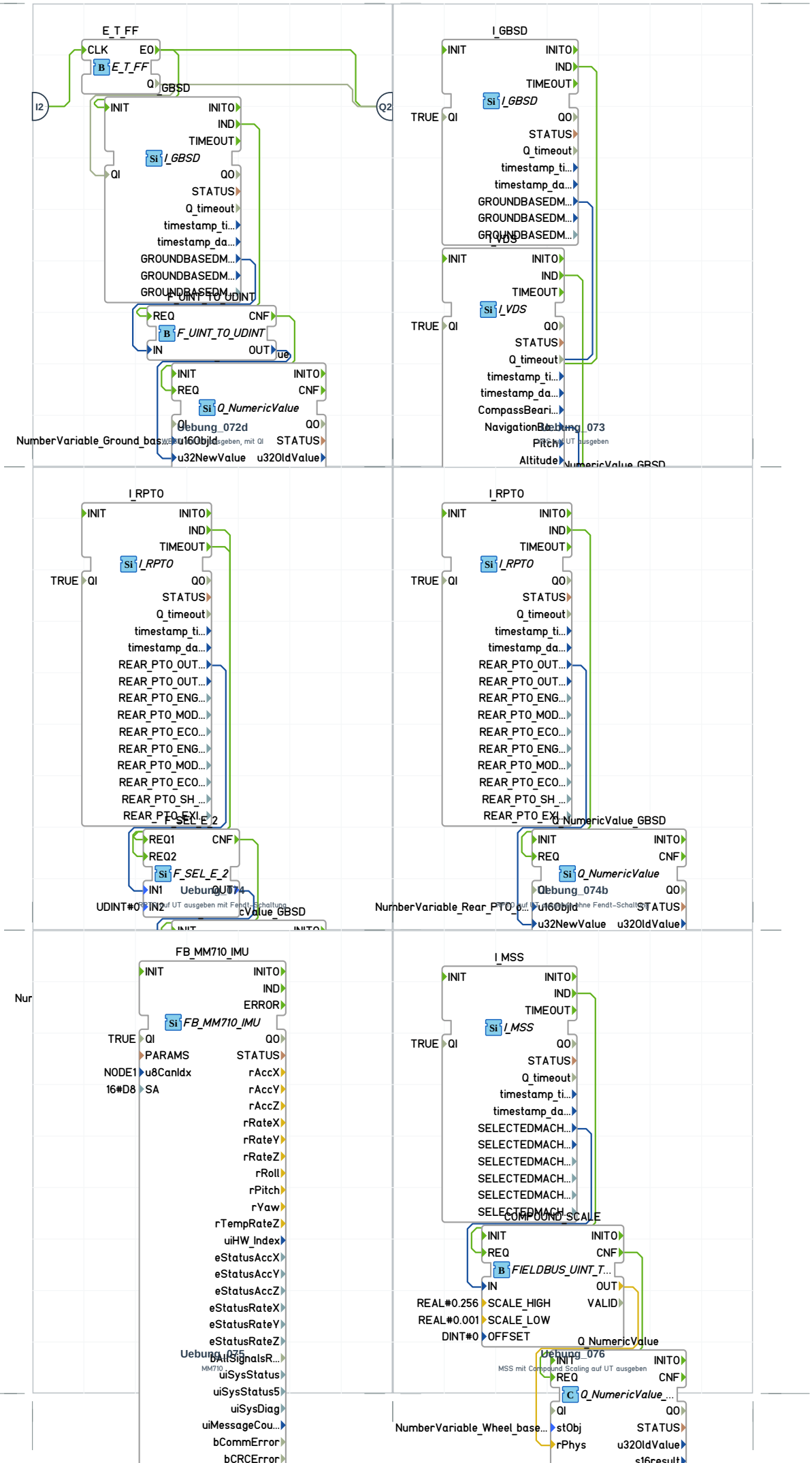
Uebung_054

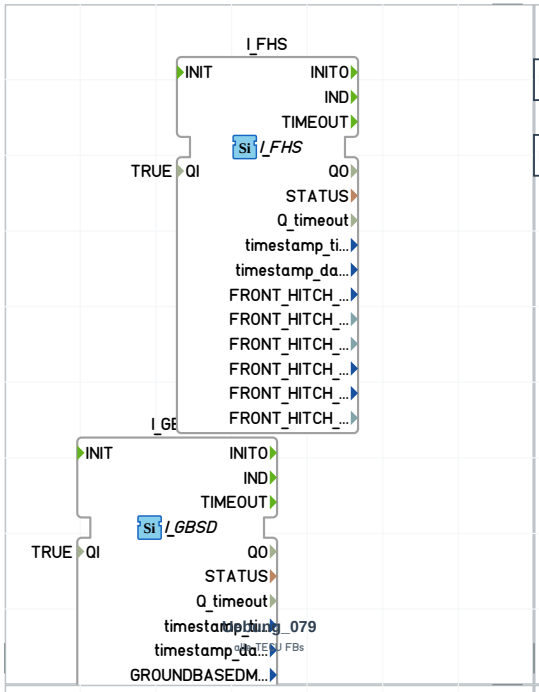
DigitalInput_I1-_I4 auf DigitalOutput_Q1-_Q4

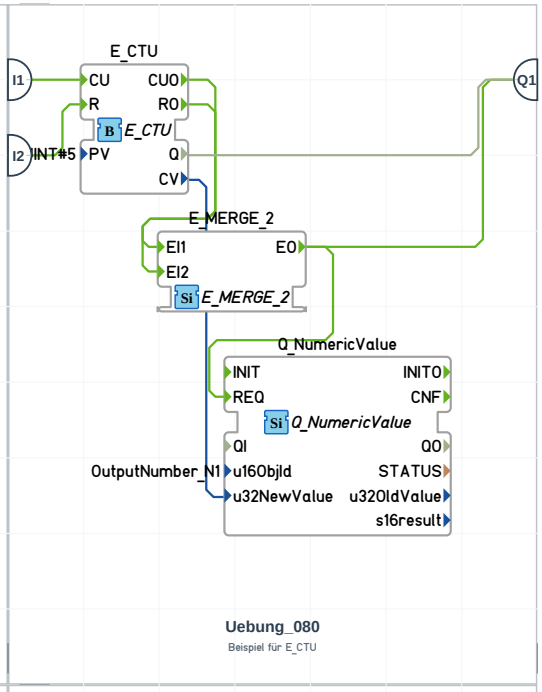

BOOL_TO_Q
BOOL_TO_QUARTER
Q_TO_BOOL
QUARTER_TO_BOOL
Q_TO_STR_STATUS
QUARTER_TO_STR_...
BOOL_TO_Q_I1
BOOL_TO_Q_I2
BOOL_TO_Q_I3
BOOL_TO_Q_I4
Uebung_055
DigitalInput_I1 auf DigitalOutput_Q1
sequence_block
sequence_E_05
sequence_E_05_I...
F_SINT_TO_UINT
Q_NumericValue
Uebung_057
Reine Event-Schrittkette 5-Kanal
Uebung_058
Reine Event-Schrittkette 5-Kanal ENDLOS
E_DEMUX_0
E_DEMUX
E_CTU_1
E_CTU
F_SUB_0
F_SUB
SETPOINT_CONDENSED_WORK_STATE_1_16
TC_ID
SECTION_CONTROL_STATE_IN
SPLIT_DWORD_INT...
Uebung_059
Reine Event-Schrittkette 8-Kanal
Uebung_060
für TC-SC also Task Controller Section Control

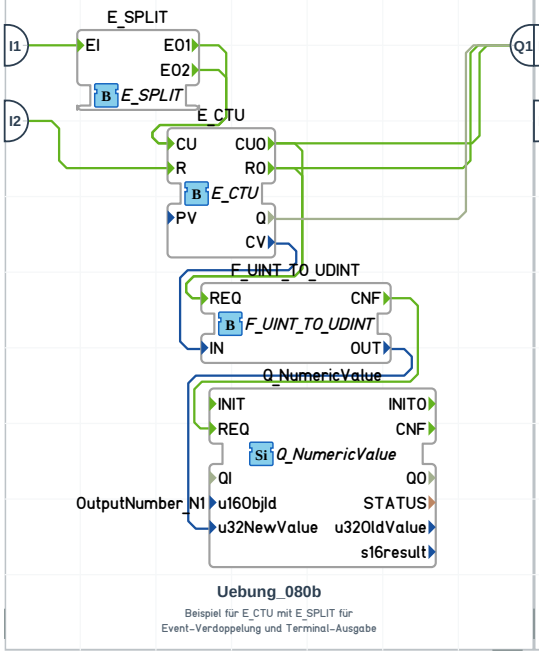

sequence_block
START_S1
RESET
CNF
EO_S1
EO_S2
EO_S3
EO_S4
sequence_T_04
T#1s DT_S1_S2
T#1s DT_S2_S3
T#1s DT_S3_S4
T#1s DT_S4_START
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
timeOut
REQ
CNF
F_SINT_TO_UINT
IN
OUT
E_TimeOut
TimeOutSocke...
Q_NumericValue
INIT
INITO
REQ
CNF
Uebung_061
QI
QO
OutputNumber_N1
u16ObjId
STATUS
u32NewValue
u32OldValue
sequence_T_05
EO_S5
T#1s DT_S5_START
DO_S5
Uebung_062
sequence_T_08
SET_DT_S1_S2
SET_DT_S2_S3
SET_DT_S3_S4
SET_DT_S4_S5
SET_DT_S5_S6
SET_DT_S6_S7
SET_DT_S7_S8
SET_DT_S8_ST...
EO_S6
EO_S7
EO_S8
T#1s DT_S6_S7
T#1s DT_S7_S8
T#1s DT_S8_START
DO_S6
DO_S7
DO_S8
I_WBSD
IND
TIMEOUT
TRUE
Q_timeout
timestamp_ti...
timestamp_da...
WHEELBASEDMA...
MAXI_TIMEOFT...
KEYSWITCHSTA...
START_STOP_S...
OPERATOR_DIR
F_UINT_TO_UDINT
Uebung_070
NumberVariable_Wheel_base...
E_T_FF
CLK
EO
Q
WBSD
FIELDBUS_UINT_TO_SIGNAL_SCALED
FIELDBUS_UINT_T...
REAL#0.001
SCALE
VALID
DINT#0
OFFSET
Uebung_070b
Uebung_070c
Q_NumericValue_...
stObj
rPhys

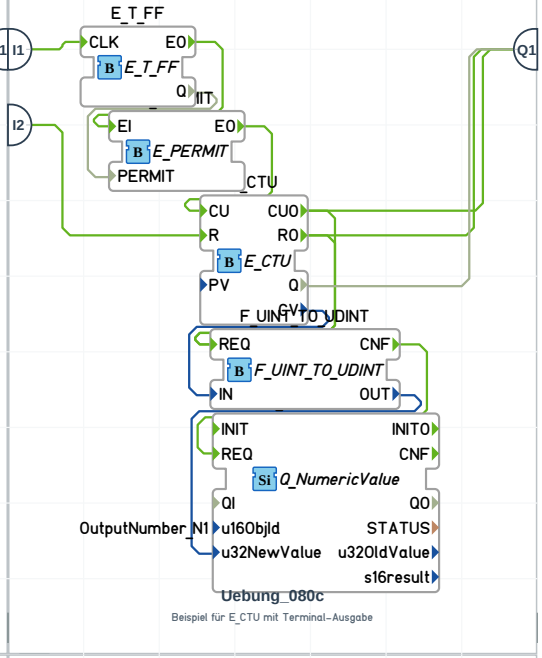

I_WBSD
F_GT
INIT
INITO
IND
TIMEOUT
I_WBSD
UINT#0
TRUE
QI
QO
STATUS
Q_timeout
timestamp_ti...
timestamp_da...
WHEELBASEDMA...
WHEELBASEDMA...
MAXI_TIMEOFT...
WHEELBASEDMA...
KEYSWITCHSTA...
START_STOP_S...
OPERATOR_DIR
REQ
CNF
F_GT
IN1
OUT
IN2
Q1
Q_NumericValue_WBSD
Q_NumericValue
NumberVariable_Wheel_base...
u160bjId
u32NewValue
u32OldValue
Uebung_071
s16result
WBSD auf UT ausgeben, Q1 Schalten wenn Speed >0
Uebung_071a
CLK
EO
I_GBSD
I_WBSD
Q_NumericValue_Wheel_based_machine_speed
NumberVariable_Wheel_base...
Uebung_071b
Q_NumericValue_Wheel_based_machine_distance
Uebung_072
Nur
Q2
Q3
Q4
NumberVariable_Ground_bas...
OFFSET_UDINT
Uebung_072c
CYCLE_TIME
FIRST
CYCLE
LAST

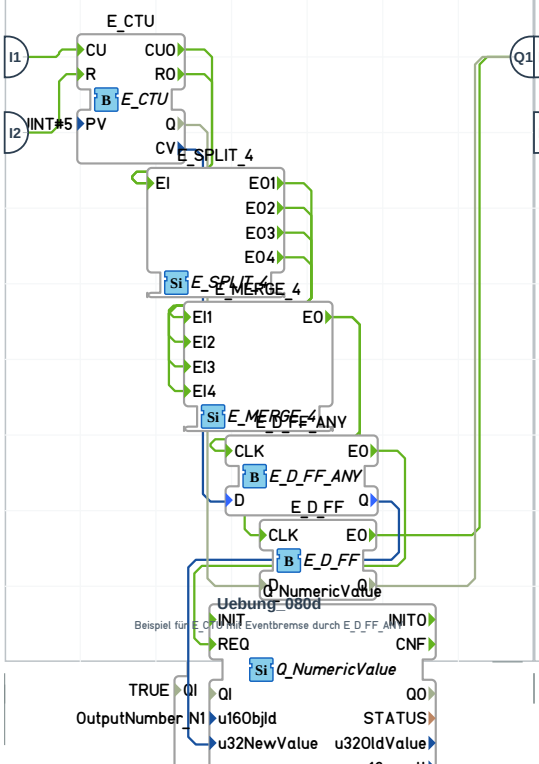

E_T_FF
CLK
EO
E_T_FF
Q
I2
Q2
GBSD
INIT
INITO
IND
TIMEOUT
I_GBSD
QI
QO
STATUS
Q_timeout
timestamp_ti...
timestamp_da...
GROUNDBASEDM...
F_UINT_TO_UDINT
REQ
CNF
IN
OUT
Q_NumericValue
Uebung_072d
u16ObjId
u32NewValue
u32OldValue
NumberVariable_Ground_bas...
I_VDS
TRUE
Uebung_073
Pitch
Altitude
I_RPT0
REAR_PTO_OUT...
REAR_PTO_ENG...
REAR_PTO_MOD...
REAR_PTO_ECO...
REAR_PTO_SH_...
REAR_PTO_EXI...
F_SEL_E_2
REQ1
REQ2
IN1
IN2
UDINT#0
Uebung_074
Q_NumericValue_GBSD
Uebung_074b
NumberVariable_Rear_PTO_o...
FB_MM710_IMU
ERROR
PARAMS
NODE1
16#D8
u8CanIdx
SA
rAccX
rAccY
rAccZ
rRateX
rRateY
rRateZ
rRoll
rPitch
rYaw
rTempRateZ
uiHW_Index
eStatusAccX
eStatusAccY
eStatusAccZ
eStatusRateX
eStatusRateY
eStatusRateZ
Uebung_075
uiSysStatus
uiSysStatus5
uiSysDiag
uiMessageCou...
bCommError
bCRCError
I_MSS
SELECTEDMACH...
COMPOUND_SCALE
FIELDBUS_UINT_T...
REAL#0.256
SCALE_HIGH
REAL#0.001
SCALE_LOW
DINT#0
OFFSET
VALID
Uebung_076
Q_NumericValue_...
NumberVariable_Wheel_base...
stObj
rPhys
s16result

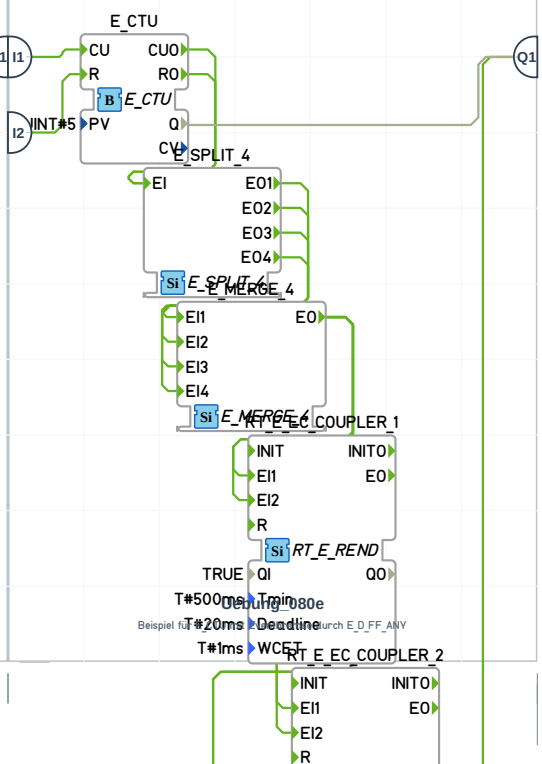

I_FHS
INIT
INITO
IND
TIMEOUT
I_FHS
TRUE
QI
QO
STATUS
Q_timeout
timestamp_ti...
timestamp_da...
FRONT_HITCH_...
FRONT_HITCH_...
FRONT_HITCH_...
FRONT_HITCH_...
FRONT_HITCH_...
FRONT_HITCH_...
I_GBSD
INIT
INITO
IND
TIMEOUT
I_GBSD
TRUE
QI
QO
STATUS
Q_timeout
timestamp_ti...
timestamp_da...
GROUNDBASEDM...
Uebung_079

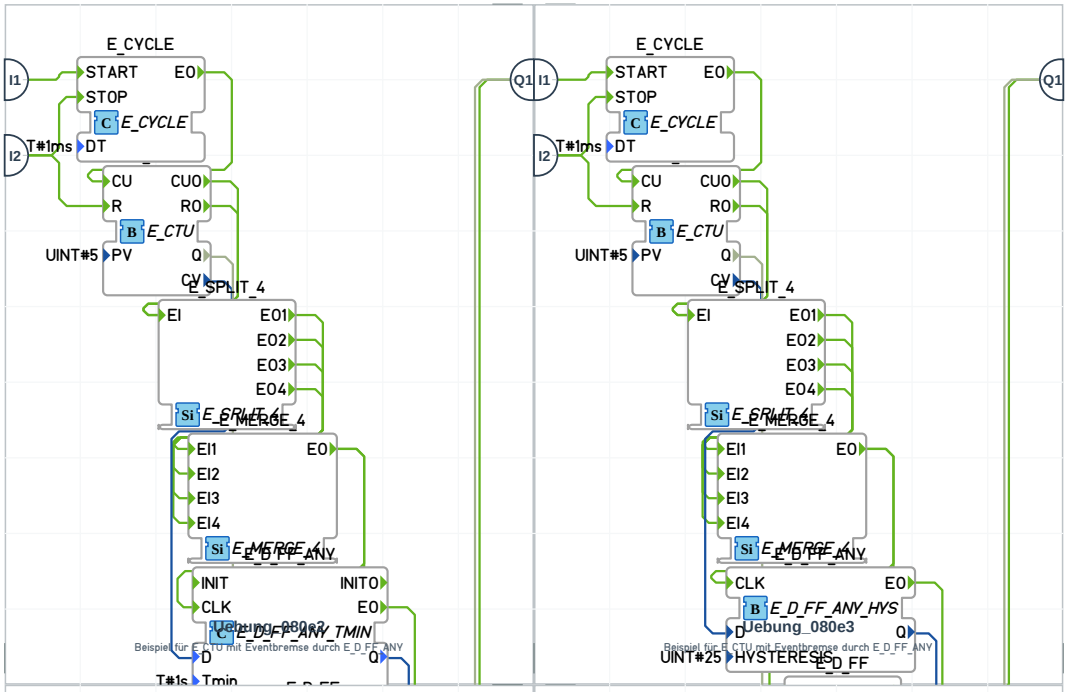

E_CTU
I1
CU
CUO
R
RO
E_CTU
I2
INT#5
PV
Q
CV
Q1
E_MERGE_2
EI1
EO
EI2
E_MERGE_2
Q_NumericValue
INIT
INITO
REQ
CNF
Q_NumericValue
QI
QO
OutputNumber_N1
u16ObjId
STATUS
u32NewValue
u32OldValue
s16result
Uebung_080

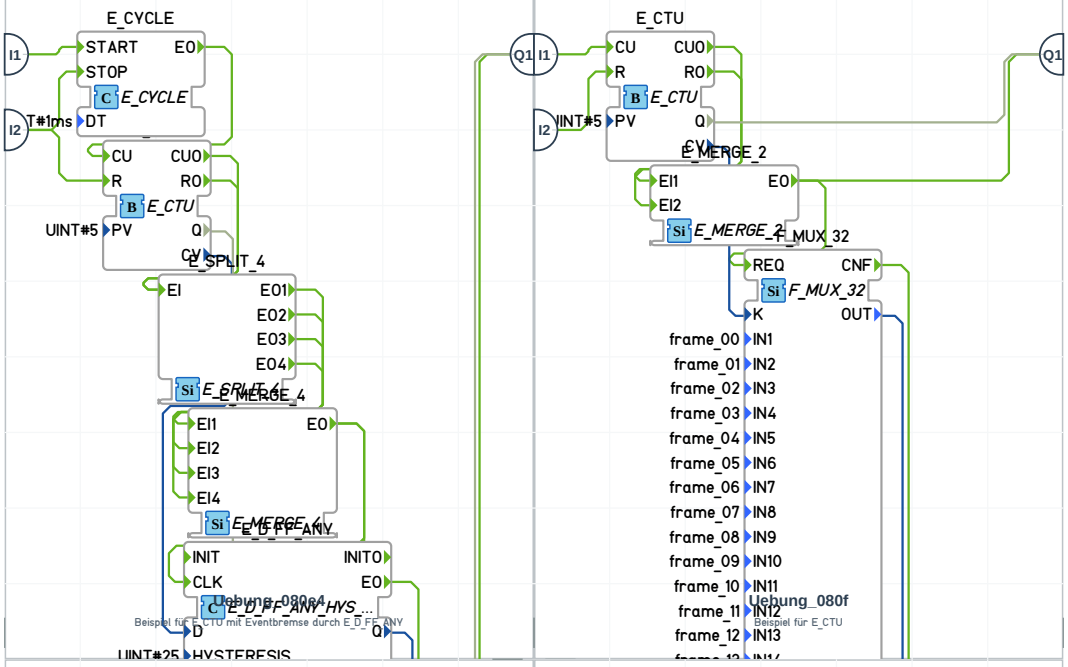

E_SPLIT
I1
EI
EO1
EO2
E_SPLIT
I2
E_CTU
CU
CUO
R
RO
E_CTU
PV
Q
CV
Q1
F_UINT_TO_UDINT
REQ
CNF
F_UINT_TO_UDINT
IN
OUT
Q_NumericValue
INIT
INITO
REQ
CNF
Q_NumericValue
QI
QO
OutputNumber_N1
u16ObjId
STATUS
u32NewValue
u32OldValue
s16result
Uebung_080b

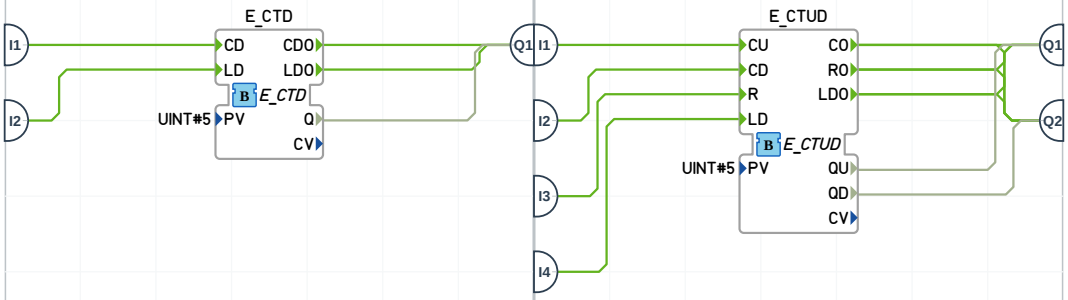

E_T_FF
I1
CLK
EO
E_T_FF
Q
IIT
Q1
I2
EI
EO
E_PERMIT
PERMIT
CTU
CU
CUO
R
RO
E_CTU
PV
Q
CV
F_UINT_TO_UDINT
REQ
CNF
F_UINT_TO_UDINT
IN
OUT
INIT
INITO
REQ
CNF
Q_NumericValue
QI
QO
OutputNumber_N1
u16ObjId
STATUS
u32NewValue
u32OldValue
s16result
Uebung_080c


E_CTU
I1
CU
CUO
R
RO
E_CTU
I2
INT#5
PV
Q
CV
E_SPLIT_4
EI
EO1
EO2
EO3
EO4
E_SPLIT_4
E_MERGE_4
EI1
EO
EI2
EI3
EI4
E_MERGE_4
E_D_FF_ANY
CLK
EO
E_D_FF_ANY
D
Q
E_D_FF
CLK
EO
E_D_FF
D
Q
Q1
Q_NumericValue
INIT
INITO
REQ
CNF
Q_NumericValue
TRUE
QI
QO
OutputNumber_N1
u16ObjId
STATUS
u32NewValue
u32OldValue
Uebung_080d


E_CTU
I1
CU
CUO
R
RO
E_CTU
I2
INT#5
PV
Q
CV
Q1
E_SPLIT_4
EI
EO1
EO2
EO3
EO4
E_SPLIT_4
E_MERGE_4
EI1
EO
EI2
EI3
EI4
E_MERGE_4
RT_E_EC_COUPLER_1
INIT
INITO
EI1
EO
EI2
R
RT_E_REND
TRUE
QI
QO
T#500ms
Tmin
T#20ms
Deadline
T#1ms
WCET
RT_E_EC_COUPLER_2
INIT
INITO
EI1
EO
EI2
R
Uebung_080e





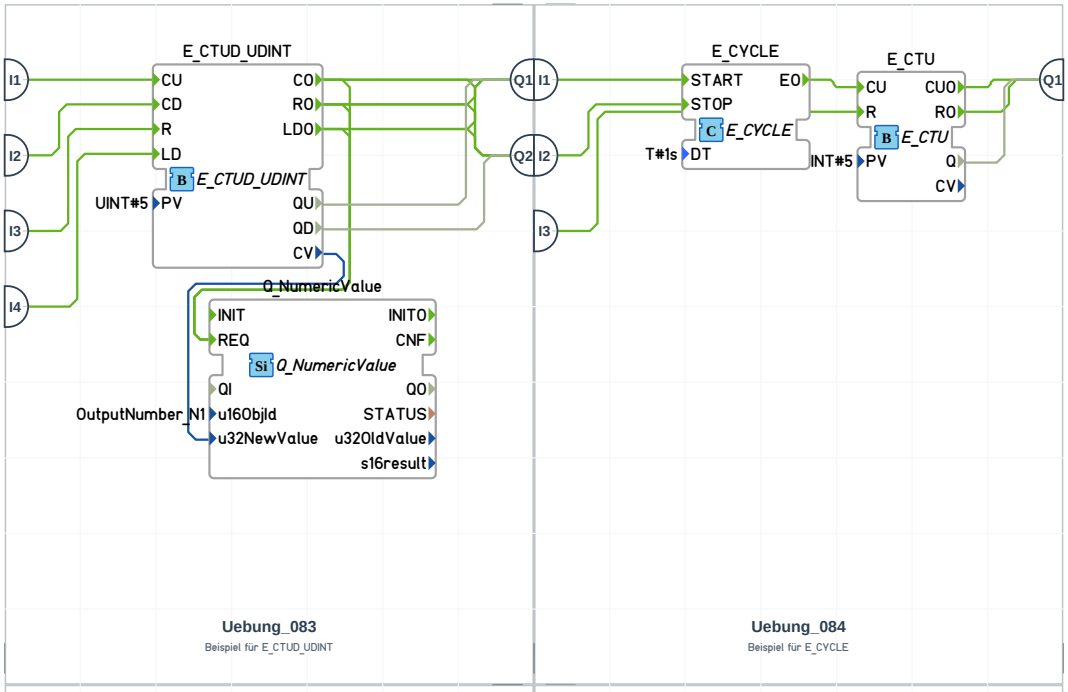


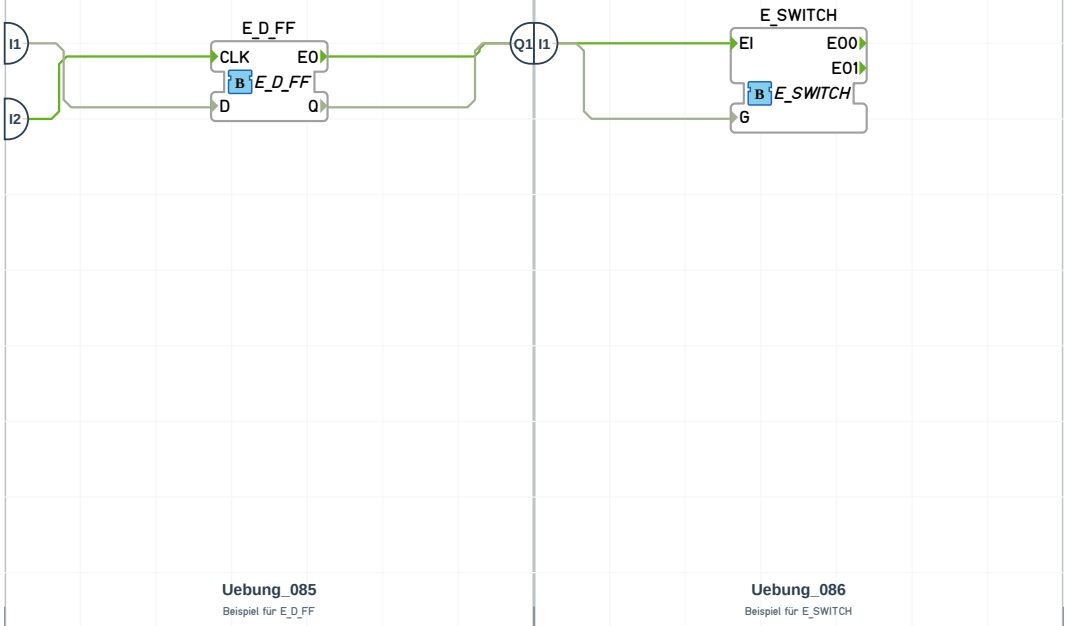
Uebung_081

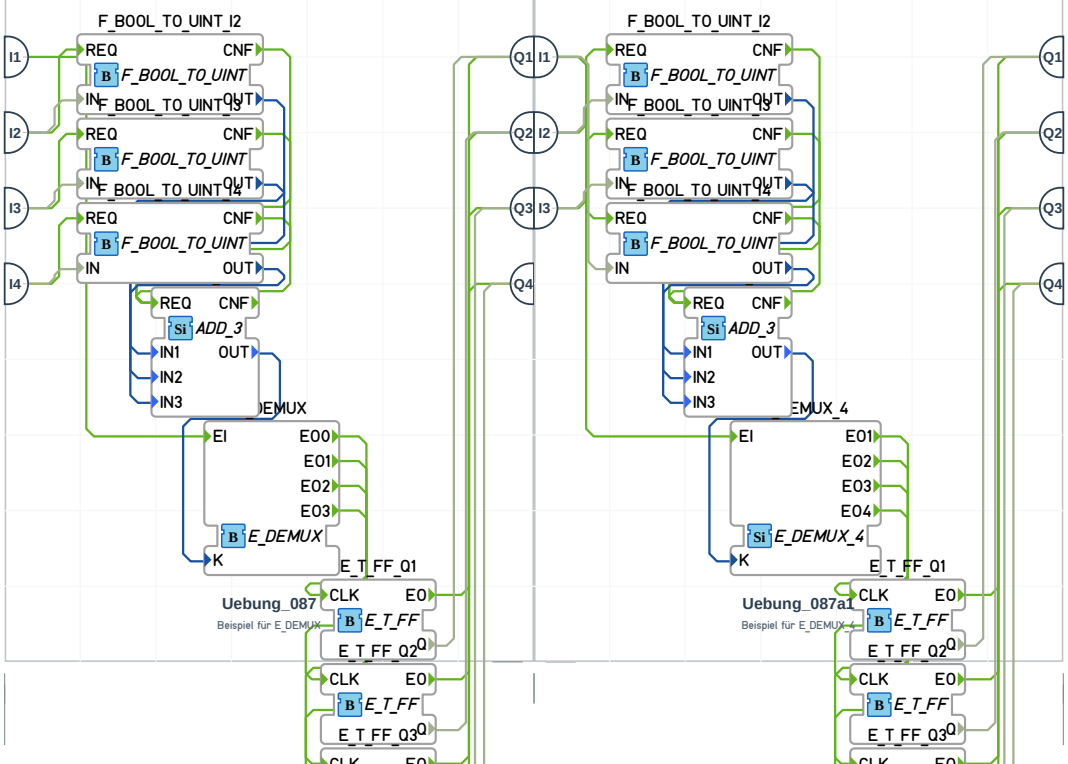
Beispiel für E_CTD

Uebung_082

Beispiel für E_CTUD

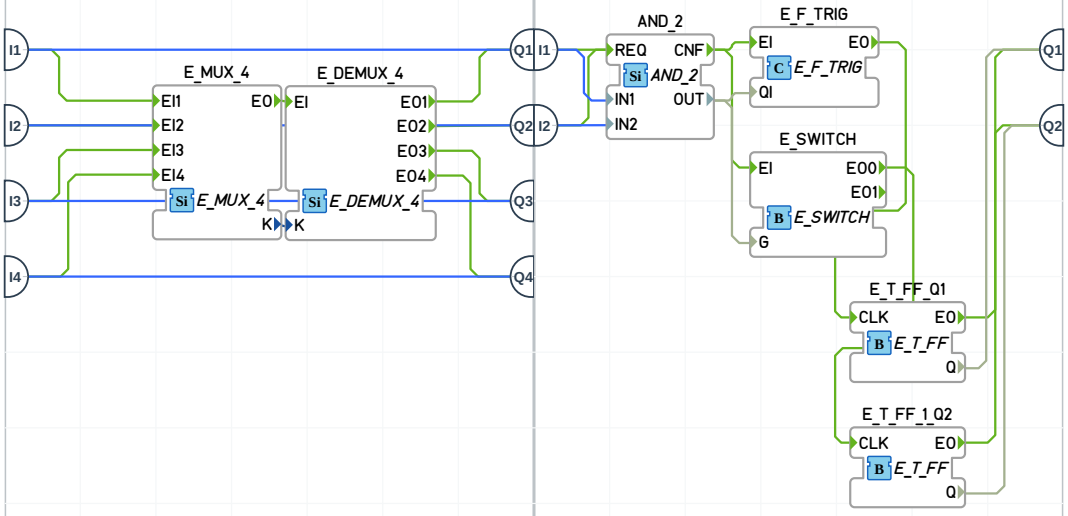

E_CTUD_UDINT
I1
I2
I3
I4
CU
CD
R
LD
PV
UINT#5
CO
RO
LDO
QU
QD
CV
Q_NumericValue
INIT
REQ
QI
OutputNumber_N1
u16ObjId
u32NewValue
INITO
CNF
QO
STATUS
u32OldValue
s16result
Q1
Q2
E_CYCLE
START
STOP
T#1s
DT
EO
E_CTU
INT#5
CUO
Q
CV
Uebung_083
Beispiel für E_CTUD_UDINT
Uebung_084
Beispiel für E_CYCLE


E_D_FF
CLK
D
EO
Q
E_SWITCH
EI
G
EO0
EO1
Uebung_085
Beispiel für E_D_FF
Uebung_086
Beispiel für E_SWITCH


F_BOOL_TO_UINT_I2
F_BOOL_TO_UINT
ADD_3
E_DEMUX
E_DEMUX_4
E_T_FF
E_T_FF_Q1
E_T_FF_Q2
E_T_FF_Q3
Uebung_087
Beispiel für E_DEMUX
Uebung_087a1
Beispiel für E_DEMUX_4

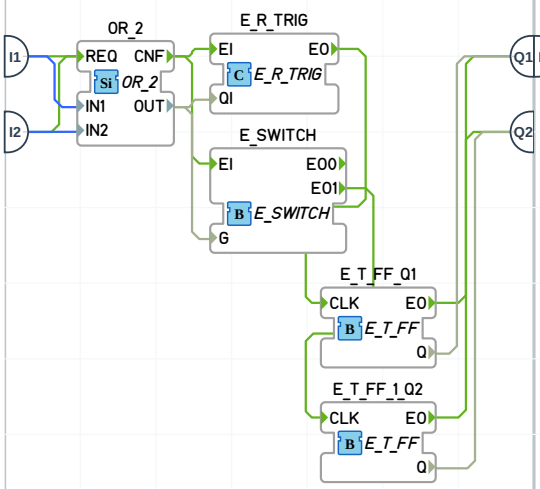
Uebung_087a2

Beispiel für E_DEMUX_4



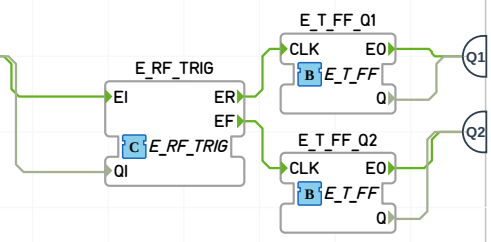
Uebung_088

Beispiel für E_F_TRIG



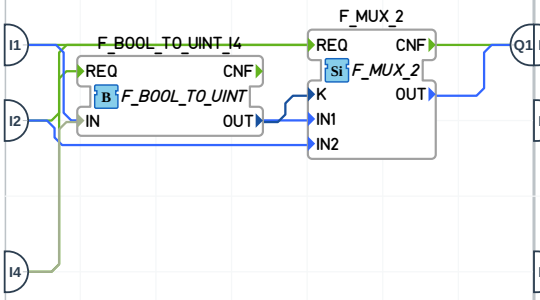
Uebung_089

Beispiel für E_R_TRIG



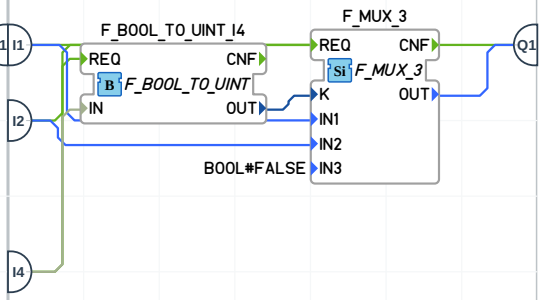
Uebung_089a

Beispiel für E_RF_TRIG



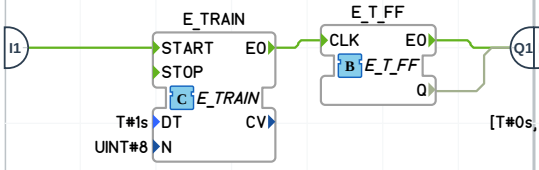
Uebung_090a1

Beispiel für F_MUX_2



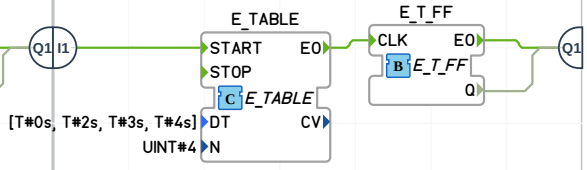
Uebung_090a2

Beispiel für F_MUX_3



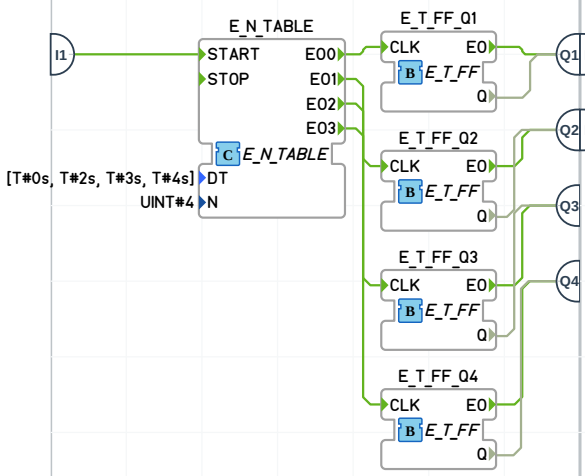
Uebung_091

Beispiel für E_TRAIN



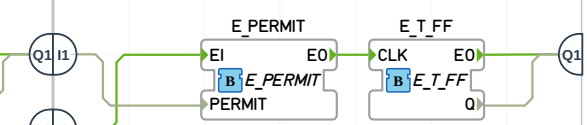
Uebung_093

Beispiel für E_TABLE



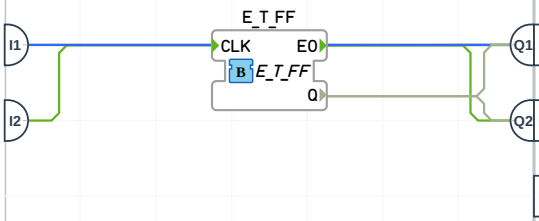
Uebung_093b

Beispiel für E_N_TABLE



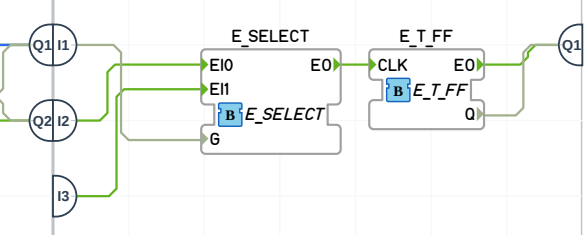
Uebung_094

Beispiel für E_PERMIT



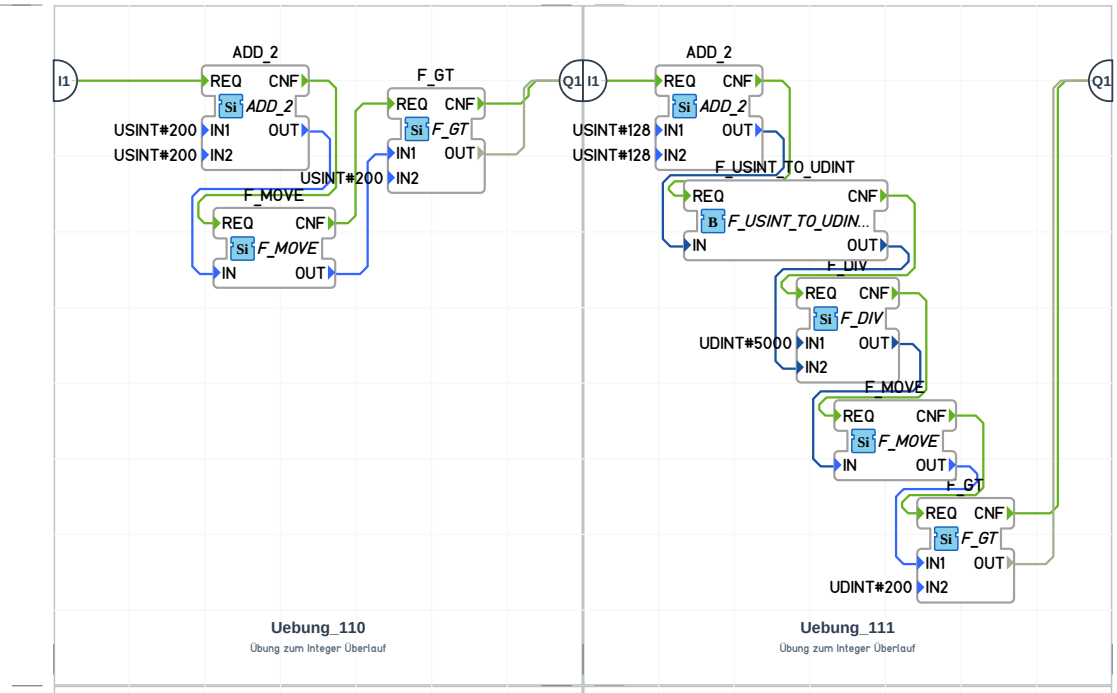
Uebung_094a

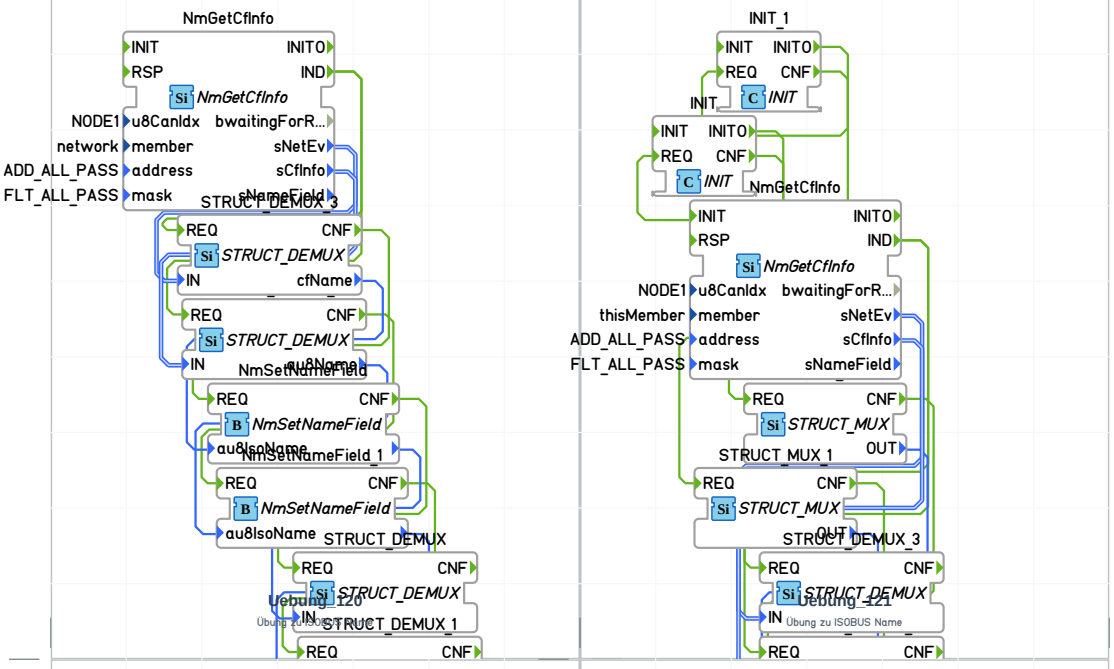
Beispiel für QI anstelle Permit

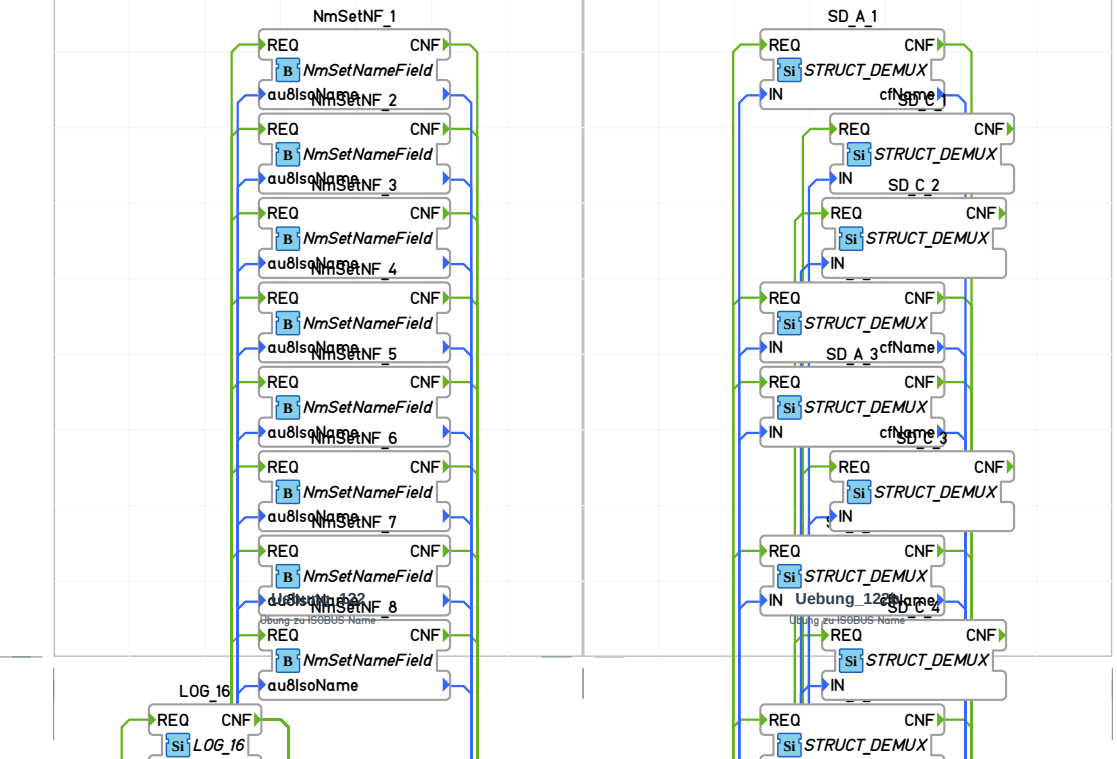


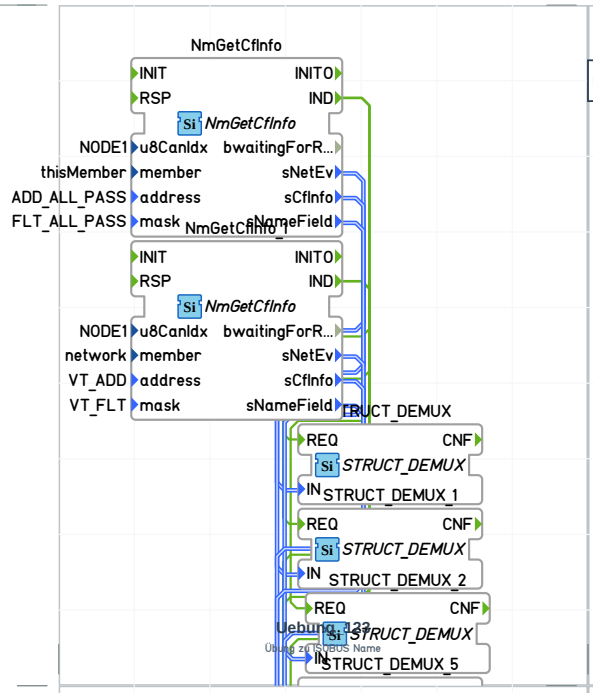
Uebung_095

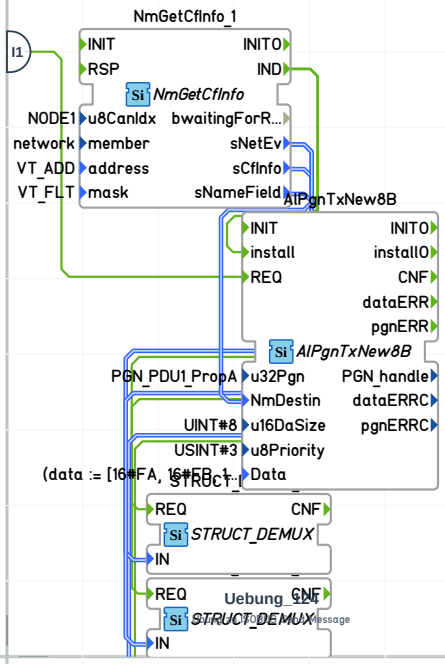
Beispiel für E_SELECT

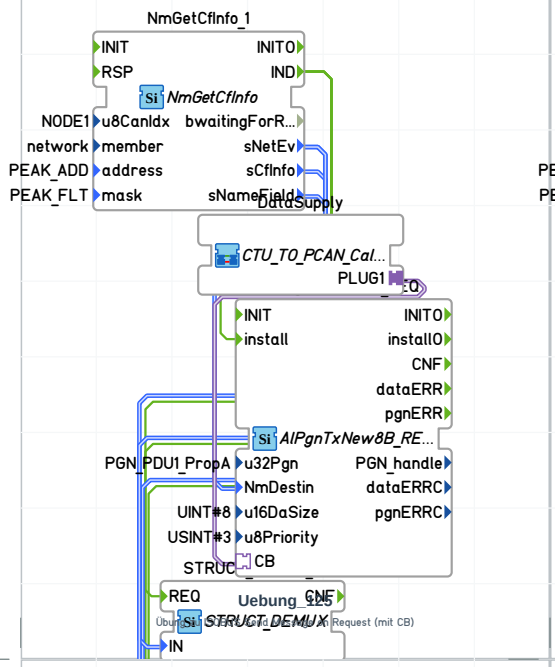

ADD_2
F_GT
F_MOVE
USINT#200
USINT#200
USINT#200
F_USINT_TO_UDINT
F_DIV
UDINT#5000
USINT#128
USINT#128
UDINT#200
Uebung_110
Übung zum Integer Überlauf
Uebung_111
Übung zum Integer Überlauf

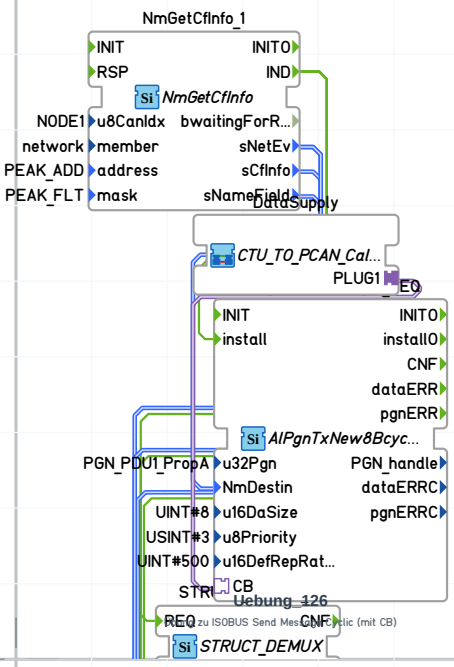

NmGetCfInfo
INIT_1
STRUCT_DEMUX_3
NmSetNameField_1
STRUCT_DEMUX
STRUCT_DEMUX_1
STRUCT_MUX_1
STRUCT_DEMUX_3
Uebung_120
Übung zu ISOBUS Name
Uebung_121
Übung zu ISOBUS Name

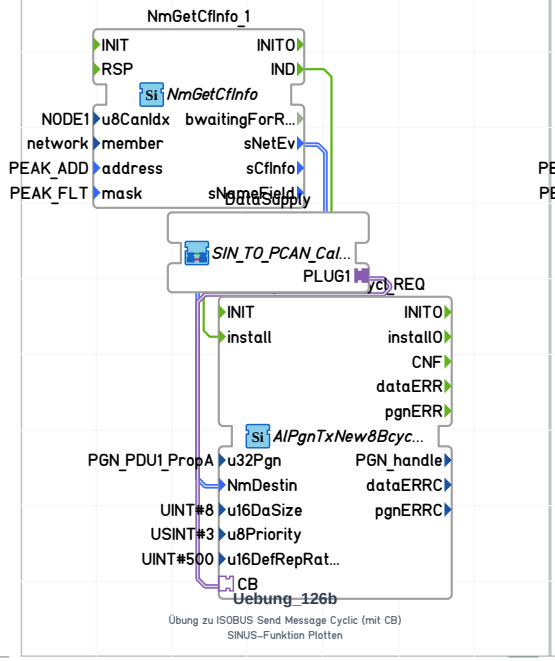

NmSetNF_1
NmSetNF_2
NmSetNF_3
NmSetNF_4
NmSetNF_5
NmSetNF_6
NmSetNF_7
NmSetNF_8
LOG_16
SD_A_1
SD_C_1
SD_C_2
SD_A_3
SD_C_3
SD_C_4
Uebung_122
Übung zu ISOBUS Name
Uebung_123
Übung zu ISOBUS Name

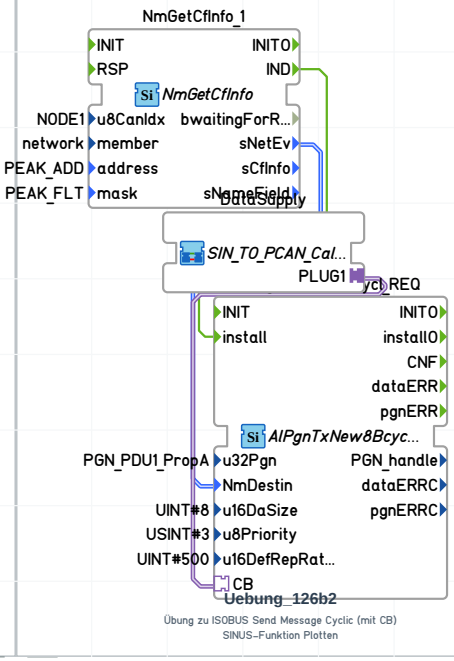

NmGetCfInfo
INIT
RSP
NmGetCfInfo
NODE1
u8CanIdx
thisMember
member
ADD_ALL_PASS
address
FLT_ALL_PASS
mask
INITO
IND
bwaitingForR...
sNetEv
sCfInfo
sNameField
NmGetCfInfo_1
network
VT_ADD
VT_FLT
STRUCT_DEMUX
REQ
CNF
STRUCT_DEMUX
IN
STRUCT_DEMUX_1
STRUCT_DEMUX_2
STRUCT_DEMUX_5
Uebung_123
Übung zu ISOBUS Name

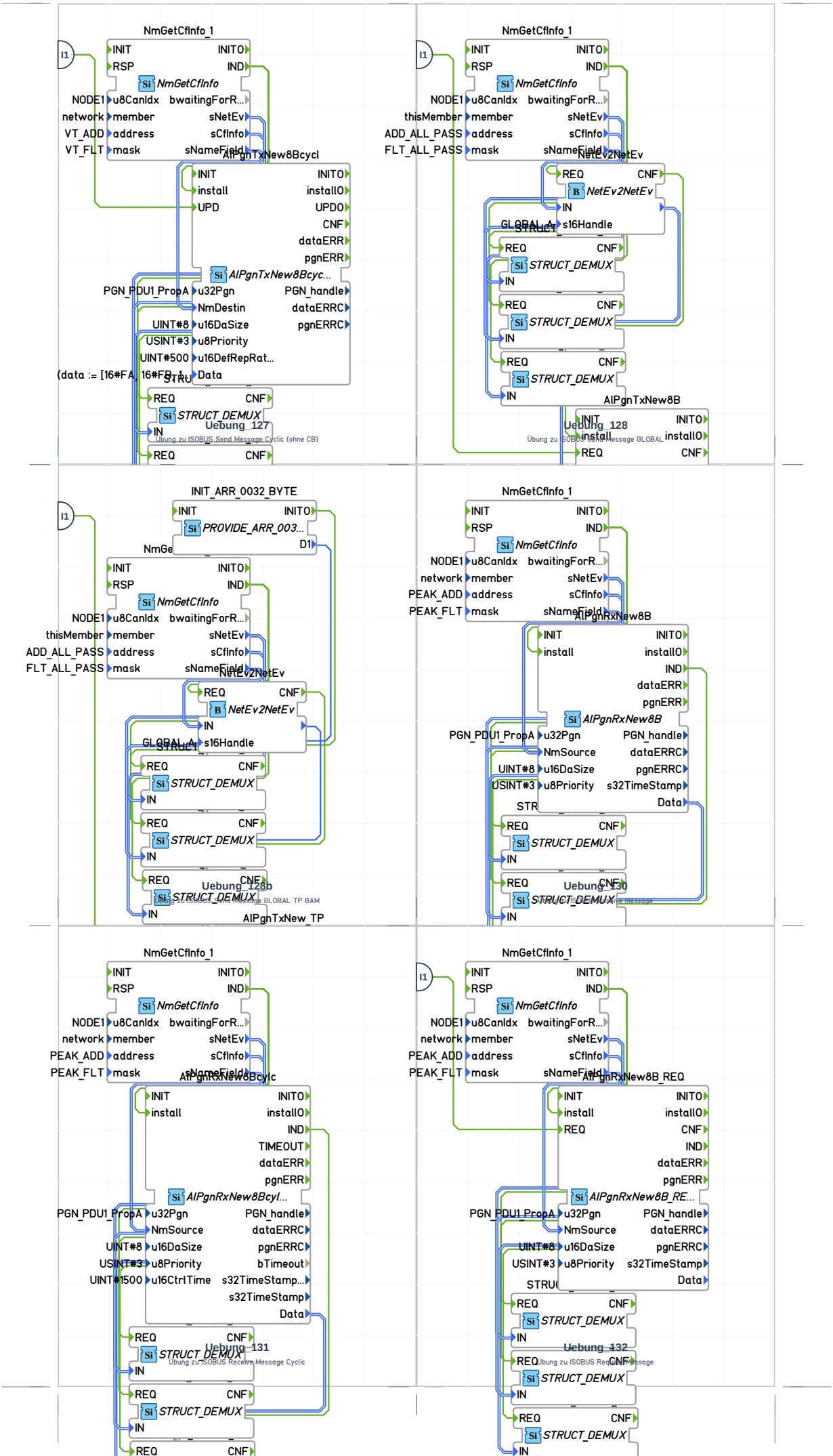

NmGetCfInfo_1
I1
INIT
RSP
NmGetCfInfo
NODE1
u8CanIdx
network
member
VT_ADD
address
VT_FLT
mask
INITO
IND
bwaitingForR...
sNetEv
sCfInfo
sNameField
AlPgnTxNew8B
install
installO
REQ
CNF
dataERR
pgnERR
PGN_PDU1_PropA
u32Pgn
PGN_handle
NmDestin
dataERRC
UINT#8
u16DaSize
pgnERRC
USINT#3
u8Priority
Data
(data := [16#FA, 16#FB,
STRUCT_DEMUX
IN
Uebung_124
Übung zu ISOBUS Send Message

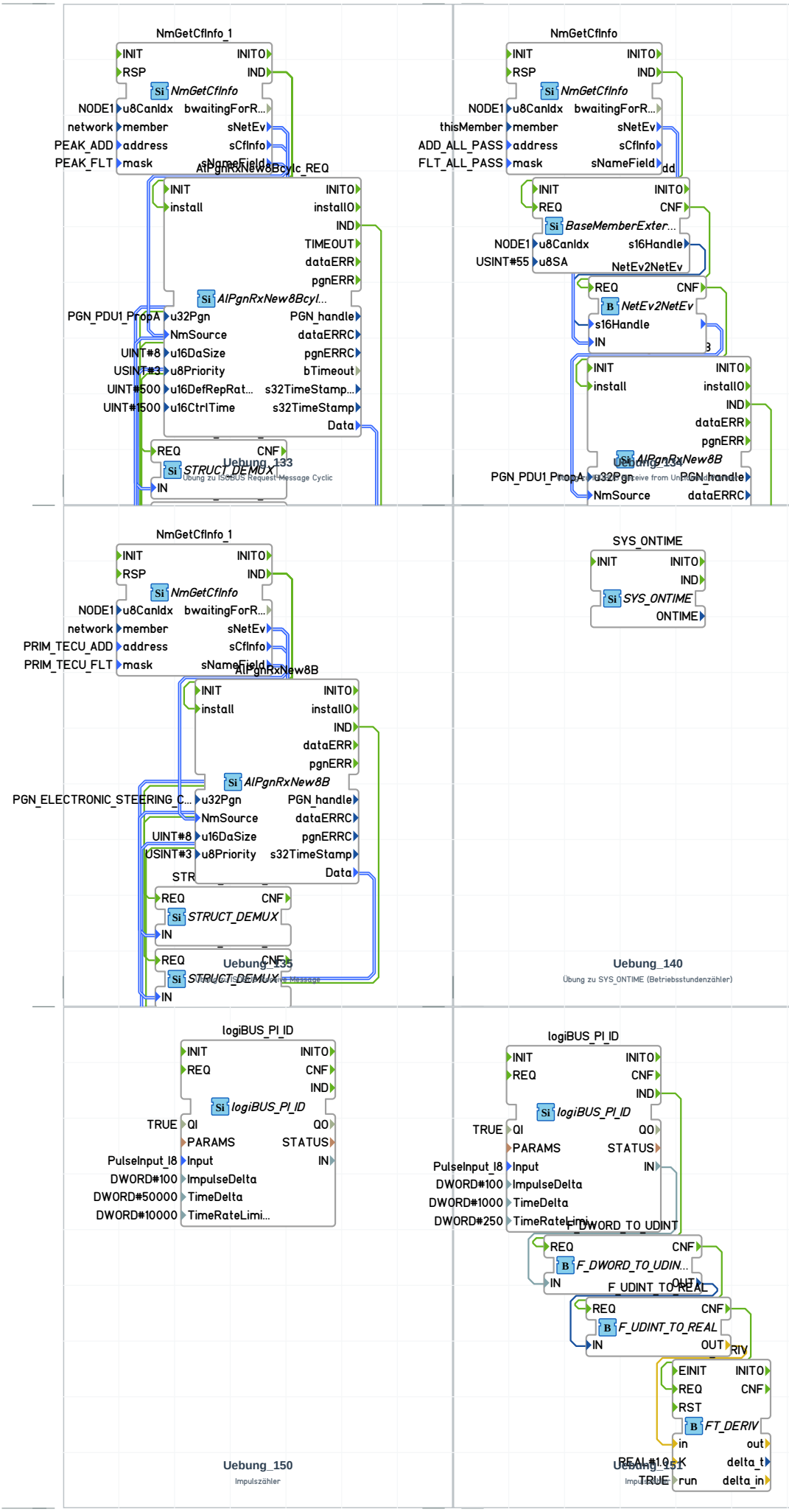

NmGetCfInfo_1
INIT
RSP
NmGetCfInfo
NODE1
u8CanIdx
network
member
PEAK_ADD
address
PEAK_FLT
mask
INITO
IND
bwaitingForR...
sNetEv
sCfInfo
sNameField
DataSupply
CTU_TO_PCAN_Cal...
PLUG1
REQ
install
installO
CNF
dataERR
pgnERR
AlPgnTxNew8B_RE...
PGN_PDU1_PropA
u32Pgn
PGN_handle
NmDestin
dataERRC
UINT#8
u16DaSize
pgnERRC
USINT#3
u8Priority
CB
REQ
CNF
STRUCT_DEMUX
IN
Uebung_125
Übung zu ISOBUS Send Message Request (mit CB)

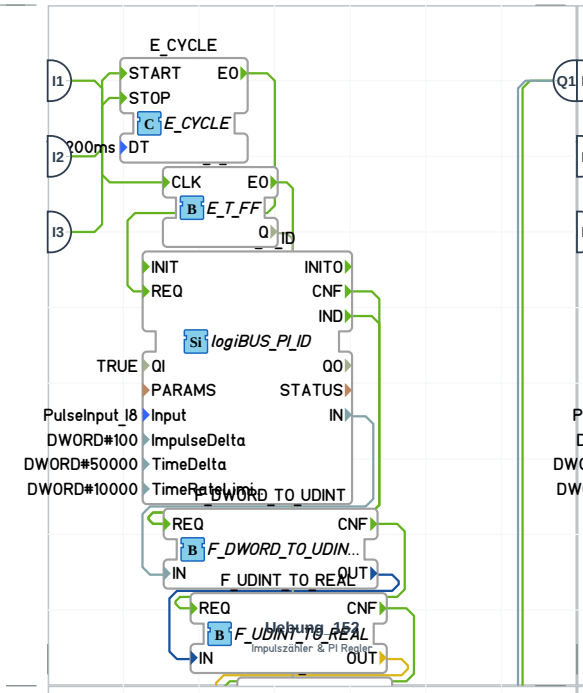

NmGetCfInfo_1
INIT
RSP
NmGetCfInfo
NODE1
u8CanIdx
network
member
PEAK_ADD
address
PEAK_FLT
mask
INITO
IND
bwaitingForR...
sNetEv
sCfInfo
sNameField
DataSupply
CTU_TO_PCAN_Cal...
PLUG1
REQ
install
installO
CNF
dataERR
pgnERR
AlPgnTxNew8Bcyc...
PGN_PDU1_PropA
u32Pgn
PGN_handle
NmDestin
dataERRC
UINT#8
u16DaSize
pgnERRC
USINT#3
u8Priority
UINT#500
u16DefRepRat...
CB
REQ
CNF
STRUCT_DEMUX
IN
Uebung_126
Übung zu ISOBUS Send Message Cyclic (mit CB)

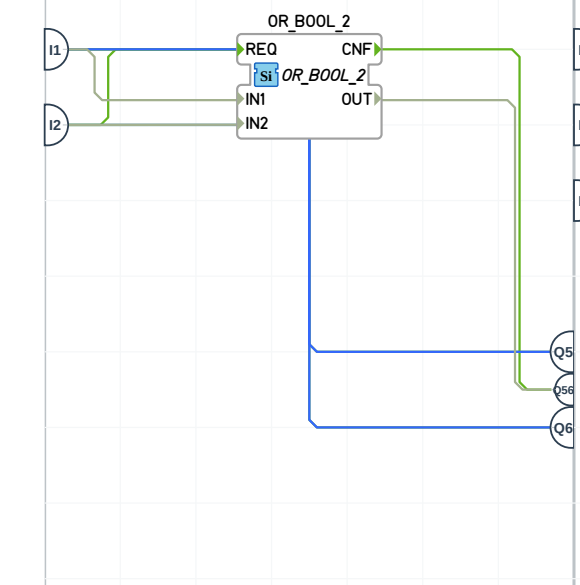

NmGetCfInfo_1
INIT
RSP
NmGetCfInfo
NODE1
u8CanIdx
network
member
PEAK_ADD
address
PEAK_FLT
mask
INITO
IND
bwaitingForR...
sNetEv
sCfInfo
sNameField
DataSupply
SIN_TO_PCAN_Cal...
PLUG1
REQ
install
installO
CNF
dataERR
pgnERR
AlPgnTxNew8Bcyc...
PGN_PDU1_PropA
u32Pgn
PGN_handle
NmDestin
dataERRC
UINT#8
u16DaSize
pgnERRC
USINT#3
u8Priority
UINT#500
u16DefRepRat...
CB
Uebung_126b
Übung zu ISOBUS Send Message Cyclic (mit CB)
SINUS-Funktion Plotten


NmGetCfInfo_1
INIT
RSP
NmGetCfInfo
NODE1
u8CanIdx
network
member
PEAK_ADD
address
PEAK_FLT
mask
INITO
IND
bwaitingForR...
sNetEv
sCfInfo
sNameField
DataSupply
SIN_TO_PCAN_Cal...
PLUG1
REQ
install
installO
CNF
dataERR
pgnERR
AlPgnTxNew8Bcyc...
PGN_PDU1_PropA
u32Pgn
PGN_handle
NmDestin
dataERRC
UINT#8
u16DaSize
pgnERRC
USINT#3
u8Priority
UINT#500
u16DefRepRat...
CB
Uebung_126b2
Übung zu ISOBUS Send Message Cyclic (mit CB)
SINUS-Funktion Plotten


NmGetCfInfo_1
INIT
RSP
NmGetCfInfo
NODE1
network
VT_ADD
VT_FLT
u8CanIdx
member
address
mask
INITO
IND
bwaitingForR...
sNetEv
sCfInfo
sNameField
AIPgnTxNew8Bcycl
install
UPD
installO
UPDO
CNF
dataERR
pgnERR
AIPgnTxNew8Bcyc...
PGN_PDU1_PropA
u32Pgn
NmDestin
UINT#8
u16DaSize
USINT#3
u8Priority
UINT#500
u16DefRepRat...
Data
PGN_handle
dataERRC
pgnERRC
REQ
STRUCT_DEMUX
IN
Uebung_127
Übung zu ISOBUS Send Message Cyclic (ohne CB)
NetEv2NetEv
s16Handle
GLOBAL_A
AIPgnTxNew8B
Uebung_128
INIT_ARR_0032_BYTE
PROVIDE_ARR_003...
D1
thisMember
ADD_ALL_PASS
FLT_ALL_PASS
Uebung_128b
Übung zu ISOBUS Send Message GLOBAL TP BAM
AIPgnTxNew_TP
PEAK_ADD
PEAK_FLT
AIPgnRxNew8B
AIPgnRxNew8B
PGN_PDU1_PropA
NmSource
UINT#8
USINT#3
s32TimeStamp
Uebung_130
AIPgnRxNew8Bcyc
AIPgnRxNew8Bcyl...
UINT#1500
u16CtrlTime
bTimeout
s32TimeStamp...
TIMEOUT
Uebung_131
Übung zu ISOBUS Receive Message Cyclic
AIPgnRxNew8B_REQ
AIPgnRxNew8B_RE...
Uebung_132

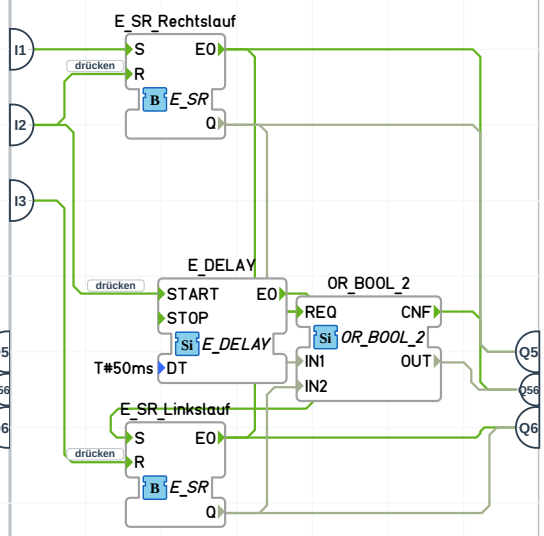

NmGetCfInfo_1
INIT
RSP
NmGetCfInfo
NODE1
network
PEAK_ADD
PEAK_FLT
u8CanIdx
member
address
mask
INITO
IND
bwaitingForR...
sNetEv
sCfInfo
sNameField
AIPgnRxNew8Bcyl...
INIT
install
INITO
installO
IND
TIMEOUT
dataERR
pgnERR
AIPgnRxNew8Bcyl...
PGN_PDU1_PropA
u32Pgn
NmSource
UINT#8
u16DaSize
USINT#3
u8Priority
UINT#500
u16DefRepRat...
UINT#1500
u16CtrlTime
PGN_handle
dataERRC
pgnERRC
bTimeout
s32TimeStamp...
s32TimeStamp
Data
REQ
CNF
STRUCT_DEMUX
IN
Uebung_133
Übung zu ISOBUS Request-Message Cyclic
NmGetCfInfo
thisMember
ADD_ALL_PASS
FLT_ALL_PASS
BaseMemberExter...
s16Handle
USINT#55
u8SA
NetEv2NetEv
B
NetEv2NetEv
AIPgnRxNew8B
Uebung_134
NmGetCfInfo_1
PRIM_TECU_ADD
PRIM_TECU_FLT
AIPgnRxNew8B
PGN_ELECTRONIC_STEERING_C...
STR
Uebung_135
SYS_ONTIME
SYS_ONTIME
ONTIME
Uebung_140
Übung zu SYS_ONTIME (Betriebsstundenzähler)
logiBUS_PI_ID
TRUE
QI
QO
PARAMS
STATUS
PulseInput_I8
Input
DWORD#100
ImpulseDelta
DWORD#50000
TimeDelta
DWORD#10000
TimeRateLimi...
Uebung_150
Impulszähler
DWORD#1000
DWORD#250
F_DWORD_TO_UDINT
F_DWORD_TO_UDIN...
F_UDINT_TO_REAL
F_UDINT_TO_REAL
EINIT
RST
FT_DERIV
in
out
K
delta_t
run
delta_in
REAL#10
Uebung_151





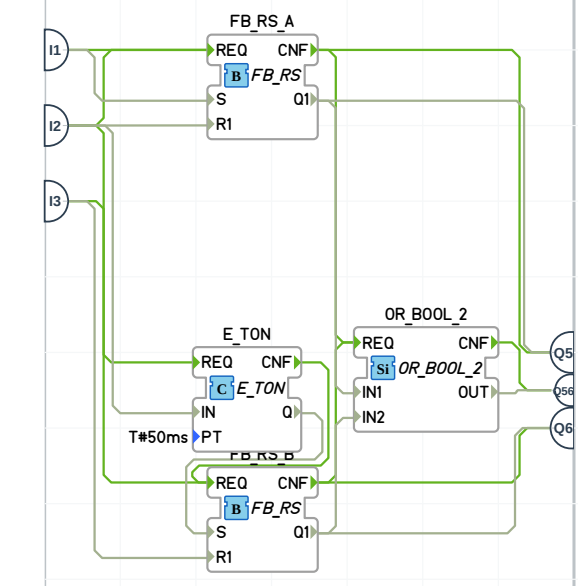
Uebung_160

Motor Links/Rechtslauf



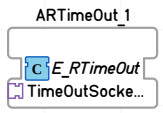
Uebung_160b

Motor Links/Rechtslauf Automatisierung via IE



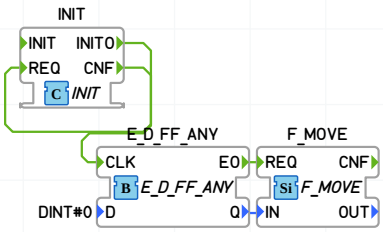
Uebung_160b2

Motor Links/Rechtslauf Automatisierung IX



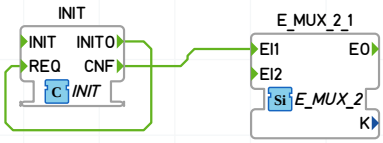
Uebung_170

Exercise for ARTimeOut



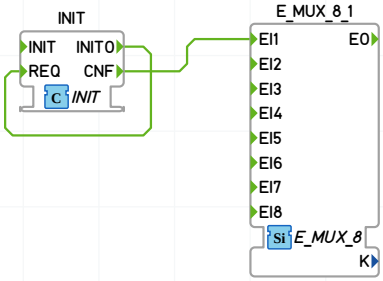
Uebung_171

Exercise for E_D_FF_ANY



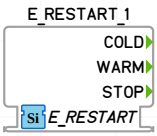
Uebung_172

Exercise for E_MUX_2



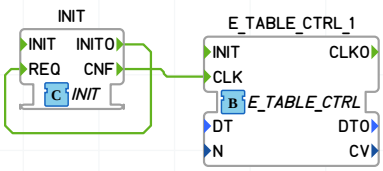
Uebung_173

Exercise for E_MUX_8



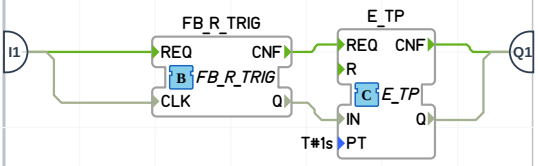
Uebung_174

Exercise for E_RESTART



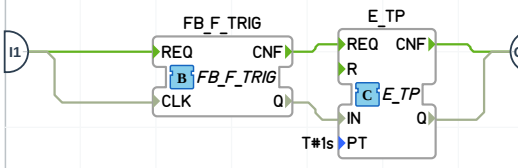
Uebung_175

Exercise for E_TABLE_CTRL



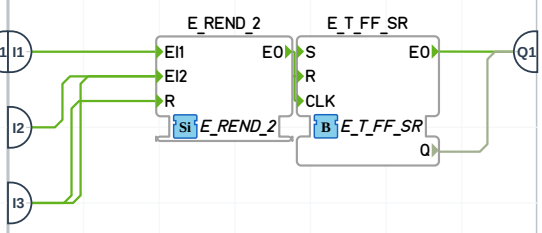
Uebung_177

Beispiel für FB_R_TRIG (Rising Edge Detection)



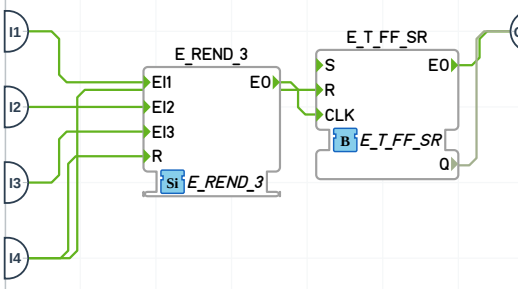
Uebung_178

Beispiel für FB_F_TRIG (Falling Edge Detection)



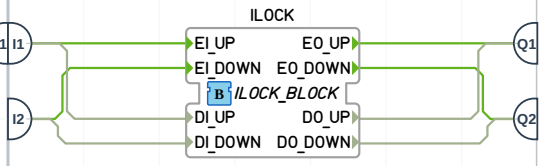
Uebung_179

Beispiel für E_REND_2 (Rendezvous von zwei Events)



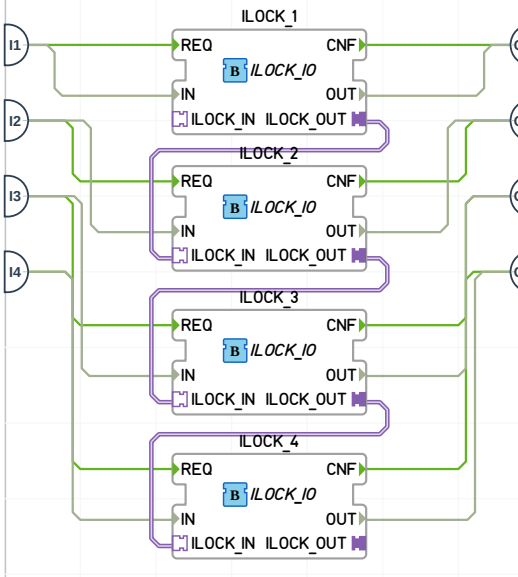
Uebung_180

Beispiel für E_REND_3 (Rendezvous von drei Events)



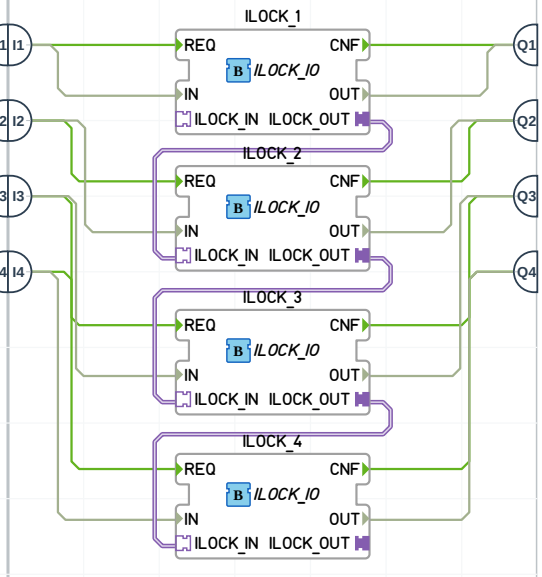
Uebung_201

Interlock: ILOCK_BLOCK (Gegenseitige Verriegelung)



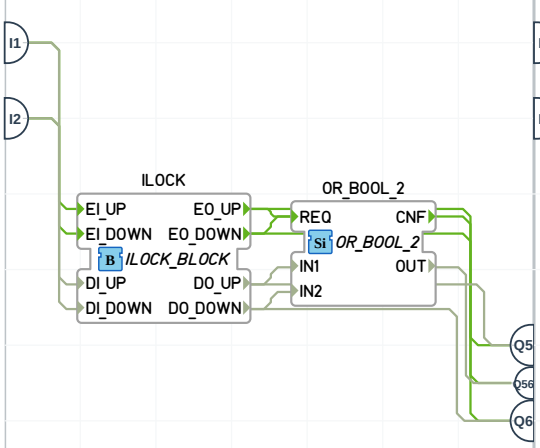
Uebung_201_Interlock

Interlock advanced exercise (AX)



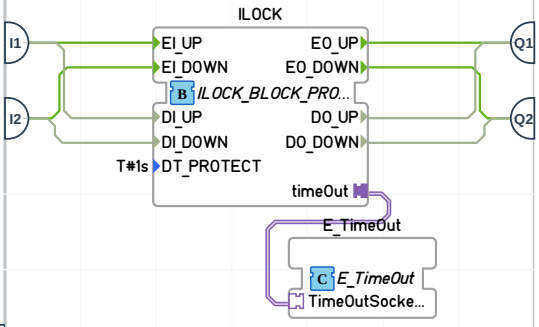
Uebung_201_Interlock_BOOL

Interlock basic exercise (BOOL)



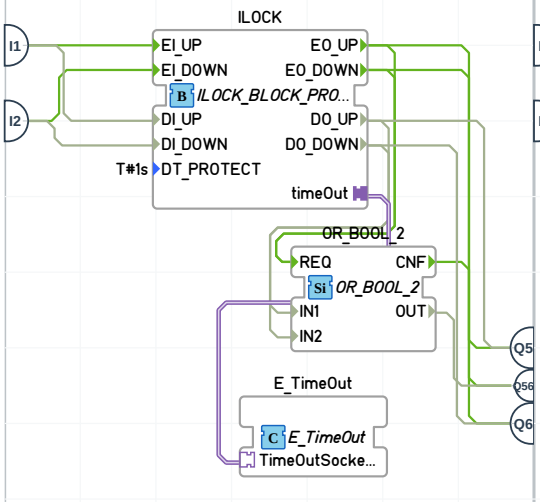
Uebung_201b

Interlock: ILOCK_BLOCK (Motor Rechts/Linkslauf mit Low-Side Treiber)



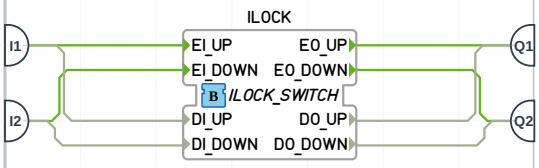
Uebung_202

Interlock: ILOCK_BLOCK_PROTECT (Verriegelung mit Schutzzeit)



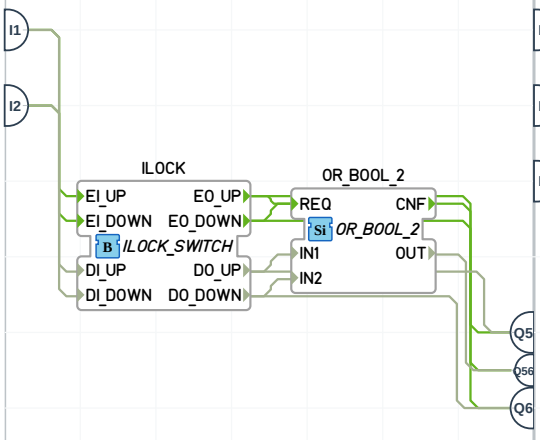
Uebung_202b

Interlock: ILOCK_BLOCK_PROTECT (Motor Reversierung mit Schutzzeit)



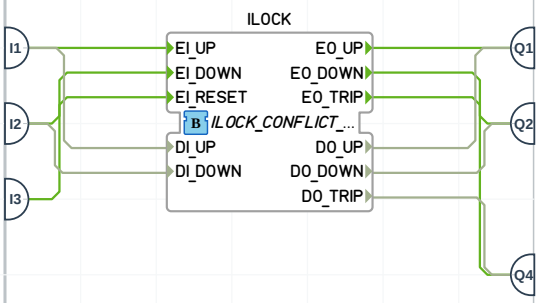
Uebung_203

Interlock: ILOCK_SWITCH (Umschalt-Priorität Last-Wins)



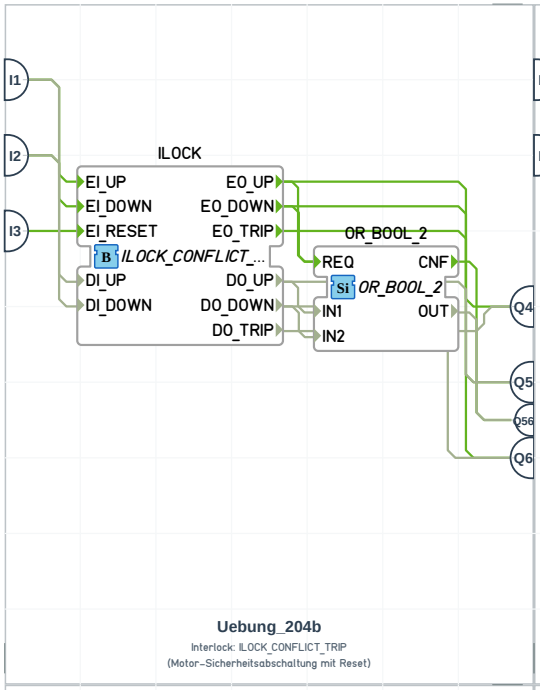
Uebung_203b

Interlock: ILOCK_SWITCH (Motor Reversierung Priorität Last-Wins)



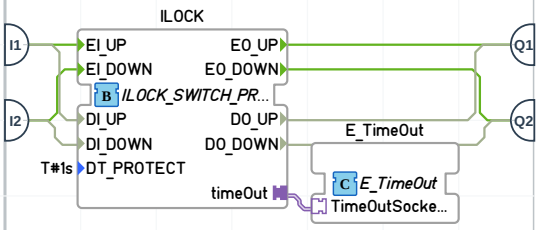
Uebung_204

Interlock: ILOCK_CONFLICT_TRIP (Trip bei Konflikt mit Reset)



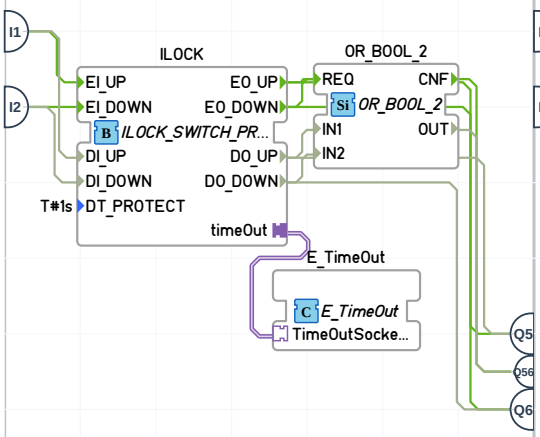
Uebung_204b

Interlock: ILOCK_CONFLICT_TRIP
(Motor-Sicherheitsabschaltung mit Reset)



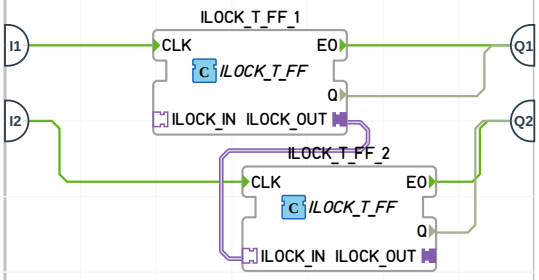
Uebung_205

Interlock: ILOCK_SWITCH_PROTECT (Umschalt-Priorität
mit Schutzzeit)



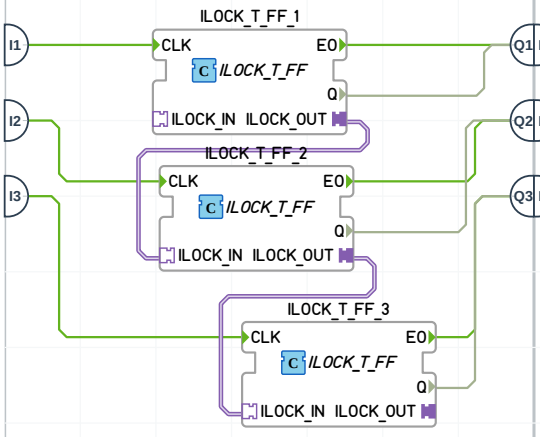
Uebung_205b

Interlock: ILOCK_SWITCH_PROTECT (Motor Reversierung
Priorität mit Schutzzeit)



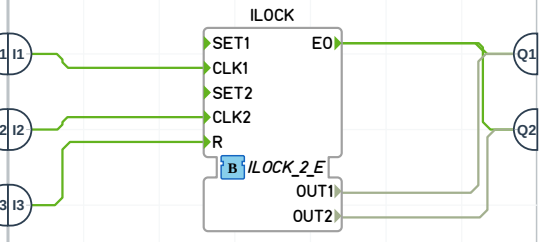
Uebung_206

Interlock: ILOCK_T_FF (Zwei gegenseitig verriegelte
Toggle-Flip-Flops via Adapter-Kette)



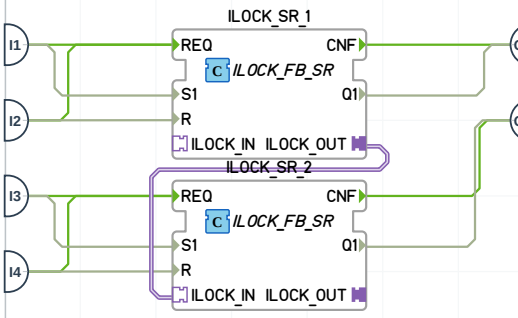
Uebung_206b

Interlock: ILOCK_T_FF (Drei gegenseitig verriegelte
Toggle-Flip-Flops via Adapter-Kette)



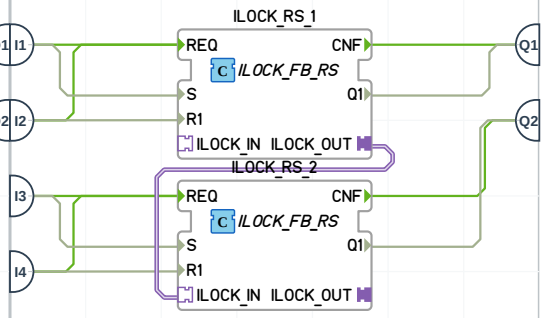
Uebung_207

Interlock: ILOCK_2_E (Event-gesteuertes bistabiles
Relais mit gegenseitiger Verriegelung)



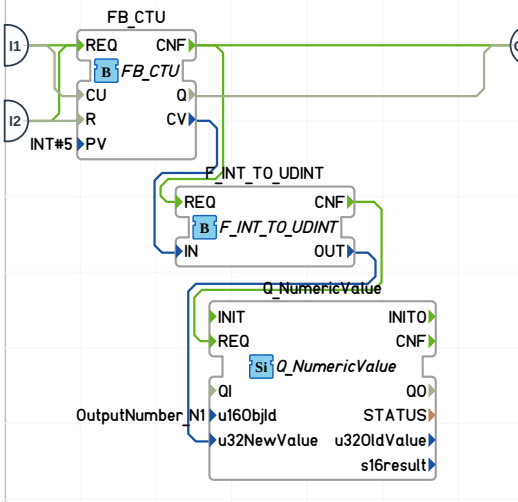
Uebung_208

Interlock: ILOCK_FB_SR (Zwei gegenseitig verriegelte Set-Dominante Latches via AX2-Adapter)



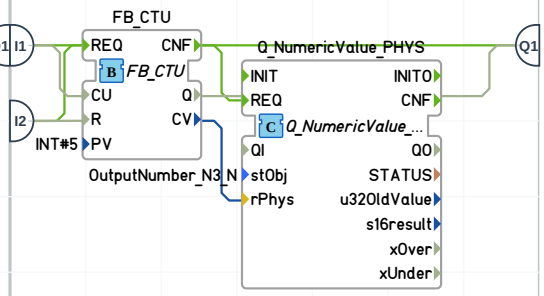
Uebung_209

Interlock: ILOCK_FB_RS (Zwei gegenseitig verriegelte Reset-Dominante Latches via AX2-Adapter)



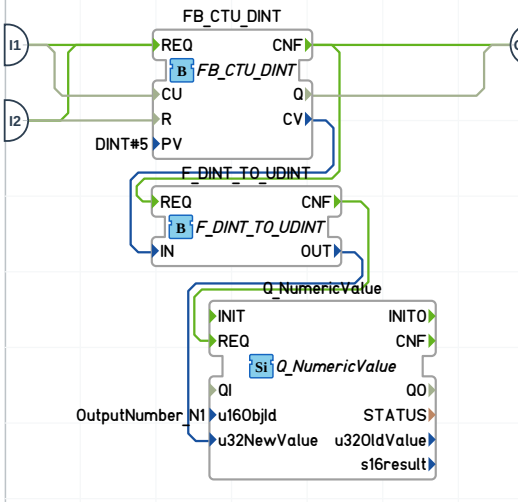
Uebung_210

Standard IEC 61131-3 FB_CTU (Vorwärtszähler, INT) mit Terminal-Ausgabe



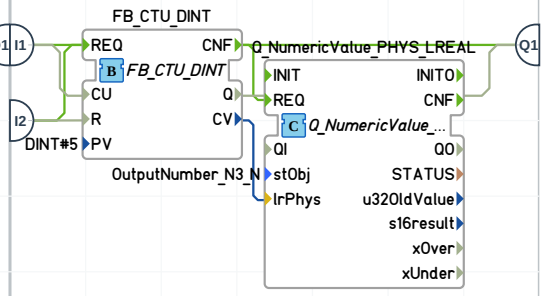
Uebung_210b

Standard IEC 61131-3 FB_CTU (Vorwärtszähler, INT) mit Terminal-Ausgabe



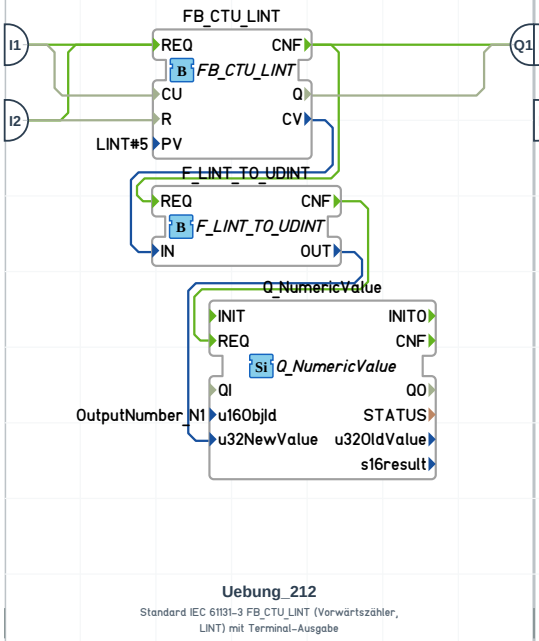
Uebung_211

Standard IEC 61131-3 FB_CTU_DINT (Vorwärtszähler, DINT) mit Terminal-Ausgabe



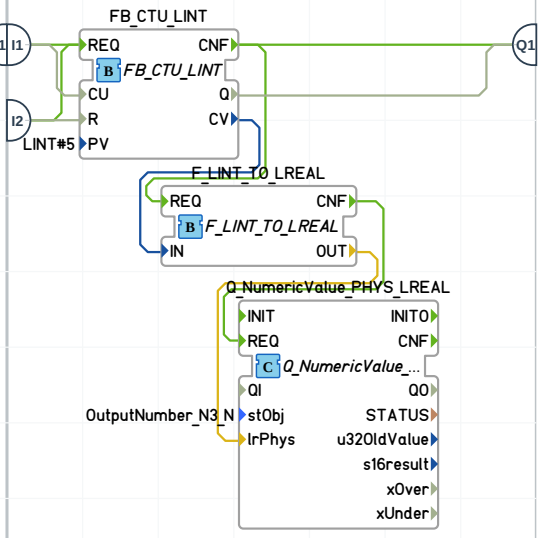
Uebung_211b

Standard IEC 61131-3 FB_CTU_DINT (Vorwärtszähler, DINT) mit Terminal-Ausgabe (PHYS_LREAL)



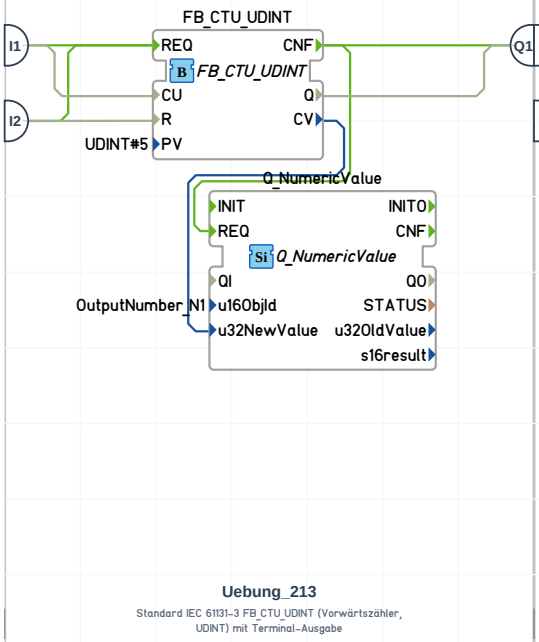
Uebung_212

Standard IEC 61131-3 FB_CTU_LINT (Vorwärtszähler, LINT) mit Terminal-Ausgabe



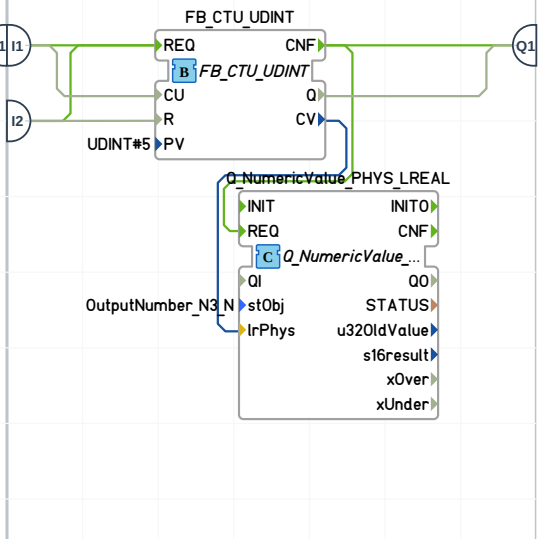
Uebung_212b

Standard IEC 61131-3 FB_CTU_LINT (Vorwärtszähler, LINT) mit Terminal-Ausgabe (PHYS_LREAL)



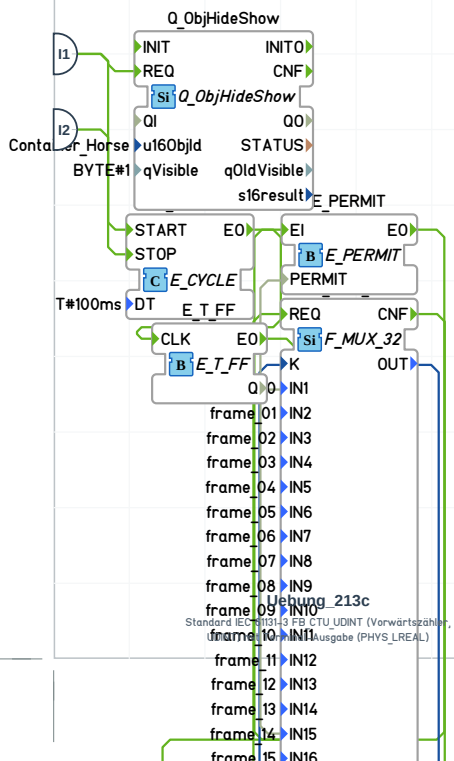
Uebung_213

Standard IEC 61131-3 FB_CTU_UDINT (Vorwärtszähler, UDINT) mit Terminal-Ausgabe



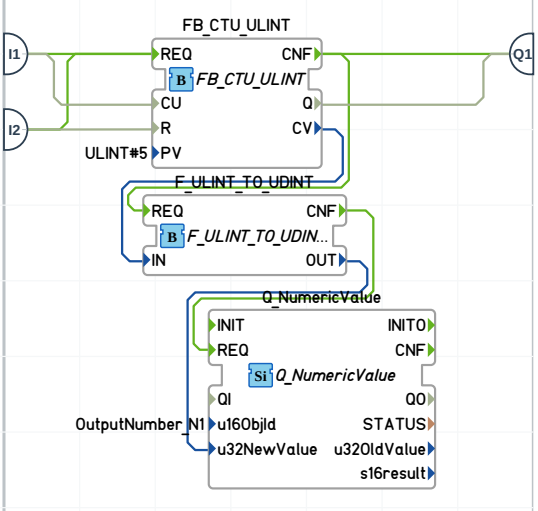
Uebung_213b

Standard IEC 61131-3 FB_CTU_UDINT (Vorwärtszähler, UDINT) mit Terminal-Ausgabe (PHYS_LREAL)



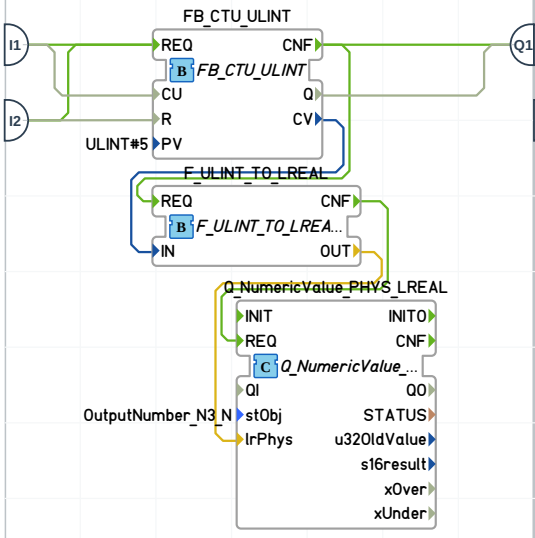
Uebung_213c

Standard IEC 61131-3 FB_CTU_UDINT (Vorwärtszähler, UDINT) mit Terminal-Ausgabe (PHYS_LREAL)



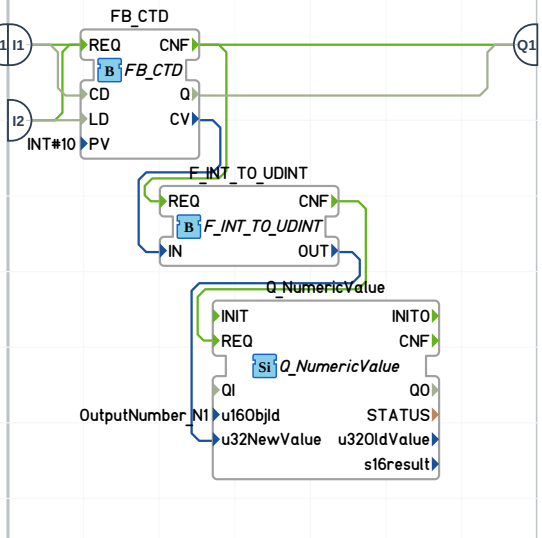
Uebung_214

Standard IEC 61131-3 FB_CTU_ULINT (Vorwärtszähler, ULINT) mit Terminal-Ausgabe



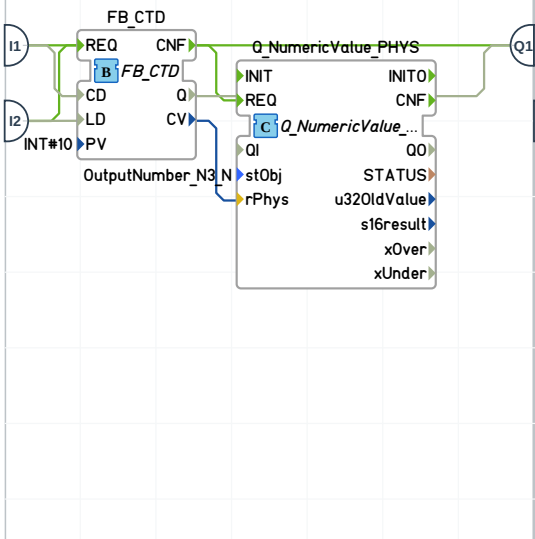
Uebung_214b

Standard IEC 61131-3 FB_CTU_ULINT (Vorwärtszähler, ULINT) mit Terminal-Ausgabe (PHYS_LREAL)



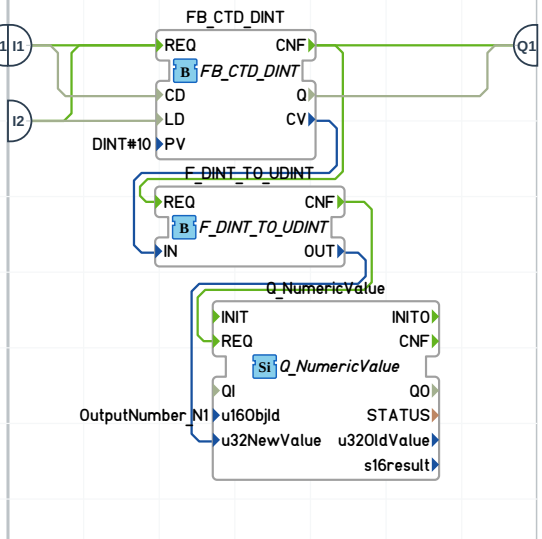
Uebung_215

Standard IEC 61131-3 FB_CTD (Rückwärtszähler, INT) mit Terminal-Ausgabe



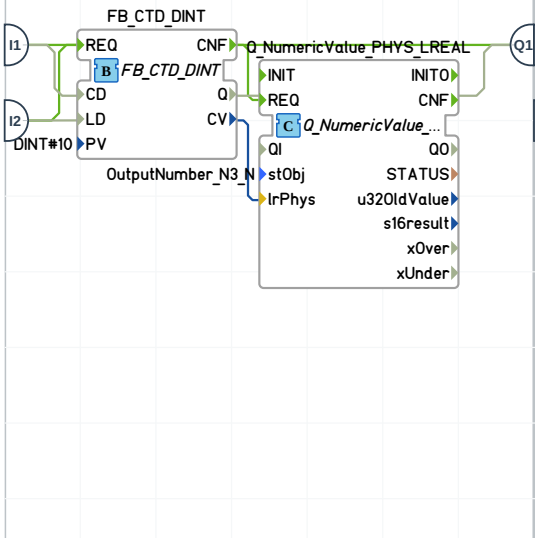
Uebung_215b

Standard IEC 61131-3 FB_CTD (Rückwärtszähler, INT) mit Terminal-Ausgabe (PHYS)



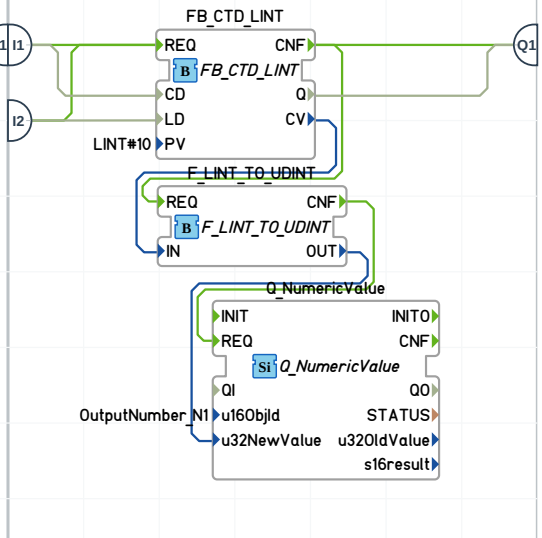
Uebung_216

Standard IEC 61131-3 FB_CTD_DINT (Rückwärtszähler, DINT) mit Terminal-Ausgabe



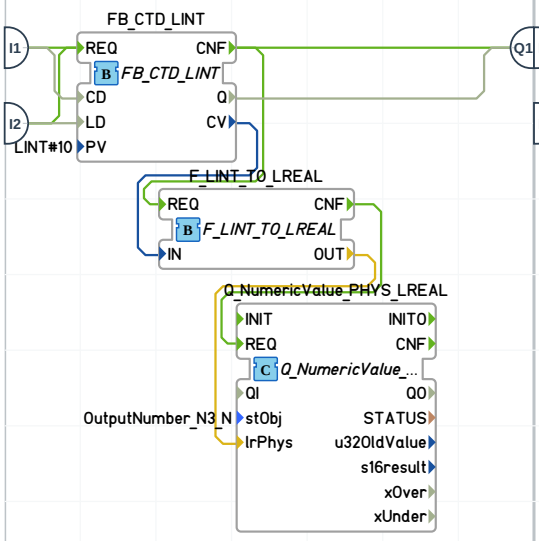
Uebung_216b

Standard IEC 61131-3 FB_CTD_DINT (Rückwärtszähler, DINT) mit Terminal-Ausgabe (PHYS_LREAL)



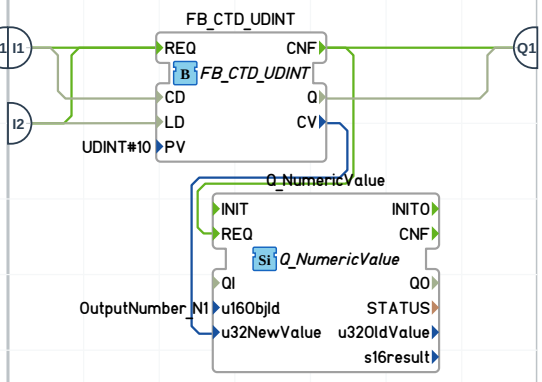
Uebung_217

Standard IEC 61131-3 FB_CTD_LINT (Rückwärtszähler, LINT) mit Terminal-Ausgabe



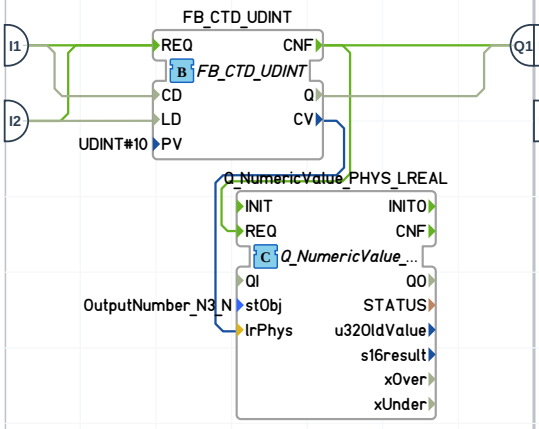
Uebung_217b

Standard IEC 61131-3 FB_CTD_LINT (Rückwärtszähler, LINT) mit Terminal-Ausgabe (PHYS_LREAL)



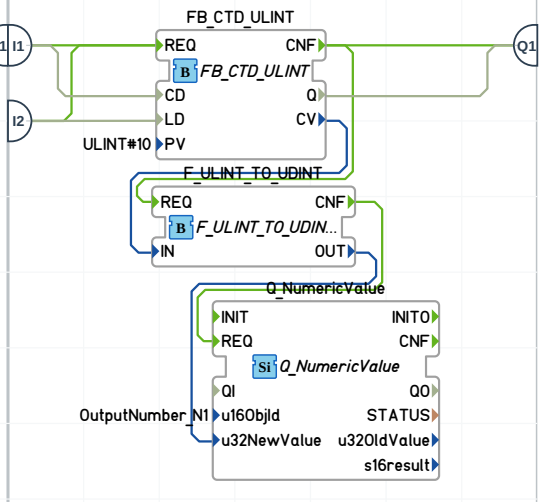
Uebung_218

Standard IEC 61131-3 FB_CTD_UDINT (Rückwärtszähler, UDINT) mit Terminal-Ausgabe



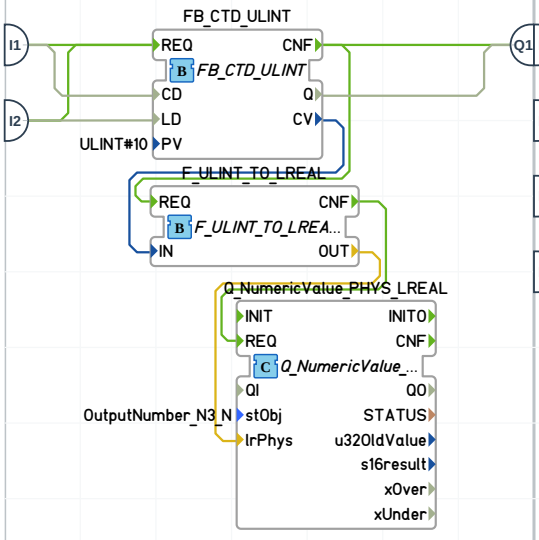
Uebung_218b

Standard IEC 61131-3 FB_CTD_UDINT (Rückwärtszähler, UDINT) mit Terminal-Ausgabe (PHYS)



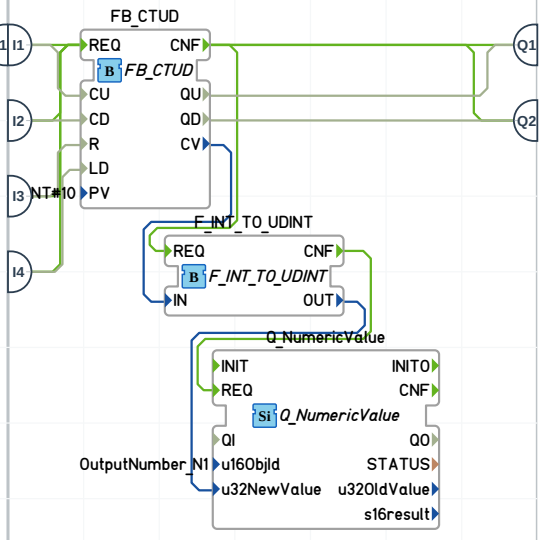
Uebung_219

Standard IEC 61131-3 FB_CTD_ULINT (Rückwärtszähler, ULINT) mit Terminal-Ausgabe



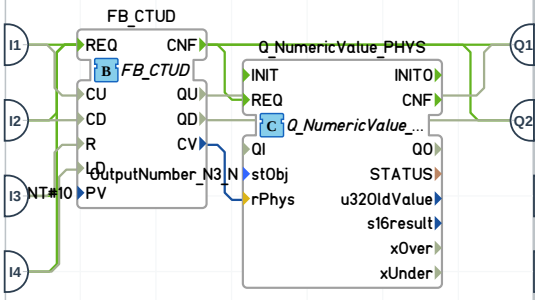
Uebung_219b

Standard IEC 61131-3 FB_CTD_ULINT (Rückwärtszähler, ULINT) mit Terminal-Ausgabe (PHYS_LREAL)



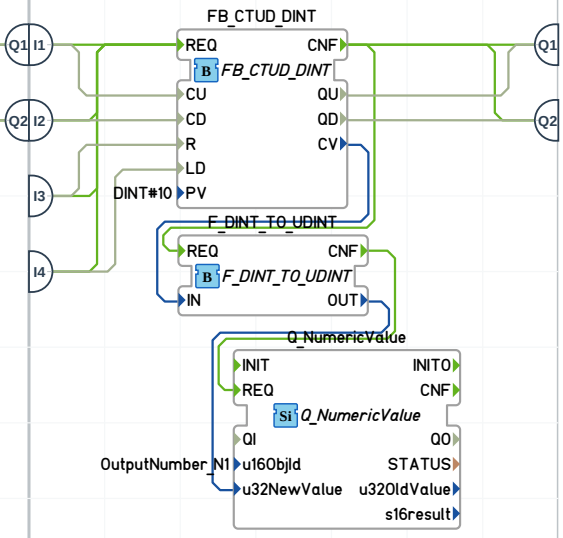
Uebung_220

Standard IEC 61131-3 FB_CTUD (Vor-/Rückwärtszähler, INT) mit Terminal-Ausgabe



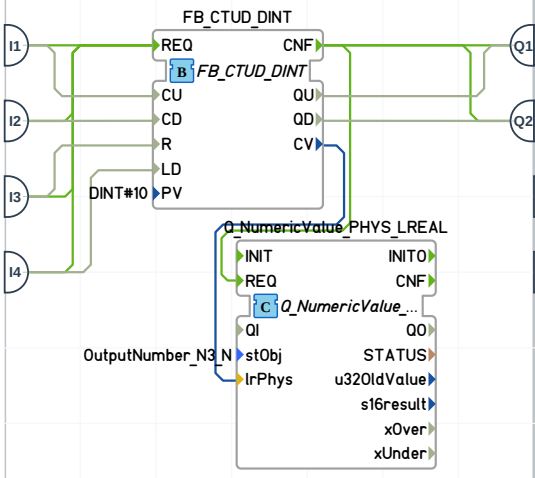
Uebung_220b

Standard IEC 61131-3 FB_CTUD (Vor-/Rückwärtszähler, INT) mit Terminal-Ausgabe (PHYS)



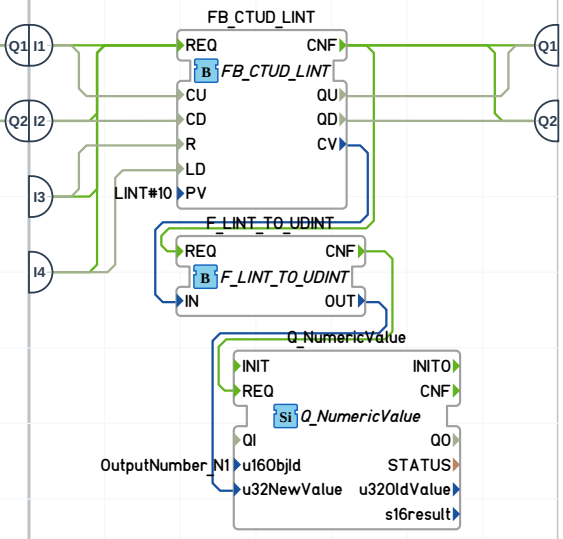
Uebung_221

Standard IEC 61131-3 FB_CTUD_DINT (Vor-/Rückwärtszähler, DINT) mit Terminal-Ausgabe



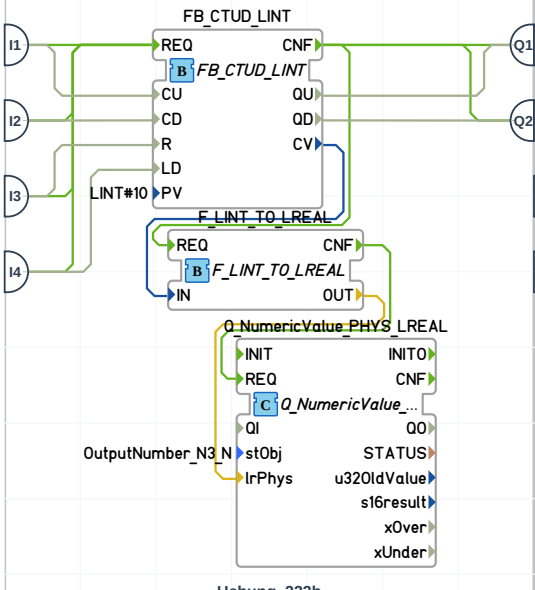
Uebung_221b

Standard IEC 61131-3 FB_CTUD_DINT (Vor-/Rückwärtszähler, DINT) mit Terminal-Ausgabe (PHYS)



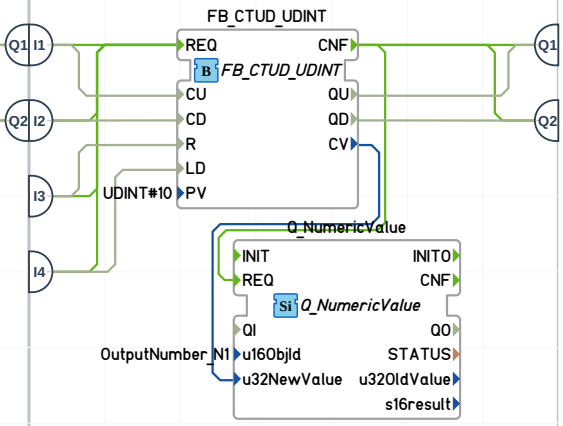
Uebung_222

Standard IEC 61131-3 FB_CTUD_LINT (Vor-/Rückwärtszähler, LINT) mit Terminal-Ausgabe



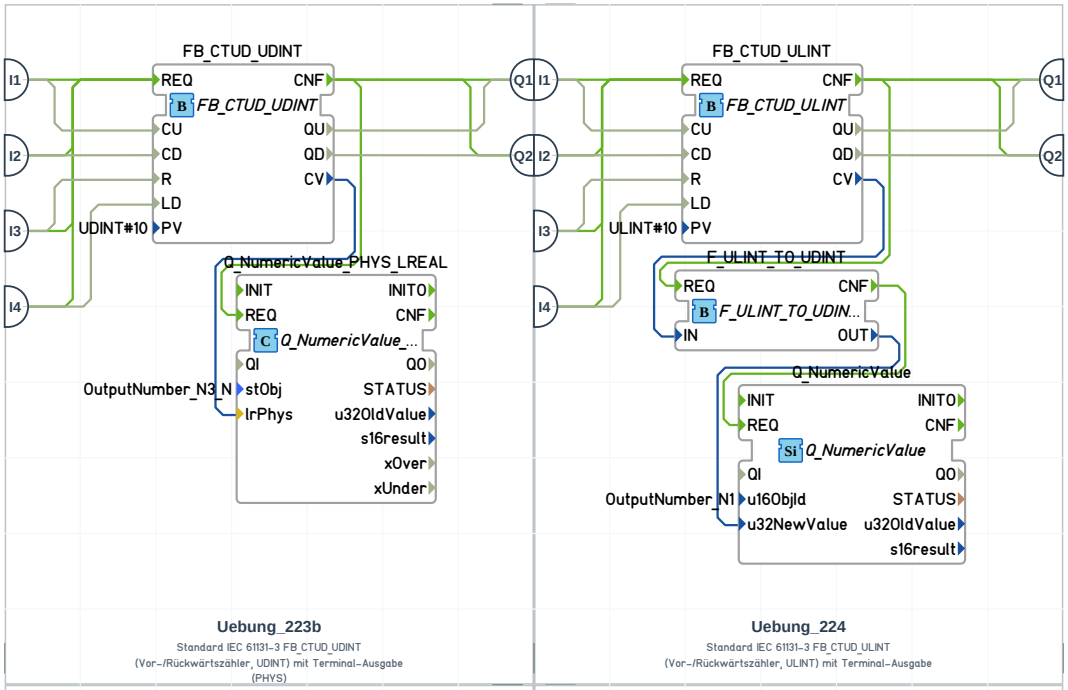
Uebung_222b

Standard IEC 61131-3 FB_CTUD_LINT (Vor-/Rückwärtszähler, LINT) mit Terminal-Ausgabe (PHYS_LREAL)



Uebung_223

Standard IEC 61131-3 FB_CTUD_UDINT (Vor-/Rückwärtszähler, UDINT) mit Terminal-Ausgabe

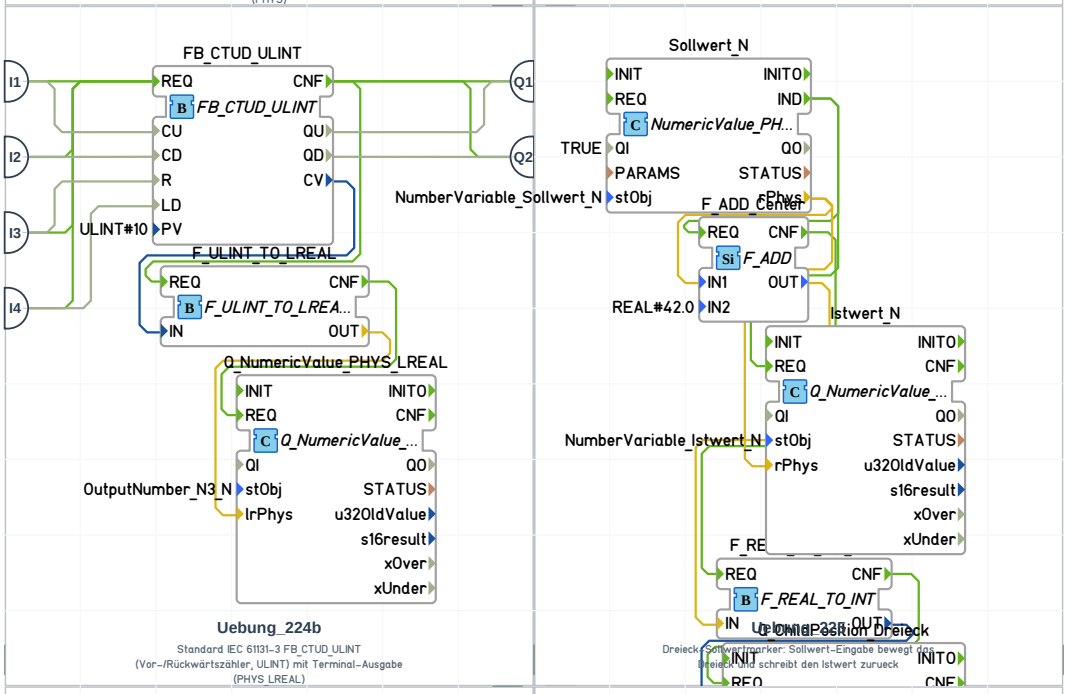


Uebung_223b

Standard IEC 61131-3 FB_CTUD_UDINT (Vor-/Rückwärtszähler, UDINT) mit Terminal-Ausgabe (PHYS)

Uebung_224

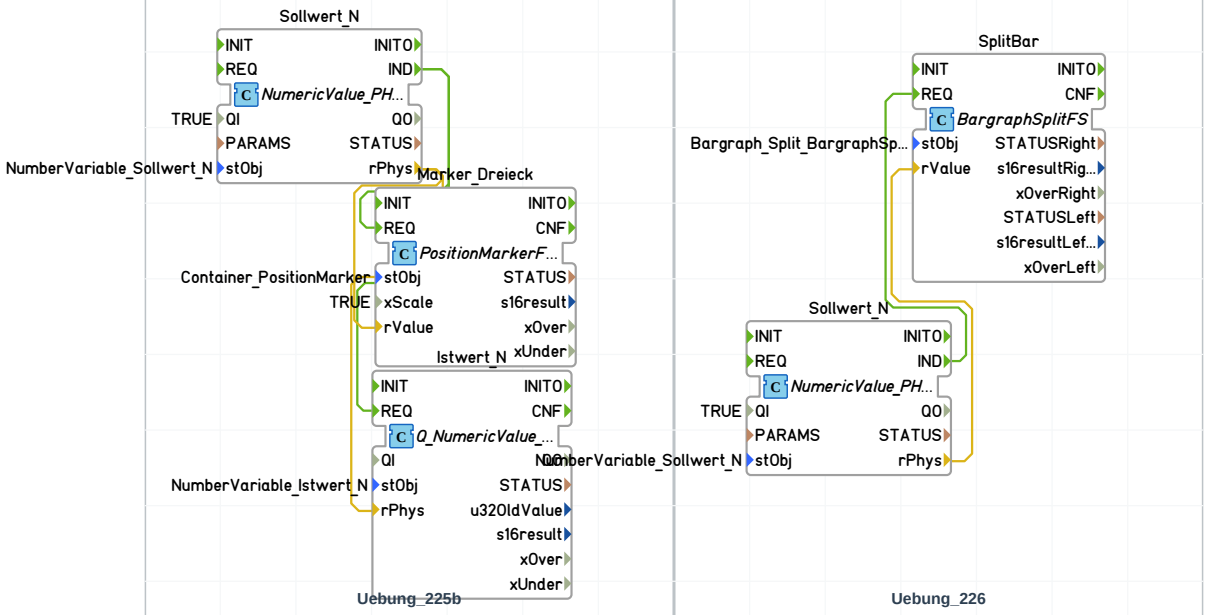
Standard IEC 61131-3 FB_CTUD_ULINT (Vor-/Rückwärtszähler, ULINT) mit Terminal-Ausgabe



Uebung_224b

Standard IEC 61131-3 FB_CTUD_ULINT (Vor-/Rückwärtszähler, ULINT) mit Terminal-Ausgabe (PHYS_LREAL)

Dreieck-Sollwertmarker: Sollwert-Eingabe bewegt das Dreieck und schreibt den Istwert zurueck

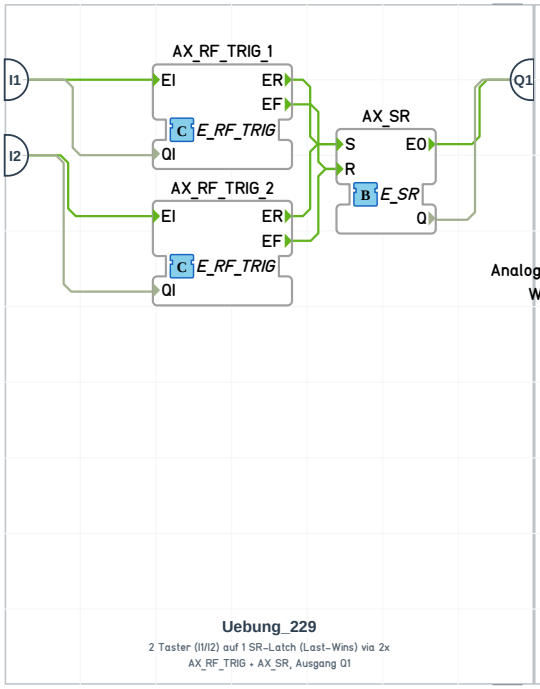


Uebung_225b

Dreieck-Sollwertmarker (wie Uebung_225, aber mit dem wiederverwendbaren PositionMarkerFS-Baustein statt F_ADD_Center/F_REAL_TO_INT_Pos/Q_ChildPosition_Dreieck)

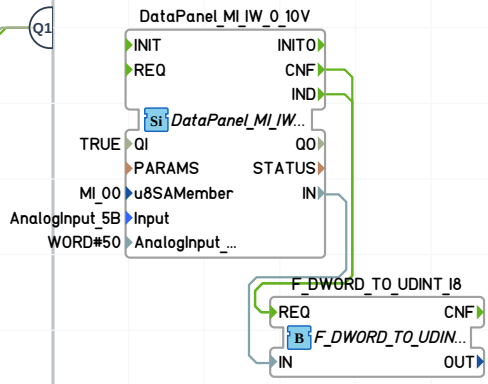
Uebung_226

Split-Bargraph: Sollwert-Eingabe treibt zwei nebeneinanderliegende Bargraphen als vorzeichenbehafteten Ausschlag



Uebung_229

2 Taster (I1/I2) auf 1 SR-Latch (Last-Wins) via 2x AX_RF_TRIG + AX_SR, Ausgang Q1



Uebung_230

Analog-Eingang